Abstraction Mechanisms for Virtual Avatars

Master's Thesis

submitted by

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Abstract

Virtual reality applications increasingly rely on avatar representations to facilitate social presence and user embodiment. However, photorealistic avatar rendering can trigger the uncanny valley effect, where near-realistic human representations evoke discomfort and reduced trust rather than engagement. Non-photorealistic rendering (NPR) offers a principled alternative, deliberately abstracting avatar appearance while preserving expressiveness and identity.

This thesis presents the design, implementation and evaluation of a real-time NPR stylisation system for virtual avatars. Eight techniques were developed iteratively: a geometry-based inverted hull outline, toon shading and its X-Toon extension, screen-space normal edge detection, Sobel edge detection with adaptive skin-to-clothing thresholding, Gaussian prefiltered Sobel, anisotropic Kuwahara painterly filtering and hierarchical multi-cue edge detection combining depth, normal and colour discontinuities. The techniques were implemented and compared across three avatar platforms: a Mixamo humanoid model, the Meta Avatars SDK and Avaturn. This revealed their visual behaviour and robustness under different mesh topologies and rendering pipelines. Avaturn was selected as the primary platform for the user study due to its facial rigging capabilities and higher photorealistic fidelity.

The user study investigates how visual abstraction affects the perceived trustworthiness of an avatar delivering advice. Participants watched short video scenarios in which a speaker recommended one of two options in a practical decision situation. The speaker was presented either as a real person, a photorealistic Avaturn avatar or an NPR-stylised avatar. After each scenario, participants rated the speaker on eeriness, likeability and three dimensions of trust. Participants registered the stylised avatar as visually distinct from its photorealistic counterpart. It was nonetheless rated no less eerie and no more trustworthy than that counterpart and both avatars fell well below the real speaker on every trust measure. Eeriness did predict lower trust within participants across all three dimensions. Abstraction and uncanny affect therefore behaved as separable properties for the composite evaluated here. This qualifies the expectation that stylisation alone relocates an avatar out of the uncanny valley.

Keywords: non-photorealistic rendering, virtual reality, avatar abstraction, trust perception, edge detection, Avaturn, uncanny valley, Kuwahara filter.

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Abbreviations

1D	One-Dimensional
2D	Two-Dimensional
3D	Three-Dimensional
ANOVA	Analysis of Variance
API	Application Programming Interface
AR	Augmented Reality
CG	Computer-Generated
ESF	Extensible Shader Framework
FPS	Frames Per Second
GPU	Graphics Processing Unit
HCI	Human-Computer Interaction
HLSL	High-Level Shading Language
HMD	Head-Mounted Display
HSV	Hue, Saturation, Value (colour space)
LOD	Level of Detail
NPR	Non-Photorealistic Rendering
PBR	Physically-Based Rendering
RGB	Red, Green, Blue
SDK	Software Development Kit
SSS	Subsurface Scattering
UI	User Interface
URP	Universal Render Pipeline
UV	Texture Coordinate Axes (U and V)
VR	Virtual Reality

1. Introduction

1.1. Motivation

Computer-generated (CG) humanoid characters have become an ordinary part of contemporary digital experiences, especially as generative AI video content is widely encountered on a daily basis. As virtual assistants, game characters and avatars in immersive environments increasingly involve users in sustained interactions with virtual humans, users are no longer limited to passive viewing [1]. Therefore, engaging with a CG human raises the value of its design, since the character is no longer a background presence but an active participant in the interaction. Its design affects how users perceive and interact with it, as well as aesthetic appeal. These effects raise several design concerns with consequences across film, games and other domains where virtual humans appear. One of the most difficult is how realistically or abstractly such a character should be rendered [2]. As a result, this tension between realism and abstraction in avatars is the starting point for the work presented in this thesis.

This design choice can be represented along a fidelity axis. Higher fidelity corresponds to greater human likeness in that axis, where human likeness refers to the degree to which an avatar visually resembles a real human. Furthermore, an avatar's perceived anthropomorphism and behaviour have been demonstrated to increase users' confidence in its guidance and their willingness to follow it [3]. However, this increase in confidence does not scale linearly with likeness across the fidelity axis. Mori termed this effect the uncanny valley. Comfort rises with human likeness only up to a point, then drops sharply for entities that appear almost but not quite human, before recovering near perfect likeness [4]. Consequently, an avatar that approaches full realism may reduce users' confidence rather than enhance it. The same pattern holds for CG characters generally [2]. Avatars fall within this category and their stylisation can affect users' behavioural choice [5]. Together, these results suggest that rendering style influences not only users' subjective confidence but also the decisions they make. This indicates that avoiding the perceptual pitfalls associated with highly realistic avatars through alternative visual styles could preserve both.

Non-Photorealistic Rendering (NPR) provides an alternative approach by deliberately abstracting an avatar's visual appearance while preserving its recognisability and expressive capabilities. Moreover, NPR allows designers to position an avatar at different locations along the fidelity axis through controlled stylisation rather than maximising realism. The central question is accordingly not whether stylised avatars are visually appealing. It is whether deliberate abstraction can maintain or even improve users' trust while avoiding the perceptual costs of near photorealistic rendering. That question is becoming increasingly relevant as virtual avatars move beyond research settings into widely used consumer applications such as gaming and advertising [6]. Thus, determining the balance between realism and abstraction is a central challenge in the design of trustworthy virtual avatars. How this tension is resolved will shape whether virtual avatars can transition from novelty to genuinely trusted participants in social and professional contexts.

1.2. Problem Statement

Although NPR provides a potential approach for controlling avatar appearance, applying NPR stylisation to modern photorealistic avatar systems is not straightforward. Current avatar platforms expose substantially different rendering architectures and each imposes its own constraints on the implementation of stylisation techniques. Specifically, some platforms provide only a single shader integration point while keeping the remainder of the rendering pipeline inaccessible. In particular, this prevents additional render passes or access to intermediate shading information. Others expose broader pipeline access but introduce different challenges arising from photogrammetric textures and asset representations. Accordingly, the applicability of an NPR approach depends on both the desired visual effect and the rendering capabilities of the underlying platform.

A recent avatar stylisation study focused on perceptual outcomes rather than the practical challenges of production integration [5]. Classical NPR techniques share this same gap, since they are typically demonstrated

in rendering environments where developers have direct control over rendering operations and asset creation [7]. It remains unclear how these techniques transfer to avatar platforms with substantially different integration constraints.

Closing this technical gap does not by itself indicate how NPR should be applied. A further, open question is which design choices remain visually and perceptually viable under the most restrictive of these constraints. This is because stronger abstraction typically reduces the perceived intensity of facial expressions, even where basic recognition is preserved [8], [9]. More broadly, the trade-off between abstraction and facial legibility has not yet been characterised for avatar rendering. Even where integration is technically feasible, this does not guarantee an effective balance between abstraction and expressive clarity.

Resolving these technical challenges naturally raises a broader empirical concern. This is due to the abstraction's ultimate goal of conveying trust rather than merely rendering successfully. Two questions remain open. The first is whether abstraction lessens the drop in comfort that the uncanny valley predicts. The second is whether it can render a stylised avatar as trustworthy as an actual human speaker in a digital environment. This concerns the perceptual and behavioural effect of stylisation rather than its technical feasibility or design. Equally, it is consequential for any domain in which avatar credibility is at stake: social VR, remote consultation, training simulation, advertising and education.

This thesis addresses all three gaps in turn. NPR techniques were developed and integrated across three avatar platforms and this resolved the technical gap. A dedicated pilot study then calibrated the resulting composite to close the gap in technique selection. Finally, a controlled user study evaluated whether the stylised avatar could escape the uncanny valley and reach a level of trust comparable to a real human speaker. This addressed the trust gap.

1.3. Research Questions

The following three research questions guide this work:

RQ1: How can NPR stylisation techniques be adapted to avatar platforms with different rendering architectures and what capabilities and limitations does each platform impose?

RQ2: Which NPR technique or combination of techniques best balances visual abstraction against facial and expressive legibility under production rendering constraints?

RQ3: How does NPR stylisation affect perceived eeriness and trustworthiness and can an NPR avatar reach the trustworthiness of a real human speaker?

1.4. Contributions

This thesis makes the following contributions:

- **Empirical characterisation of NPR behaviour across distinct avatar platforms.** Classical NPR operators were found to behave differently under production rendering constraints across three avatar platforms: Mixamo, the Meta Avatars SDK and Avaturn. They were chosen for their differing feature sets and as candidates for the user study described later in this section. Some techniques required recalibration depending on the texture of the avatar. Others became unusable on the most restrictive platform due to its more limited pipeline access. The observed platform differences indicate which NPR techniques transfer across production rendering pipelines and which require adaptation tailored to the platform.
- **Iteratively developed stylisation techniques.** Eight progressively refined techniques were produced for production avatar platforms, each addressing limitations observed in its predecessor. They range from a geometry-based outline through edge detection to a painterly post-processing filter, and all are selectable at runtime without recompilation.

- **Preprocessing workflow for reliable edge detection on photogrammetric avatar textures.** A preprocessing workflow was developed to restore facial feature legibility on photogrammetrically reconstructed avatar textures. The workflow enables reliable NPR edge detection where gradient-based techniques would otherwise fail.
- **Perceptual calibration of NPR parameters.** A pilot study determined the final NPR parameter values through preference testing rather than visual inspection alone. Participants ranked rendered avatar variants by perceived trustworthiness and these results also determined which techniques were retained in the final composite.
- **A measurement framework combining self-reported and behavioural trust indicators.** The study questionnaire combined validated instruments from prior trust, likeability and eeriness research with a behavioural advisor ranking measure. The combination addresses a methodological gap identified in prior reviews, where reported trust ratings and behavioural choice have rarely been measured together for stylised avatars.
- **Experimental design that isolates rendering style from confounding factors.** Three realistic decision scenarios were constructed so that each targets a different dimension of interpersonal trust: competence, benevolence or integrity. Confounds were controlled by alternating which option each scenario recommends and alternating the order in which options are discussed. The same script structure was applied uniformly across all three scenarios. Any difference in trust ratings therefore reflects rendering style rather than variation in scenario content or argumentation pattern.
- **Evidence on NPR stylisation and perceived avatar trust.** A within-subjects user study was conducted using prerecorded video scenarios in which the same avatar and speaker appear across three rendering conditions. Comparing the three conditions yields empirical evidence on how NPR stylisation affects the perceived trustworthiness of an avatar's advice relative to both a photorealistic counterpart and a real human speaker.

1.5. Thesis Outline

The remainder of this thesis is structured as follows:

Chapter 2 introduces the technical and theoretical foundations that the later chapters assume, spanning rendering concepts, the image processing operators behind the stylisation techniques and the constructs of trust and eeriness.

Chapter 3 surveys prior work relevant to avatar realism, trust and NPR rendering, then identifies the specific gap that this thesis addresses.

Chapter 4 details the research design and system architecture, elaborating on the research questions above and the user study design.

Chapter 5 describes the full implementation of the NPR shader system across all three avatar platforms.

Chapter 6 reports the technical evaluation of each stylisation technique and presents the results of the user study, followed by a discussion of what those outcomes indicate and the limitations that qualify them.

Chapter 7 summarises the contributions of this thesis and identifies directions for future work.

2. Background

This chapter introduces the technical and theoretical foundations that the remainder of the thesis assumes. It opens with the notation used throughout, then presents the rendering concepts on which the stylisation techniques depend. Virtual avatars follow, together with the constructs of trust and eeriness that the user study measures. The final two sections cover the image processing operators from which the stylisation techniques derive and the structure-preserving filter behind painterly abstraction.

2.1. Notation and Conventions

Scalars are denoted by italic letters x. Vectors appear in bold, as with the surface normal $\mathbf{N}$. Convolution kernels take an italic uppercase letter and a directional subscript, as in G_x. The convolution of an image I with a kernel K is written $K * I$. Colour values are treated as linear RGB triples in the range $[0, 1]$ unless stated otherwise.

An image is treated as a function $I(x, y)$ over a discrete pixel grid, where x increases towards the right of the screen and y increases upwards, matching the coordinate system that the GPU exposes to a shader.

2.2. Rendering Concepts

A render pass is one complete traversal of scene geometry. Its result is written into a render target, an off-screen image buffer. A fragment shader that operates within a single pass sees only the surface point it is shading. Any operator needing a spatial neighbourhood must therefore gather that neighbourhood itself. Additional passes remove this restriction. A post-processing pass renders one full-screen quad and samples the render target left by the preceding pass, which gives every fragment access to the whole rendered image. Screen-space depth and normal textures reach later passes by the same route. Unity's Universal Render Pipeline (URP) schedules such injected work through a renderer feature [10]. Every extra pass costs a further traversal of the geometry or of the full screen. The number of passes a platform permits is thus an engineering constraint rather than a free parameter.

Despite this restriction on passes, the material data itself remains unaffected. It is held in separate texture maps under physically based rendering (PBR). The albedo map is the one these techniques analyse. This map stores the base colour of the surface with no lighting baked into it. However, the way it was produced matters. An albedo painted by an artist carries material colour alone. A photographic albedo is instead reconstructed from images of a real subject. Accordingly, it retains the illumination present when those images were captured. No operator examining the texture in isolation can separate that illumination from genuine material variation. This all assumes the material bound to the surface stays fixed. A level of detail (LOD) system substitutes a simplified mesh as the character moves further away [11]. Some implementations swap the material along with it. A stylisation applied through a custom material assignment can consequently be replaced at runtime.

Runtime change is not confined to the material. The geometry also moves, since an animated character never holds a fixed pose. Vertex positions are displaced by one of two mechanisms. A skeleton deforms the body and every vertex carries bone weights that blend the transforms acting on it. Blendshapes handle finer motion such as facial expression. They hold alternative vertex positions that are interpolated against the base mesh [12].

2.3. Avatars, Trust and Eeriness

An avatar is a digital body that represents a person within a virtual space. The term spans forms ranging from abstract geometric markers to fully articulated humanoid figures. Its visual appearance is the primary cue from which viewers form an impression of the person that the avatar represents. Consequently, the central social

judgment an observer makes about an avatar is one of trust: the degree to which they are willing to rely on the person it portrays.

This judgment is not a single quantity. Perceived trustworthiness decomposes into three independently evaluable components: ability, benevolence and integrity [13]. *Ability* denotes competence within a relevant domain, *benevolence* the extent to which the trustee appears to care about the observer's wellbeing and *integrity* adherence to acceptable principles. McKnight and Chervany extended this framework for human-computer interaction by adding predictability as a fourth trusting belief dimension [14]. Together, these four components describe what an observer believes about the trustee and they leave the perceptual response to appearance unaddressed.

A complementary family of instruments targets that perceptual dimension directly. The Godspeed questionnaire provides five subscales totalling 24 semantic differential items, each rated on a five-point scale between two bipolar anchors [15]. Its likeability subscale, which comprises five of those 24, is adopted in this thesis. However, eeriness is not captured by any of the five Godspeed subscales. Ho and MacDorman addressed this gap with a dedicated instrument: eight items for eeriness alongside six for perceived humanness [16]. Eeriness, as they operationalise it, is the unsettled perceptual response that arises when a stimulus approaches but does not reach the peak of human likeness. Yet the two subscales proved orthogonal in validation, with $r \approx .02$ and this means that high human-likeness and strong discomfort can co-occur in the same stimulus. Both instruments are used in the user study described in Chapter 6.

2.4. Edge Detection

A digital image consists of a grid of pixels, each storing an intensity value. An edge is conventionally defined as a location at which this intensity changes rapidly [17]. Detecting such locations involves estimating the image gradient and applying a threshold to identify pixels whose gradient magnitude is large.

2.4.1. Gradient-Based Detection

Given an image $I(x, y)$, the gradient is the vector $\nabla I = (\partial I/\partial x,\ \partial I/\partial y)$, whose two components measure the rate of intensity change along the horizontal and vertical axes. Its magnitude

$$M = \|\nabla I\| = \sqrt{\left(\frac{\partial I}{\partial x}\right)^2 + \left(\frac{\partial I}{\partial y}\right)^2} \tag{2.1}$$

provides a scalar measure of edge strength at each pixel. This magnitude stays small on uniform surfaces and grows large at boundaries. Moreover, it merges both directional derivatives into a single value, so it detects edges irrespective of their orientation. The partial derivatives are approximated in practice by discrete convolution with finite difference kernels, small filters that estimate local variation from the difference between neighbouring pixel values.

The Sobel operator is a widely used kernel of this kind. It estimates the two directional derivatives with a pair of 3×3 kernels [17]:

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}, \qquad G_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}. \tag{2.2}$$

G_x responds to horizontal intensity transitions and thereby detects vertically oriented edges, while G_y measures vertical transitions and identifies the horizontal ones. Each kernel is convolved with the image separately and the two responses combine pointwise into the gradient magnitude

$$M = \sqrt{(G_x * I)^2 + (G_y * I)^2}. \tag{2.3}$$

The central row and column of each Sobel kernel carry weights of ± 2, whereas the simpler Prewitt operator weights all three rows and columns equally at ± 1 [17]. This doubling averages the samples nearest the centre

before the difference is taken. The Sobel operator consequently smooths slightly as it differentiates and its robustness to noise improves.

A smaller alternative is the Roberts Cross operator, which differentiates along the two diagonals of a 2×2 neighbourhood [18]:

$$R_1 = \begin{bmatrix} +1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad R_2 = \begin{bmatrix} 0 & +1 \\ -1 & 0 \end{bmatrix}. \tag{2.4}$$

Its edge magnitude is commonly approximated as $|R_1 * I| + |R_2 * I|$, which avoids the square root of Equation (2.3) [17]. The operator has no centre sample and reads four pixels rather than nine. Its response is therefore more local and cheaper than that of Sobel, at the cost of greater sensitivity to noise.

Both operators ultimately act on the image intensity function $I(x, y)$ rather than on the three colour channels of a rendered frame. Applying them per channel would allow a difference in hue carrying no change in brightness to register as an edge. Gradient computation is accordingly performed on a single luma channel formed as a weighted sum of the red, green and blue components:

$$L = 0.2126\,R + 0.7152\,G + 0.0722\,B, \tag{2.5}$$

following the coefficients defined in ITU-R Recommendation BT.709 for high-definition video [19]. The weighting reflects the differing sensitivity of the human visual system to each primary. Differentiating this channel aligns edge strength with perceived brightness contrast rather than with an arbitrary numerical difference between colour channels.

2.4.2. Screen-Space Normal Derivatives

The kernels above require several texture samples per pixel. GPUs offer a cheaper alternative for any value already interpolated across the surface. Since fragments are rasterised in 2×2 quads, the hardware can return the finite difference of such a value between adjacent fragments. This is exposed by the `ddx` and `ddy` intrinsics at no sampling cost.

Applying these intrinsics to the world-space surface normal $\mathbf{N}$ gives

$$M_N = \sqrt{\|\mathtt{ddx}(\mathbf{N})\|^2 + \|\mathtt{ddy}(\mathbf{N})\|^2}, \tag{2.6}$$

which responds to abrupt changes in surface orientation such as creases, mesh boundaries and regions of high curvature. Equally important, the operator measures change per pixel rather than per unit of surface area. A crease of fixed world size spans fewer pixels as the camera retreats, so the orientation change recorded across one pixel step grows with viewing distance. The extent of the detected edge therefore varies with the viewpoint instead of remaining tied to the surface.

2.4.3. Gaussian Prefiltering

The operators introduced so far are sensitive to small fluctuations in their input. Differentiation amplifies these high-frequency components. Therefore, a noisy input yields an unstable gradient. Marr and Hildreth established that reliable edge detection requires low-pass smoothing before the gradient is computed [20]. This smoothing need not run as a separate step, since both convolution and differentiation are linear operations. The smoothed gradient may thus be written as

$$\nabla(G_\sigma * I) = (\nabla G_\sigma) * I, \tag{2.7}$$

where G_σ is a Gaussian kernel with standard deviation σ. Canny built an influential detector on this principle [21]. The detector smooths the image with a Gaussian first. It then computes the gradient of the smoothed result. Non-maximum suppression comes last and the surviving edge map is thin and stable.

The Gaussian is separable. A 2D convolution accordingly decomposes into one horizontal pass and one vertical pass and this is what makes the operator inexpensive. The saving depends on storing the horizontal result so

that neighbouring pixels can share it. However, a shader that must produce its output in one execution has nowhere to store it. Every fragment consequently repeats the full 2D neighbourhood itself and the redundant work grows with the size of the kernel.

2.5. Structure-Preserving Filtering

Gradient operators mark boundaries by drawing lines over an otherwise unchanged surface. A second family abstracts the surface itself and the Kuwahara filter is the member of that family used in this thesis. It removes detail inside high-contrast regions while protecting shape boundaries in low-contrast ones [22]. The filter divides the neighbourhood around each pixel into k overlapping subregions and computes the mean μ_i and variance σ_i^2 of each. It then emits the mean of whichever subregion is most homogeneous:

$$F_{\text{Kuwahara}}(x, y) = \mu_j, \qquad j = \arg\min_i \sigma_i^2. \tag{2.8}$$

Any subregion straddling a boundary carries a higher variance and loses this comparison. Interior regions therefore flatten towards uniform fills while boundaries stay sharp and the result resembles painterly brushwork [22]. The classical formulation partitions the neighbourhood into four quadrants of equal size and it accordingly treats every orientation alike.

Orientation is the property that the classical formulation discards. Kyprianidis et al. recovered it by aligning the sampling window with local image structure, which they estimate through the structure tensor [22]. For an image $\mathbf{f}$ with colour-valued partial derivatives $\mathbf{f}_x$ and $\mathbf{f}_y$, the smoothed tensor is the 2×2 matrix

$$\mathbf{S} = G_\rho * \begin{bmatrix} \mathbf{f}_x \cdot \mathbf{f}_x & \mathbf{f}_x \cdot \mathbf{f}_y \\ \mathbf{f}_x \cdot \mathbf{f}_y & \mathbf{f}_y \cdot \mathbf{f}_y \end{bmatrix}, \tag{2.9}$$

where each dot product runs over the colour channels and the Gaussian G_ρ smooths every component so that the estimate reflects a neighbourhood rather than one pixel. Its eigenvectors give the dominant local orientation and the eigenvalues indicate how strong that orientation is. An elliptical kernel derived from the two aligns the brush marks with salient contours. That smoothing step is itself a convolution across the image, so it requires a render pass of its own.

3. Related Work

This chapter examines the prior work most relevant to the central research questions of this thesis. That examination spans three established areas. The first is the perceptual and affective consequences of avatar realism and the uncanny valley phenomenon. The second is trust, credibility and social presence in virtual agent interactions. The third is the technical design space of NPR rendering for animated human figures. The chapter concludes by identifying the specific gap that this thesis addresses.

3.1. The Uncanny Valley and Avatar Realism

The uncanny valley hypothesis was first articulated by Mori. He proposed that human comfort towards an artificial figure increases with its human likeness. This rise holds only up to a critical threshold. Beyond it, subtle imperfections cause a sharp drop in comfort. The drop persists until the figure appears almost fully human. Only then is comfort recovered [4]. This relationship is captured in the curve shown in Figure 3.1.

Mori plotted separate curves for still and moving figures. Both follow the same general shape. The critical observation is that the curve for moving figures descends far deeper into negative comfort than the curve for still ones. Moreover, the reference points such as a corpse, a zombie and a prosthetic hand, that Mori placed inside the valley share a common characteristic. Each looks human in most respects, but each has one flaw that people notice instantly [4]. A figure at roughly sixty percent human likeness, the region occupied by a stuffed animal or a humanoid robot, sits on the safe rising slope before the drop. A figure at eighty to ninety-five percent is realistic enough to raise expectations of full fidelity, but not realistic enough to satisfy them. Hence, it occupies the most problematic region of the curve. Motion makes that positioning considerably worse.

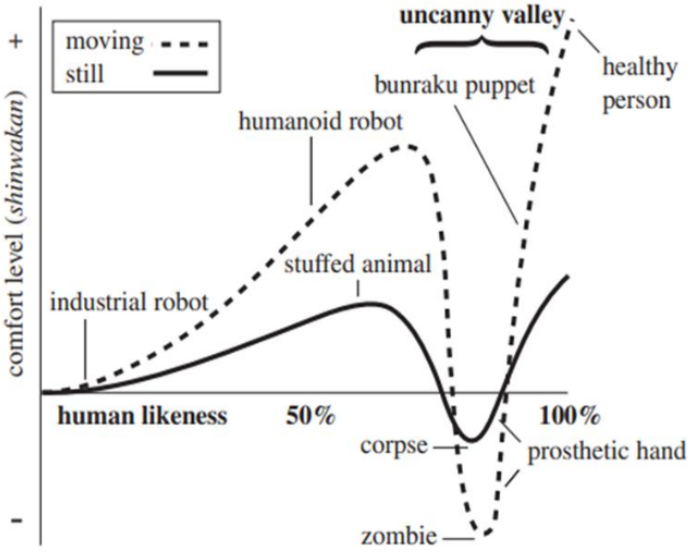


Figure 3.1.: The vertical axis represents comfort level (*shinwakan*). The horizontal axis represents degree of human likeness. The solid curve describes affective response to still figures, while the dashed curve describes figures in motion [4].

Furthermore, MacDorman and Ishiguro advanced the uncanny valley from a qualitative hypothesis into an empirically tractable research programme. They extended the theory to androids, which are robots designed to be indistinguishable from humans in appearance and behaviour, as shown in Figure 3.2. It has been argued that these androids offer a uniquely controllable stimulus for cognitive and social science research. This is because androids can be positioned at any point along the human likeness continuum with experimental control [23]. An android positioned at such an ambiguous point falls into the region Mori's curve identifies as the sharpest

drop in comfort [4]. Consequently, any near human avatar risks the same response if its rendering leaves this ambiguity unresolved.

Figure 3.2.: An android (left, Replicee R1) alongside a mechanical-looking humanoid robot (right, Eveliee P1), developed at Hiroshi Ishiguro's Intelligent Robotics Laboratory [23].

This risk became the subject of direct empirical study. McDonnell et al. first demonstrated that viewers perceive emotional actions in animated characters largely independent of avatar body style. This independence remained consistent across photorealistic, cartoon and stylised skeletal representations [24]. Stylised rendering can accordingly preserve the communicative legibility of avatar behaviour. The same research group later investigated the perceptual consequences of render style choice more directly. Participants were exposed to video stimuli of the same animated character, rendered across eleven distinct styles spanning the range from photorealistic to fully stylised. The results showed a valley-shaped pattern across appeal, reassurance and trustworthiness alike: styles at the most abstract and the most realistic ends of the range scored significantly higher on all three measures than styles in the mid-realism range. The authors thereby established that stylised rendering is not merely a technical shortcut but a principled perceptual strategy for sidestepping the uncanny valley [2]. Taken together, these findings suggest that stylised rendering can avoid the perceptual and trust costs concentrated in the mid-realism range without requiring full photorealistic fidelity.

Nevertheless, Dill et al. validated the uncanny valley response specifically for CG characters drawn from contemporary commercial films and games. Familiarity ratings followed Mori's predicted curve, with the strongest negative response in the near-human range. Motion amplified this effect for every stimulus tested and participants named the eyes and the mouth as the facial regions producing the most discomfort [25]. This confirmation that the valley extends to digital characters from popular media, not only to physical robots or androids, motivates its consideration across all virtual avatar contexts. Modern avatar systems fall within this same problematic region of the uncanny valley curve. They achieve enough visual realism to raise viewers' expectations, but cannot fully deliver on them in motion timing, expressions or gaze behaviour. As a result, a photorealistic avatar may undermine the very trustworthiness it was designed to convey. Conversely, Seymour et al. tested this directly using a professionally produced, real-time photorealistic avatar in a field study. None of the eighteen participants who interacted with it reported an uncanny or aversive response and the avatar was also rated as having greater affinity, trustworthiness and preference than a caricature counterpart [26]. This result suggests that a sufficiently high-fidelity, real-time avatar can escape the uncanny valley entirely rather than merely offset its cost with other advantages.

This escape, however, appears conditional on production quality. Kuffner dos Anjos and Pereira observed that realistic mesh avatars in third-person VR perspective produced pronounced uncanny effects when animation quality was constrained by sensor resolution, while abstract representations avoided the valley entirely in the same viewing condition [27]. This pattern is consistent with the escape by design principle Mori originally

proposed. A larger, more recent test of this pattern comes from a network meta-analysis by Tao et al. It pooled 13 independent studies and 2,343 participants across three realism levels. High realism avatars were rated significantly more trustworthy than low realism avatars, while medium realism differed significantly from neither [28]. None of the eeriness comparisons reached significance, although the cumulative ranking placed medium realism as the most eerie. The pooled evidence therefore favours realism for trust and leaves the eeriness pattern unresolved.

3.2. Trust and Credibility in Virtual Agents

Trust in virtual avatars is multidimensional, not a single attitude. Mayer et al. established this structure by decomposing perceived trustworthiness into ability, benevolence and integrity. They further separated trustworthiness as an attribute of the trustee from trust as an attitude held by the trustor [13]. McKnight and Chervany subsequently carried that distinction into a typology separating trusting beliefs from trusting intentions in technology-mediated contexts [14]. Both frameworks are set out in Section 2.3. Their consequence for the present review is that trust should not be reduced to a single score, as rendering style could plausibly affect the three components differently.

Trust measurement has since been examined more broadly across the literature. Alipour et al. conducted a systematic review of 53 empirical studies to determine whether and how the uncanny valley effect influences trust [29]. They found three different operationalisations of trust across this literature. The first is a direct rating of trust or trustworthiness. The second is a related construct such as competence or credibility. The third is a trust-related intention such as advice following. The reported relationship between the uncanny valley effect and trust was inconsistent across these studies, with some reporting a direct negative effect and others finding no relationship at all. This inconsistency suggests that the effect of appearance on trust may depend on other perceptual factors.

One such factor is anthropomorphism, the degree to which an avatar is perceived as human-like. Nowak and Biocca demonstrated that anthropomorphism and perceived agency contribute independently to social presence and copresence. The highest levels of this cue do not consistently elicit greater social presence [30]. The implication is that avatar realism operates under diminishing perceptual returns. Beyond a certain threshold of human likeness, additional anthropomorphic cues do not increase social engagement and may even reduce it. This pattern is structurally analogous to the uncanny valley, where affinity likewise ceases to scale linearly with human resemblance and inverts before recovering at full fidelity [4]. Nowak and Rauh later extended this into credibility specifically. In a study where 255 participants evaluated a series of avatars, perceived anthropomorphism and avatar gender expression both predicted credibility judgments. More anthropomorphic avatars were rated as more credible. Avatar design choices therefore function as meaningful social signals that shape participants' willingness to engage and trust [31].

Canales et al. tested this signalling role directly. They manipulated avatar stylisation and measured its effect on multi-component trust. They presented participants with virtual doctors rendered at three stylisation levels: realistic, semi-realistic and caricatured.

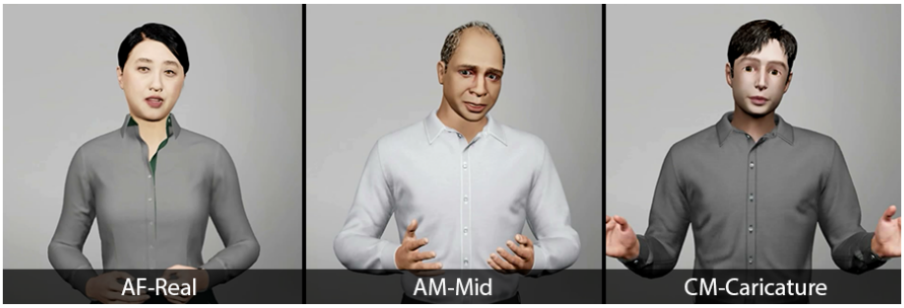


Figure 3.3.: The three avatar stylisation levels used by Canales et al.: realistic (AF-Real), semi-realistic (AM-Mid) and caricatured (CM-Caricature) [5].

Trust was measured across competence, benevolence and integrity dimensions before and after a short interaction. Competence here corresponds to what Mayer et al. term ability. In addition, participants were asked which doctor they would consult for a follow-up. This provided a behavioural indicator alongside the self-report scale. Together these span two of the three measurement approaches in the Alipour et al. taxonomy. The missing category is a direct rating of trust itself [29]. The results showed that self-reported trust ratings were broadly equivalent across all three style levels. However, the semi-realistic condition attracted the highest rate of behavioural selection [5]. This suggests that moderate stylisation can match or exceed the behavioural trust of full photorealism. Other studies by Gao et al. and Alimardani et al. supported this pattern. Gao et al. found comparable self-reported trust between stylised and realistic virtual agents [32] and Alimardani et al. found no significant change in trust even when appearance and voice were manipulated independently [33]. Overall, these studies demonstrate that self-report alone does not fully capture how render style shapes trust.

Dubosc et al. extended this evidence beyond trust to eeriness specifically. They conducted a VR face-to-face interaction study and found that stylised avatars were rated significantly less eerie than photorealistic avatars. Trustworthiness also trended higher for the stylised avatars, though this difference did not reach statistical significance [34]. This pattern suggests that stylisation reduces the uncanny valley response without compromising social credibility. Cihodaru-Ştefanache and Podina reviewed 29 empirical studies on embodied conversational agents. They identified consistent stylisation across all visual features as the most reliably recommended design strategy for avoiding uncanny valley responses, alongside accurate lip synchronisation and consistency between visual and voice realism levels [35]. Combined with the trust findings above, these results form a consistent picture across independent studies.

Testing this pattern reliably requires measurement across more than a single trust rating. The Godspeed Questionnaire Series, developed by Bartneck et al., offers one such option, providing validated semantic differential scales for five perceptual dimensions [15]. Even though the series was originally developed for human-robot interaction, it has since been widely adapted for virtual avatar studies. Section 2.3 sets out its structure and the subscale adopted in this work.

The eeriness-specific instruments introduced by MacDorman and Ishiguro complement the Godspeed series by targeting the negative affective response to near-human imperfection directly [23]. Ho and MacDorman later refined this into a two-subscale instrument that distinguishes perceived eeriness from perceived human likeness and their validation established that the two dimensions are orthogonal [16]. This orthogonality matters for avatar evaluation. Eeriness and perceived human likeness could in principle move together if NPR abstraction reduced both at once and measuring them separately is the only way to tell whether a drop in eeriness reflects reduced discomfort or merely reduced human likeness.

3.3. NPR for Virtual Humans

NPR is the process of transforming an image into abstracted, artistic or stylised output rather than reproducing it exactly. Kumar and Poornima classified these approaches into four categories: stroke-based rendering, image analogy, region-based techniques and image filtering [36]. Stroke-based rendering builds an image from discrete brush strokes that vary in size, colour, texture and orientation. Placing these strokes typically requires a dedicated analysis pass to decide where and how each stroke should be oriented. This is a drawback for pipelines with only one shading stage per pixel. Image analogy instead transfers colour and texture style from an example pair of source and target images onto a new image. This transfer depends on a precomputed exemplar or a patch-matching search rather than information available within a single frame. By comparison, region-based techniques segment an image into distinct regions before abstracting each one individually. Segmentation is an iterative process that considers the whole image. Its result cannot be computed within a single, independently executed per-pixel operation. Finally, image filtering applies local mathematical operations, such as gradients or bilateral smoothing, directly to the image to suppress detail while preserving structure. These operations examine only a small neighbourhood around each pixel. As a result, they remain computable within a single shading pass. Figure 3.4 summarises this classification.

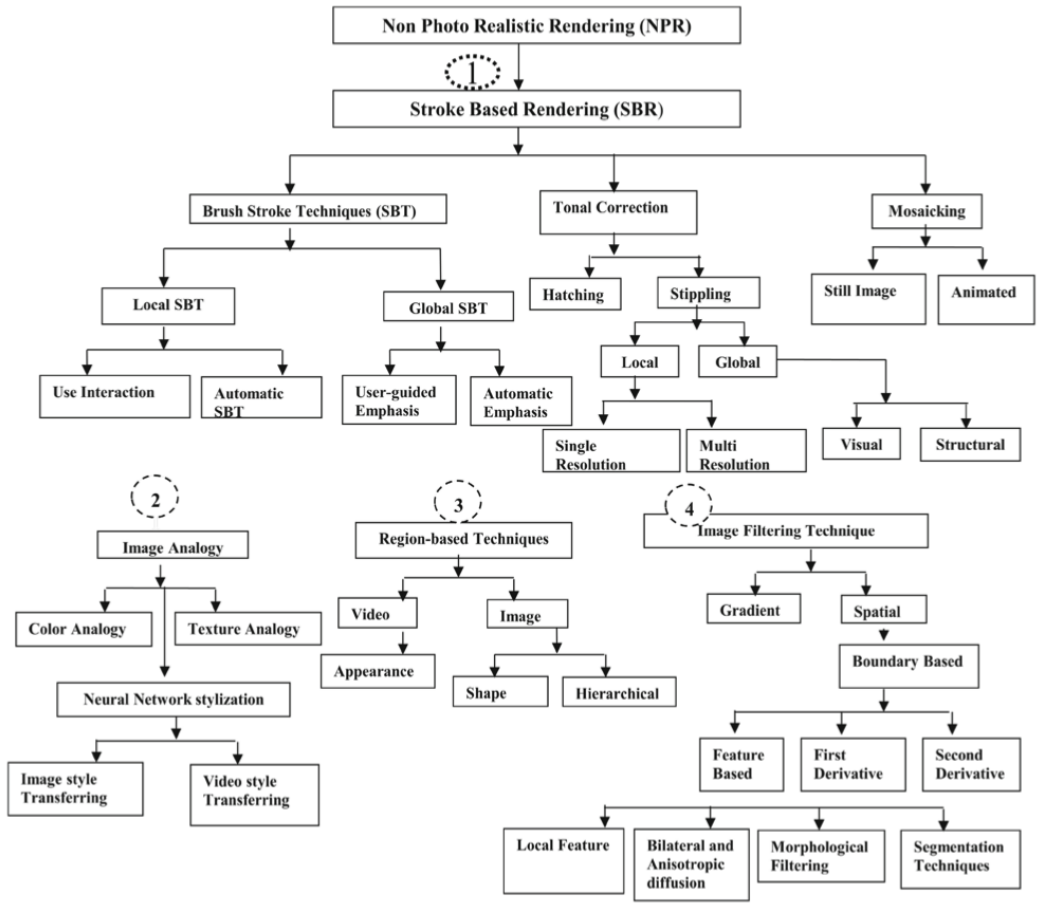


Figure 3.4.: Kumar and Poornima's classification of NPR techniques into four families: stroke-based rendering, image analogy, region-based techniques and image filtering [36].

However, this survey's scope is limited to two-dimensional (2D) image processing. A separate tradition of real-time three-dimensional (3D) rendering techniques falls outside it.

Figure 3.5.: A cartoon character rendered with toon shading, showing flat tonal bands separated by hard edges [37].

Toon shading is one strand of this tradition. It quantises the continuous diffuse lighting term into a small number of discrete bands [37]. Consequently, this approach produces flat-shaded regions separated by hard edges and gives characters a cartoon appearance, as depicted in Figure 3.5. That separation depends on a single input, the lighting term. Hence, artists have little independent influence over its placement. Barla et al. extended toon shading's quantisation into a richer parameter space with X-Toon [38]. It parameterises the shading tone ramp by both view angle and depth. This enables a 2D lookup that encodes artistic intent unavailable in a one-dimensional (1D) quantisation step.

Another approach that Kumar and Poornima's categorisation does not cover is silhouette rendering. Raskar and Cohen proposed processing the mesh twice from the same viewpoint [39]. Initially, it draws the front-facing polygons in the usual way. The back-facing polygons are then rendered and translated towards the camera. Thus, part of this back-facing geometry appears in front of the occluding front-facing surface and the visible rim left at this boundary forms the outline. Moreover, Isenberg et al. provided a systematic survey of silhouette detection algorithms. They distinguish silhouette edges from feature lines such as creases, borders and self-intersections [7]. Silhouette

edges mark only an object's outer boundary, whereas feature lines lie within it. This distinction matters for avatar rendering specifically, where facial features, clothing folds and skin creases all constitute expressive visual cues that lie within, rather than on, the silhouette boundary.

Image-space methods address this limitation directly by operating on the rasterised output rather than on the mesh. This makes them sensitive to any visible discontinuity on screen, regardless of whether it falls on the silhouette or in the interior of the figure. Screen-space normal derivatives, computed via hardware-accelerated `ddx/ddy` intrinsics available on all modern graphics processing units (GPUs), detect abrupt changes in surface orientation without additional draw calls or intermediate render targets. This approach captures surface creases and mesh boundaries that geometry expansion cannot reach. The fundamental limitation is that edge thickness is measured in screen pixels rather than world-space units. This causes edges to scale with camera distance. Users in an immersive VR context move relative to other participants and edges accordingly thin out at close range and coarsen as the viewer withdraws.

This scaling behaviour follows directly from measuring edges in screen space rather than on the mesh itself. Texture-based edge detection targets the painted surface layer. The Sobel operator, a discrete gradient approximation applied to the diffuse texture via a 3×3 convolution kernel, identifies luminance boundaries corresponding to clothing patterns, facial contours and painted surface details [17]. It responds to what was painted onto the surface rather than to the surface shape itself. In contrast, geometry-based methods respond to shape rather than surface. Marr and Hildreth demonstrated that low-pass prefiltering is a prerequisite for stable gradient computation [20]. This observation motivates the Gaussian prefiltered variant explored in this work.

Nonetheless, a single gradient operator remains fragile on its own, producing noisy and discontinuous edges even after prefiltering. Roshaan combined depth, normal and colour cues into a silhouette, structure and texture hierarchy, merged through max-pooling and scaled by scene luminance for consistent line quality without manual retuning [40].

A separate line of work abstracts texture through spatial filtering rather than gradient-based edge detection. The Kuwahara filter applies this approach to NPR and yields region-based painterly abstraction without explicit edge detection. Kyprianidis et al. extended the isotropic Kuwahara with an anisotropic formulation aligned to local image structure. This yields brush-stroke effects oriented along salient contours [22]. The anisotropic extension requires a structure tensor computed from a prior render pass, a constraint directly relevant to the integration target of this thesis. A multiscale variant of this filter, also by Kyprianidis, propagates orientation estimates and filter outputs through a resolution pyramid [41]. This removes artefacts in low-contrast regions while maintaining real-time performance. Figure 3.6 compares both variants against the unfiltered input.

(a) *Original image* (b) *Anisotropic Kuwahara filter* (c) *Proposed method*

Figure 3.6.: Painterly abstraction produced by Kuwahara filtering. The proposed method in panel (c) is the multiscale variant, which suppresses the artefacts the anisotropic filter leaves in low-contrast regions [41].

Praun et al. demonstrated real-time hatching as a complementary branch of this design space [42]. Tonal regions in this branch are conveyed by oriented stroke textures rather than flat fills or edge lines.

This technical catalogue establishes what NPR can render, but not whether the result changes how an avatar is perceived. The perceptual impact of NPR style choices on avatar social reception has been studied through a combination of psychophysical and VR experiments. Wisessing et al. demonstrated that toon shading reduces

the perceived intensity of facial emotional expression while leaving recognition accuracy intact [43]. Characters rendered in this style therefore convey the correct emotional content, without the perceptual amplification that photorealistic rendering provides. A follow-up study identified lighting parameters as a further modulator of appeal and eeriness [44]. Higher brightness made characters more appealing regardless of the emotion portrayed. Darker shadows raised eeriness and reduced the appeal of cartoon characters, while leaving the appeal of realistic characters unaffected. Together these findings show that render style and lighting both shape how a character is received, independently of the emotion being performed. Zibrek et al. conducted a within-participants VR study in which the same animated character was presented in photorealistic and non-photorealistic render conditions [45]. Participants evaluated emotional recognition, social presence and eeriness across both conditions. Photorealism did not significantly improve emotional recognition or social presence. In some conditions the NPR condition was rated more positively. This challenges the assumption that rendering quality scales linearly with social utility. Zibrek and McDonnell extended this investigation to embodied VR [46]. Photorealism significantly increased both place illusion, the sense of physically inhabiting the virtual space and self-reported social presence. The behavioural measure of interpersonal proximity was unaffected by render style. This split between self-report and behaviour indicates that a rendering change can shift how present an avatar feels without changing how a viewer acts towards it.

This raises the question of how far avatar realism carries into embodied social interaction. Latoschik et al. addressed it in an immersive social VR context. They compared an abstract wooden mannequin against high fidelity avatars generated from photogrammetric scans. The realistic avatars were rated significantly more human-like and evoked stronger virtual body ownership, alongside some indication of an uncanny valley response [47]. Realism therefore brings measurable gains for embodiment and it carries the uncanny valley risk at the same time.

Realism level is not the only dimension a designer controls. Zell et al. separated a character's shape from its material and varied each independently across five stylisation levels. Shape governed perceived realism, whereas material governed appeal and eeriness [8]. Appeal also fell and eeriness rose whenever the two were mismatched and matching them maximised appeal at any absolute level of realism [8]. Consistency between shape and material, rather than the degree of abstraction alone, therefore governs the uncanny response.

What connects nearly all of this body of work is a shared implicit assumption: the developer controls the rendering pipeline. Techniques are designed and validated against custom meshes, open rendering frameworks or research engines where shader code, render passes and material systems can be freely modified. The most directly related prior work on the avatar side, Canales et al., applied stylisation to custom assets in an open pipeline and compared coarse-grained style levels rather than specific technical approaches [5]. This assumption of full pipeline access does not hold for production avatar SDKs, where the rendering stack is managed by the SDK provider and exposes only constrained integration points.

3.4. Research Gap

The reviewed literature establishes three relevant findings. First, NPR stylisation is a principled strategy for avoiding the uncanny valley, with clear empirical support from McDonnell et al. and Zibrek et al. [2], [45]. Second, stylised avatars do not carry a trust penalty relative to photorealistic avatars in controlled evaluations, as demonstrated by Canales et al. and Gao et al. [5], [32]. Third, all prior work on both NPR rendering and avatar trust perception has assumed open pipeline access. This work relied on custom assets and research engines rather than production SDK systems.

Three specific gaps remain unaddressed. First, no published work has applied fine-grained NPR stylisation within a closed production avatar SDK that manages its own skinning and level of detail, where the developer cannot modify the core lighting pipeline, add intermediate render passes or alter material management. The NPR techniques surveyed above were all developed and validated under full rendering stack access. Whether the same techniques can be effectively integrated under production SDK constraints and whether their perceptual properties hold in that context, remains an open question. This gap is addressed by the technical contribution of this thesis (RQ1).

Second, while prior work compared coarse stylisation levels, the literature has not examined the technical design space within NPR under production SDK constraints. It remains unclear which technique families stay viable when an SDK controls render-pass access and material management or which best preserves avatar legibility while achieving perceptual abstraction. This gap is addressed by the iterative technique development and selection described in Chapters 4 to 6 (RQ2).

Third, no prior study has measured the trust and eeriness consequences of NPR stylisation applied to a photorealistic production avatar under a controlled evaluation protocol that uses validated multi-component instruments and a behavioural measure. Canales et al. is the closest prior work, but compared coarse style categories rather than a specific NPR shader pipeline, used a between-subjects design and did not measure eeriness or anthropomorphism as separate dimensions [5]. This gap is addressed by the user study of this thesis (RQ3). The study tests four hypotheses: that NPR reduces eeriness relative to photorealism (H1), that NPR does not reduce self-reported trust relative to the photorealistic avatar (H2), that eeriness negatively predicts trust across all three conditions (H3) and that NPR does not reduce self-reported trust relative to a real human speaker (H4).

This thesis addresses all three gaps. It integrates a suite of NPR techniques into the Meta Avatar SDK under real production constraints, evaluates the visual and perceptual trade-offs across technique variants. It also measures the trust and eeriness implications of NPR abstraction in a within-subjects, video-based user study. The study presents three conditions: a recording of a real person serving as the trust ceiling baseline (C1), an Avaturn avatar rendered at full photorealistic quality (C2) and the same Avaturn avatar rendered with the selected NPR shader pipeline (C3). Holding the avatar platform constant between C2 and C3 isolates the perceptual effect of NPR stylisation from platform variation. The evaluation instrument draws on several validated sources. It adopts the likeability subscale of the Godspeed questionnaire [15]. Eeriness is measured using scales grounded in MacDorman and Ishiguro and refined by Ho and MacDorman [16], [23]. A multi-component trust battery is structured around the ability, benevolence and integrity dimensions of the Mayer et al. integrative trust model, operationalised within the trusting-beliefs typology of McKnight and Chervany [13], [14]. A forced-choice behavioural preference task supplements the self-report battery, following the paradigm of Canales et al. and the methodological prescription of Alipour et al. [5], [29].

4. Methodology

This chapter describes the research questions guiding this work, the three avatar platforms used as development and evaluation targets, the NPR stylisation techniques developed across those platforms and the design of the user study that evaluates their perceptual consequences.

4.1. Research Questions

Three research questions structure the contributions of this thesis.

RQ1: How can NPR stylisation techniques be adapted to avatar platforms with different rendering architectures and what capabilities and limitations does each platform impose?

This question is primarily technical. It asks how NPR techniques can be adapted for distinct platforms that differ substantially in mesh topology, shader accessibility and pipeline structure. It also asks what each platform permits and what it prevents, since the architectural constraints of each platform determine which NPR techniques are realisable and how they appear in practice.

RQ2: Which NPR technique or combination of techniques best balances visual abstraction against facial and expressive legibility under production rendering constraints?

This question drives the iterative technique development documented in Section 4.4. Each version introduces a different abstraction approach and addresses a limitation identified in its predecessor and the overall goal is to identify a configuration that achieves a stable, readable stylised appearance without an excessive cost to facial and expressive legibility. The visual quality assessment and the parameter adjustment pilot study, both reported in Chapter 6, provide the empirical basis for this selection.

RQ3: How does NPR stylisation affect perceived eeriness and trustworthiness and can an NPR avatar reach the trustworthiness of a real human speaker?

Uncanny valley theory places a real human speaker at the peak of the affinity curve, since maximal human likeness corresponds to maximal comfort [4]. A photorealistic avatar instead sits in the trough immediately before that peak, where negative affect is strongest. Perceived eeriness is the response by which a character's position on that curve is registered and the question spans both the discomfort a stylised avatar evokes and the trust it receives. If the composite selected under RQ2 relocates the avatar onto the stable rising slope below the trough, the resulting trust level may approach that of a real human speaker even though the avatar's human likeness remains lower. RQ3 tests this possibility directly and C1, the real human speaker, accordingly serves as the reference point against which both avatar conditions are compared. The formal hypotheses tested, including the equivalence test against this reference point, are stated in Section 4.8.1.

4.2. User Study Overview

To address RQ3, a within-subjects study was conducted in which participants viewed video stimuli of an avatar delivering a spoken recommendation across three everyday decision scenarios. Three conditions were presented to every participant: a real human speaker (C1), a photorealistic avatar (C2) and the same avatar rendered with an NPR composite (C3). Holding the avatar platform constant between C2 and C3 isolates the perceptual effect of the stylisation from any variation in avatar identity or animation quality.

After each video, participants rated the advisor on perceived eeriness, likeability and trust across three dimensions: competence, benevolence and integrity. At the close of the session, participants ranked all three advisors by how much they would want to consult each one, providing a behavioural measure of trust alongside the self-report scales. The full study design, questionnaire structure and analysis plan follow in Section 4.8.

4.3. Avatar Platforms and Rendering Constraints

The NPR techniques developed for this thesis were evaluated across three distinct avatar platforms, each offering a specific combination of capabilities and integration constraints relevant to NPR development. Mixamo offers the most flexible environment for initial shader implementation due to its unrestricted access to URP shaders, but its exported characters lack facial rigging. In contrast, the Meta Avatars SDK provides native Unity integration and built-in body tracking and this supports evaluation of NPR behaviour on animated avatars in a VR environment. However, it restricts shader customisation to a proprietary injection interface and NPR shader development is hence the most constrained on this platform. Finally, Avaturn combines unrestricted URP access with high fidelity facial reconstruction and full blendshape animation, though it requires an additional import step for its glTF assets.

Mixamo.

The Mixamo platform provides an online library of rigged humanoid characters and skinned animations that can be exported to Unity as standard FBX assets [48]. Since Mixamo avatars are standard skinned meshes without proprietary shader restrictions, all NPR techniques can be implemented directly within the URP rendering pipeline. Consequently, Mixamo is the least constrained of the three platforms and it is therefore the most appropriate environment for the initial exploration and development of each proposed approach before adaptation to the more constrained platforms. Nevertheless, Mixamo character exports do not include facial rigging and the avatar's face remains static throughout all renders generated on this platform. As a result, Mixamo is unsuitable for the user study, although the Mixamo character (Figure 4.1) served as the primary prototyping target during the initial development phase, where unrestricted shader access facilitated rapid prototyping and iterative refinement.

Figure 4.1.: Mixamo: baseline avatar for NPR shader development.

Meta Avatars SDK.

Figure 4.2.: Meta avatar: baseline avatar for NPR integration within the Meta Avatars SDK.

Avatar rendering, face tracking and body tracking are handled internally by the Meta Avatars SDK, Meta's avatar framework for the Meta Quest 3 [49]. Since one objective of this thesis is to evaluate NPR avatars in VR on this device, the Meta Avatars SDK was examined for the shader customisation capabilities it provides for NPR integration (Figure 4.2). Its built-in face and body tracking eliminates the need for external motion capture systems and the proposed NPR shaders can thus be evaluated on naturally animated avatars in a VR environment. In contrast, custom NPR shaders cannot be integrated through standard URP material slots due to the SDK's internal rendering architecture. Instead, NPR rendering logic must be injected through hook functions exposed by the SDK at defined points in the shader pipeline. This constrains both the range of achievable NPR effects and the structural requirements of the shader code. The technical details of this integration are described in Chapter 5.

Avaturn.

Avaturn is an avatar creation service that reconstructs a personalised rigged avatar with facial blendshape targets from three uploaded photographs (Figure 4.3) [50]. The service exports its output as `.glb` files, which implement the glTF 2.0 standard and bundle geometry, PBR materials and skeletal rigging within a single self-contained file [51]. Since Unity does not natively support this format at runtime, an additional import package is required to load the avatar, as described in Chapter 5. Once loaded, it functions as a standard skinned mesh with PBR materials and unrestricted access to the URP rendering pipeline and the proposed NPR shaders can be implemented directly. Accordingly, Avaturn served as the primary platform for the final user study, since it is the only platform among the three that combines unrestricted shader access with personalised facial reconstruction and full blendshape animation. This complemented the prototyping workflow established with Mixamo and the VR evaluation performed using the Meta Avatars SDK.

Table 4.1.: Summary comparison of the three avatar platforms across dimensions relevant to NPR integration.

	Mixamo	**Meta Avatars SDK**	**Avaturn**
Shader access	Full URP, unrestricted	Hook injection only (SDK post lighting)	Full URP, unrestricted
Transparency	Explicit alpha cutout	SDK managed	Explicit alpha cutout
Facial animation	None	SDK managed face tracking	Blendshape animation
Key challenge	No facial rigging	Shader manipulation	Initial Unity import
Strength	Full shader freedom	Native Quest 3 integration	High fidelity facial quality

4.4. NPR Technique Design

NPR techniques in computer graphics encompass a wide range of abstraction levels [7]. Choosing a level that is too low may place an avatar within the uncanny valley, whereas excessive stylisation can reduce visual detail and compromise expressive readability [8], [9]. Thus, selecting the appropriate degree of abstraction requires calibrating the balance between realism and visual expressiveness.

Figure 4.3.: Avaturn: baseline avatar for the user study.

This calibration was performed through an iterative design process in which several NPR techniques were developed and evaluated. Each version addressed a specific limitation identified in the preceding version and the final selection was validated through a formal user study. Main techniques are labelled V1 through V6. Extended variants carry a decimal suffix, as in V2.2 and V4.2, indicating a directed modification of the preceding version.

Development proceeded in sequence across the three platforms described in Section 4.3. The SDK's shader hook architecture imposed implementation constraints that shaped several architectural decisions and the final transfer to Avaturn served as a cross-platform verification of each technique's visual output. The following sections describe the motivation, design rationale and visual characteristics of each approach. Full implementation details are provided in Chapter 5.

4.4.1. Shared Rendering Structure

All proposed NPR techniques share a common two-pass rendering structure. The first pass generates the silhouette outline using the inverted hull method, which is applied consistently across all three platforms. The

second pass computes the NPR surface effect for each technique within the forward rendering pass. Separating outline generation from surface shading allows the outline width, colour and depth behaviour to be adjusted independently while ensuring that the outline geometry is rendered at a consistent depth regardless of the transparency mode of the surface material.

4.4.2. V1 — Inverted Hull Outline

Object silhouettes are a defining visual feature of nonphotorealistic rendering [7]. Accordingly, a robust outline generation technique was required as the foundation for all NPR techniques developed in this thesis. The inverted hull method, also known as the shell method, was selected because it produces stable geometric silhouettes without requiring screen-space processing [7]. It generates an outline shell by enlarging back-facing polygons and relies on GPU face culling to form the outline, eliminating the need for edge detection or additional rendering passes. The method operates across two dedicated stages:

1. **Outline Pass.** The mesh is rendered with front-face culling enabled, displacing each vertex outward along its world-space surface normal by a configurable distance w:

$$\mathbf{P}' = \mathbf{P} + \mathbf{N} \cdot w \tag{4.1}$$

   where $\mathbf{P}$ is the original vertex position and $\mathbf{N}$ is the world-space surface normal. Only the back-facing geometry is rasterised, forming an enlarged shell around the mesh.

2. **Forward-Lit Pass.** The mesh is subsequently rendered with standard back-face culling. The front-facing surfaces overwrite the interior of the previously rendered shell, leaving only the protruding rim visible as a silhouette border with controllable thickness.

This approach is well suited to real-time NPR avatar rendering because it requires no offline mesh preprocessing and executes entirely within the GPU vertex transformation stage [7]. As a result, its computational cost scales predictably across assets of varying complexity. However, the expanded shell and the original surface may share similar depth values, which can introduce z-fighting artefacts along the silhouette boundary. To prevent this, a depth bias is applied to shift the outline geometry slightly towards the camera in clip space.

Despite its stability and low rendering cost, the inverted hull method has several limitations when applied to avatar rendering. Figure 4.4 shows the results across all three platforms. Surfaces with degenerate or coincident normals, such as flat planes, yield no visible outline because displacing vertices along identical surface normals does not generate an expanded shell. In addition, the method does not register interior surface boundaries. Consequently, creases between body regions, clothing seams and facial contours receive no edge marking, as the method captures only the external silhouette of the mesh [7]. This limitation is shared by geometry-based outline methods and motivates the interior edge detection techniques introduced in V3.

Alpha Cutout for Transparency

Avatars on the Mixamo and Avaturn platforms contain alpha-masked elements such as eyelashes and hair that are represented as flat cards. Their visible shape is defined by discarding fragments below an opacity threshold. Without alpha testing in the outline pass, the expanded hull would be generated from the full geometric extent of each card. This produces solid rectangular borders rather than outlines that follow the visible hair silhouette, as shown in the left column of Figure 4.5. Alpha testing is therefore applied using the GPU's built-in fragment clip operation to discard fragments whose texture alpha value falls below a configurable cutoff threshold α_{cutoff}:

$$\text{if } \alpha_{\text{texture}} < \alpha_{\text{cutoff}} \quad \text{then discard} \tag{4.2}$$

This test executes in both the outline pass and the forward-lit pass to maintain silhouette consistency. Figure 4.5 shows the resulting outline with the threshold applied. Unlike alpha blending, this binary approach does not require depth sorting and introduces negligible additional fragment processing cost.

Conversely, the Meta Avatar platform handles hair and eyelash rendering through its internal pipeline. Transparent fragments are consequently already discarded internally by the SDK, meaning that manual alpha

(a) Mixamo: inverted hull silhouette outline.

(b) Meta SDK: inverted hull silhouette outline.

(c) Avaturn: inverted hull silhouette outline.

Figure 4.4.: Inverted hull geometry outline applied to all three avatar platforms. The slight floor penetration visible at the Meta Avatar's feet in (b) reflects a tracking boundary offset during capture and is not a property of the outline technique.

testing is not required during outline generation. The same two-pass rendering structure is hence maintained on this platform, although the underlying outline masking approach differs from the explicit alpha cutout used on Mixamo and Avaturn. The full technical details are discussed in Section 5.4.1.

4.4.3. V2 — Toon Shading

Toon shading, also known as cel shading, is one of the most widely adopted techniques in NPR, where stylisation is achieved by simplifying the tonal representation of a surface rather than reproducing physically accurate illumination [37]. Discretising the illumination response into uniform tonal regions produces the characteristic flat appearance associated with hand-drawn animation. Correspondingly, the approach is well suited to interactive character rendering [37]. Accordingly, toon shading was selected as the V2 baseline because it provides a controlled starting point from which each subsequent technique introduces a targeted increase in stylisation. However, implementing this stylisation across the three avatar platforms requires adaptations that account for differences in each platform's rendering architecture.

On Mixamo and Avaturn, the continuous illumination response is converted into discrete tonal regions using the following quantisation function:

$$t = \frac{\lfloor s \cdot n \rfloor}{n} \tag{4.3}$$

where s represents the surface lighting response and n defines the number of shading bands. This operation replaces continuous illumination gradients with discrete tonal transitions, producing the characteristic stepped appearance of toon shading [37]. Figure 4.7 shows the resulting appearance at the default band configuration used in this thesis, with the specific value and its selection criteria given in Section 5.4.2. The resulting tonal structure consists of highlight, midtone, shadow and deep shadow regions.

Several refinements were introduced to improve the visual quality and temporal stability of the baseline toon shading model. Tonal region transitions were softened at each band boundary. This prevents normal jitter near a boundary from snapping a whole surface patch unpredictably between adjacent bands. Additional ambient illumination was introduced to prevent heavily shadowed regions from becoming visually disconnected from the avatar. A rim highlight was also added to improve visual separation between the avatar and its background where flattened shading reduces depth cues (Figure 4.6). Finally, scene cast shadows were deliberately excluded

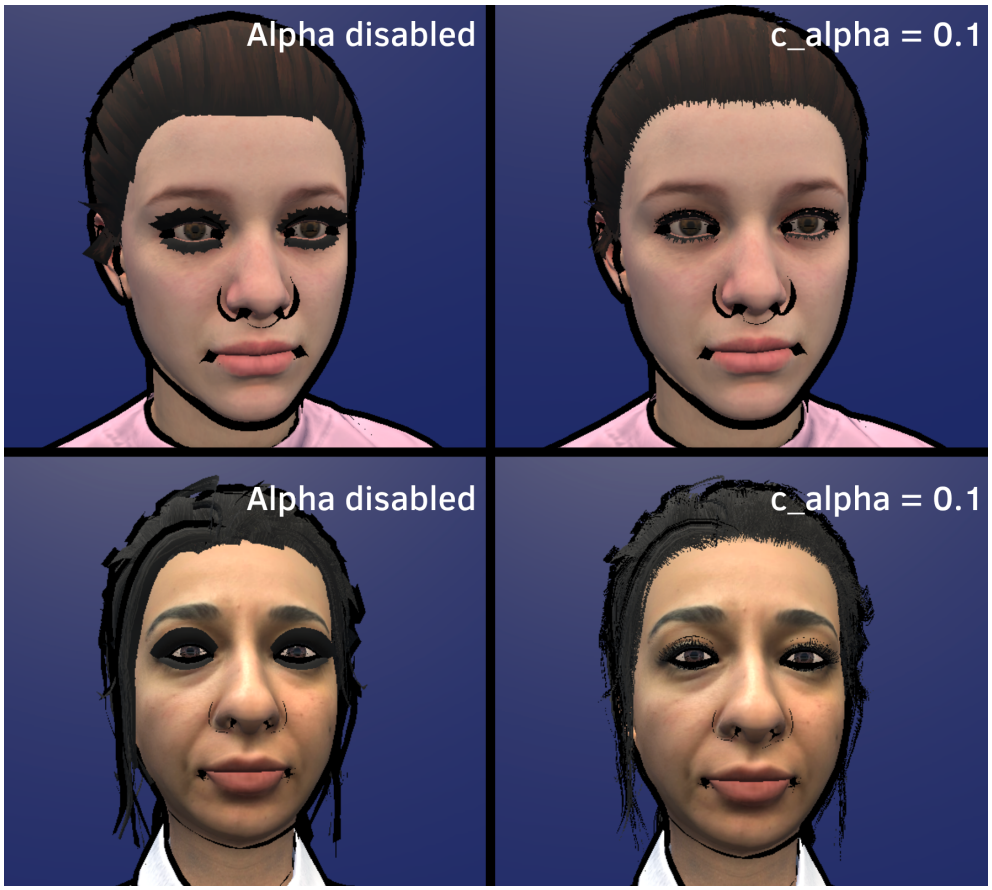


Figure 4.5.: Alpha cutout on the inverted hull outline pass (Mixamo, top; Avaturn, bottom). Left: alpha testing disabled. Right: alpha cutout enabled at $\alpha_{\text{cutoff}} = 0.1$.

because they introduce an additional visual variable that could interfere with evaluating the effect of tonal quantisation in isolation.

Unlike Mixamo and Avaturn, where intermediate lighting values are directly accessible, the Meta Avatars SDK exposes only the final composited appearance after its internal rendering process. Consequently, quantisation is applied to the available colour output rather than to the surface lighting response itself. Despite the different implementation, both approaches achieve the same intended stylisation by replacing smooth illumination gradients with discrete tonal regions.

Figure 4.6.: Rim highlight effect in V2 toon shading.

Across all three platforms, V2 successfully establishes a simplified graphic appearance through tonal quantisation. Nevertheless, V2 has two important limitations. First, the distribution of tonal regions remains closely coupled to the lighting response. This leaves little independent artistic control over the tonal transitions. Second, V2 does not introduce additional edge information for interior features such as facial contours, clothing folds or structural boundaries, as a technique driven purely by shading. These constraints motivate the subsequent development stages. V2.2 extends the tonal representation through X-Toon shading to provide greater control over the tonal distribution, while V3 introduces interior edge detection to enhance the visibility of local avatar features beyond what tonal quantisation alone can convey.

(a) Mixamo: four discrete shading bands with ambient and rim.

(b) Meta SDK: four discrete shading bands applied to the fully composited colour produced by the SDK renderer.

(c) Avaturn: four discrete shading bands with ambient and rim.

Figure 4.7.: V2 (Toon Shading) across all three avatar platforms.

4.4.4. V2.2 — X-Toon Shading

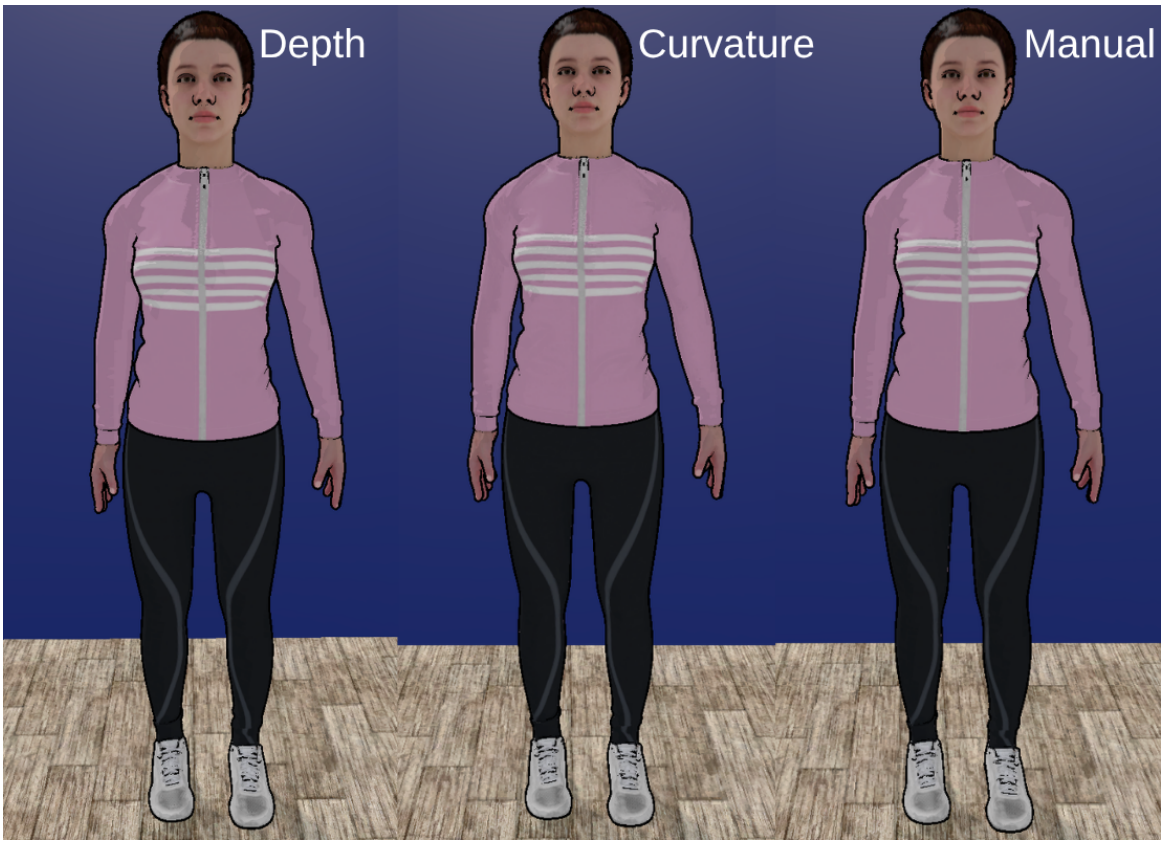
X-Toon shading extends conventional toon shading by replacing the 1D toon ramp with a 2D ramp texture, where the horizontal axis represents the lighting response and the vertical axis controls the abstraction level [38]. This second axis allows the surface appearance to be adjusted independently of the lighting conditions and this provides finer stylisation control. A higher abstraction level reduces local tonal contrast and softens transitions between shading regions towards a flatter, more uniform appearance. Figure 4.8 illustrates the ramp texture used in this work.

Figure 4.8.: 2D ramp texture used for X-Toon shading.

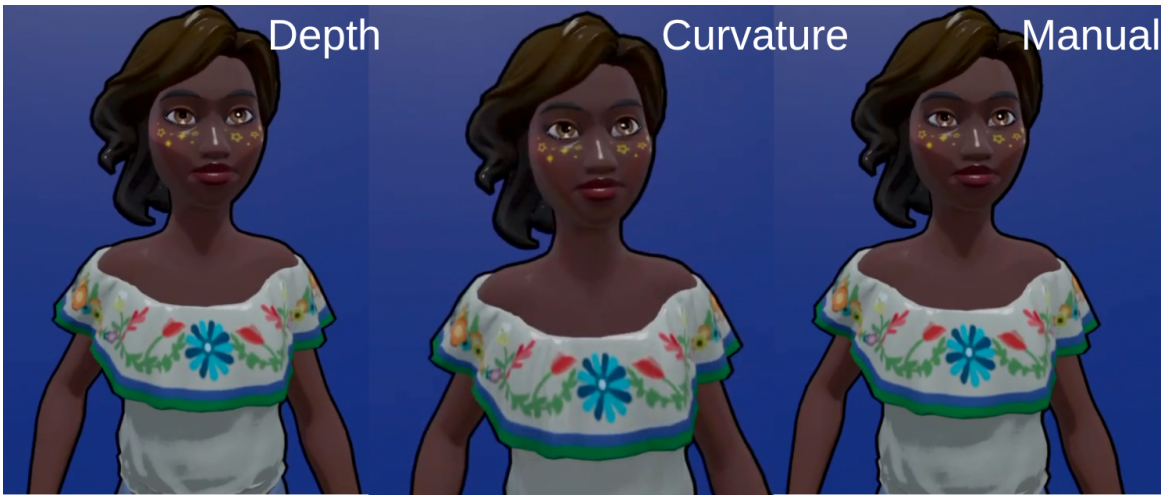
In addition to tonal abstraction, Barla et al. described a shape detail extension that modifies the surface normal field by blending the original mesh normals with normals derived from a geometrically simplified shape [38]. This blending was incorporated into the Mixamo and Avaturn implementations. It reduced local surface variation and gave more coherent stylised shading.

Barla et al. further described an attribute mapping mechanism that controls the abstraction level through three driving modes: depth, curvature and manually controlled [38]. All three were evaluated in this work. The user study used a fixed camera distance much closer to the avatar than the general platform comparison setup and at that distance the depth and curvature modes described above produce almost no variation, as Section 4.5 explains in detail. In comparison with V2, X-Toon produced a softer and more controllable stylised appearance by allowing the stylisation level to be adjusted independently of the illumination response and was consequently retained as the toon shading component in all later technique combinations.

Despite this additional control, X-Toon remains driven only by shading and therefore does not introduce edge information for interior features. Facial contours, clothing folds and texturally encoded details thus remain visually unmarked unless they produce sufficient variation in the tonal response. This limitation is inherited from V2 and motivates the edge-based techniques introduced in V3.

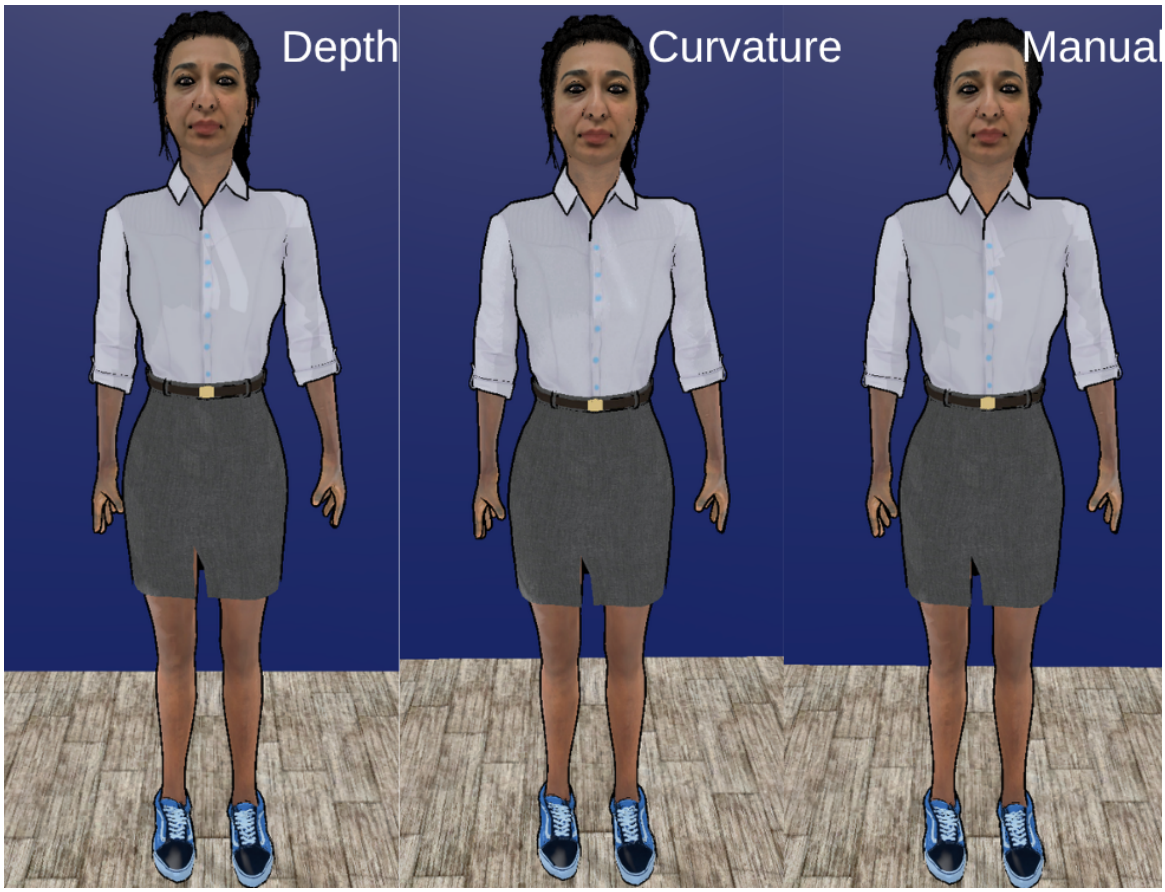


(a) Mixamo: depth, curvature and manual abstraction modes.



(b) Meta SDK: depth, curvature and manual abstraction modes.

Figure 4.9.: V2.2 (X-Toon) across the three avatar platforms. From left to right: depth-driven, curvature-driven and manually controlled abstraction.



(c) Avaturn: depth, curvature and manual abstraction modes.

4.4.5. V3 — Screen Space Normal Edge Detection

Silhouette extraction techniques can be classified into image space, object space and hybrid approaches [7]. Unlike the object-space outline method introduced in V1, image-space techniques analyse per-pixel information generated during rendering to identify visually significant boundaries. One commonly used cue is the surface normal, whose orientation changes abruptly at creases, folds and other geometric features. In prior work, per-pixel surface normals stored in G-buffer channels have been used as a cue for edge detection [52]. Discontinuities across these same channels detect silhouette, border and crease edges in real-time NPR rendering. Crease edges in particular are recovered from the normal buffer [53]. It therefore gives image-space techniques access to the interior geometric structure that silhouette methods alone leave undetected.

V3 builds on this image-space principle and introduces the first interior edge signal into the proposed rendering pipeline. It detects changes in the interpolated world-space surface normal across neighbouring screen pixels and the resulting edge marks trace geometric contours in the interior of the mesh where tonal variation alone would not identify boundaries. This detection is realised through GPU screen-space derivative functions that evaluate the difference between adjacent fragments directly within the fragment pass. Consequently, they require neither a stored normal texture nor an additional rendering pass.

The edge signal is the normal gradient magnitude defined in Equation (2.6). This magnitude runs high along geometric creases and clothing folds. Conversely, it falls close to zero over smooth surfaces. A smoothstep function then converts this raw magnitude into a continuous edge response and this prevents abrupt temporal changes as animated surfaces cross the detection threshold. The resulting edge mask is blended with the X-Toon shaded surface using the same compositing formulation adopted throughout the remaining edge detection techniques.

Across all three platforms, V3 complements the silhouette outline established in V1 by introducing interior contours derived from surface geometry. Figure 4.10 illustrates this across the three avatar platforms, where the edge output varies with both the normal information each platform exposes and the viewing distance. Mixamo and Avaturn both compute derivatives from shading normals. This produces a rich edge network that improves the readability of clothing folds, seams and other regions with pronounced surface curvature. The Meta Avatars SDK, by contrast, exposes only interpolated geometric normals through the shader hook. This restricted input results in considerably sparser interior contours, most noticeably on clothing surfaces where finer surface curvature would otherwise be detected.

Despite these improvements, V3 has two important limitations. The first is the viewpoint dependence established in Section 2.4.2. Since that operator measures change per pixel rather than per unit of surface, interior edges thicken as the camera retreats. The second concerns the cue itself. Surface orientation is the only signal the method reads and texture boundaries carrying no normal discontinuity thus remain undetected.

These limitations prevented V3 from being selected for the final rendering pipeline. Edge thickness scales with the viewing distance and this produces inconsistent results whenever a viewer moves relative to the avatar. A fixed camera distance avoids this specific problem. The user study recordings used exactly this fixed distance (Section 5.6). However, the reliance on surface orientation remains regardless of distance. Subsequently, this dependence leaves many perceptually important facial and material boundaries unmarked, since they are defined primarily by texture rather than geometry. Addressing this second limitation required an edge detector sensitive to colour boundaries rather than surface orientation. This need motivates the Sobel-based approach introduced in V4.

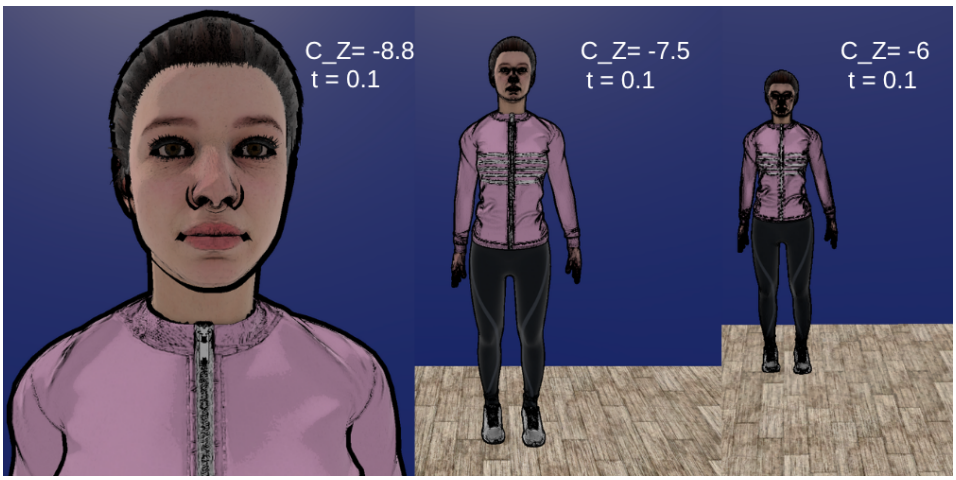
4.4.6. V4 — Sobel Edge Detection

V4 applies the Sobel operator of Section 2.4.1 to the diffuse albedo texture rather than to the surface normal field. This shift targets colour boundaries in the surface appearance, including clothing patterns, facial details and painted surface structures. It also avoids the edge thickness growth with viewing distance observed in V3.

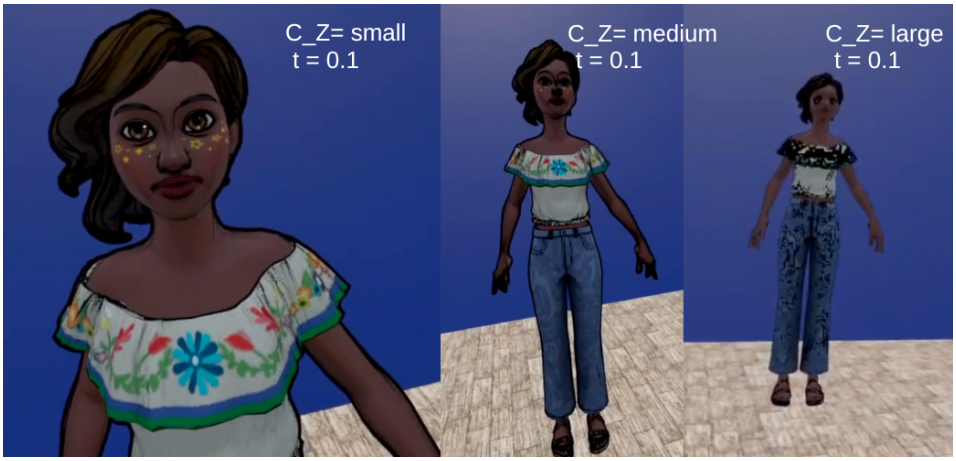
The gradient magnitude M follows from the two kernels of Equation (2.2). Pixels for which M exceeds a configurable threshold are classified as edge pixels. These are then composited over the X-Toon shaded surface to form the final stylised appearance.

Across the three avatar platforms, this output quality depends strongly on the available texture source. On Mixamo, the relatively clean painted albedo enables the detection of clothing boundaries and texture transitions with limited interference from unrelated intensity variations. In contrast, Avaturn uses photographic textures that contain illumination information captured during the reconstruction process. Consequently, the Sobel operator responds not only to material boundaries but also to baked lighting variations. As shown in Figure 4.11, this effect is particularly visible around the nose bridge, where the transition between highlight and shadow produces a strong response despite not representing a meaningful texture boundary. On the Meta Avatars SDK, the Sobel operator was applied to the available composited rendering output rather than the original diffuse texture. Although this introduces responses tied to scene lighting, the rendered texture contains less high-frequency photographic variation. Edge structures are thus cleaner and more consistent than on Avaturn, particularly along clothing boundaries.

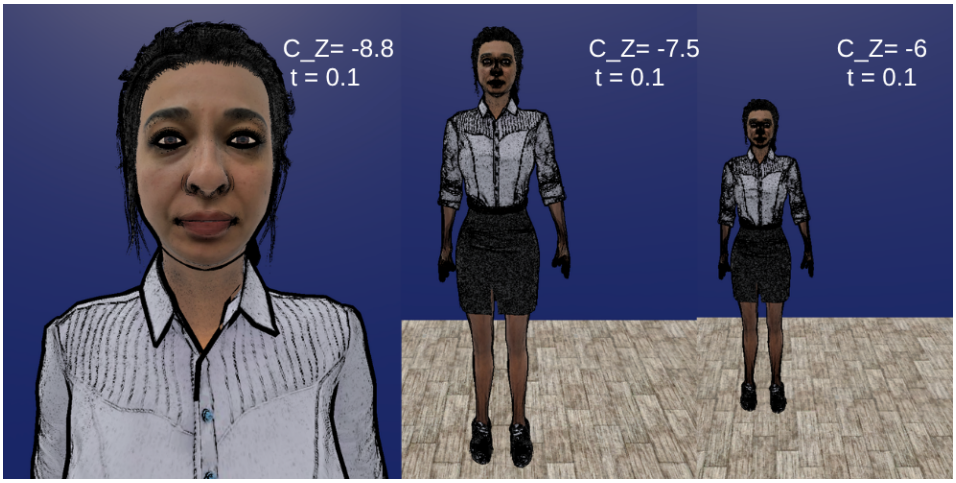
These input differences directly affect the threshold calibration required on each platform. Mixamo produced almost no unwanted edge responses at $t = 0.2$, whereas Avaturn yielded markedly more artefacts at the same threshold. This arose from its photographic texture content. Increasing the threshold on Avaturn suppressed these responses, but it also removed meaningful facial contour lines and eliminated segment boundaries present at the lower setting, as Figure 4.11d shows. The Meta Avatars SDK requires lower threshold values because its composited output yields weaker gradients at material boundaries. This same property reduces sensitivity to fine texture variations and results in comparatively cleaner edge patterns. These observations confirmed that the Sobel response depends on the available appearance representation rather than solely on underlying geometry. The threshold must therefore be calibrated to each platform's signal characteristics rather than treated as a fixed operator property.



(a) Mixamo: normal edge detection at increasing viewing distances.



(b) Meta Avatars SDK: normal edge detection at increasing viewing distances.

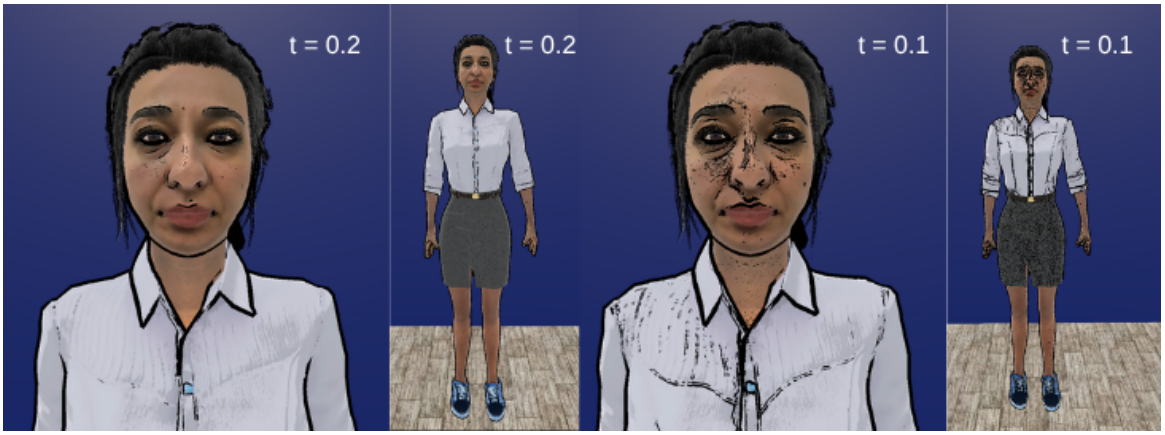


(c) Avaturn: normal edge detection at increasing viewing distances.

Figure 4.10.: V3 (Screen Space Normal Edge Detection) across the three avatar platforms. C_Z: camera-to-avatar viewing distance; t: detection threshold.

(a) Mixamo: Sobel edge detection at two threshold values.

(b) Meta Avatars SDK: Sobel edge detection at two threshold values.



(c) Avaturn: Sobel edge detection at the baseline threshold ($t = 0.2$).

(d) Avaturn at an increased threshold. Noise is reduced relative to (c), but meaningful facial contour lines are also eliminated.

Figure 4.11.: V4 (Sobel Edge Detection) across the three avatar platforms. t: Sobel threshold.

Compared with V3, Sobel edge detection successfully captures texture-defined boundaries that remain invisible to normal-based methods, including clothing patterns, facial details and painted surface structures. The operator responds to all local intensity variations, however. Specifically, fine texture details, photographic illumination and compression artefacts introduce unwanted responses alongside genuine structural boundaries. A further source of false detections arises from UV seams, which are cuts where the mesh surface is split apart so it can be unwrapped into a flat 2D texture. Neighbouring pixels on opposite sides of such a cut can correspond to distant locations in texture space and the Sobel operator reads these discontinuities as strong edges. Both categories of false detection motivate V4.2, where Gaussian smoothing is introduced before gradient computation to reduce sensitivity to high-frequency variation.

4.4.7. V4.2 — Gaussian Prefiltered Sobel

V4.2 inserts the low-pass stage that Section 2.4.3 identifies as a prerequisite for stable gradient estimation. A discrete 3×3 Gaussian kernel smooths the texture before the Sobel operator computes the image gradients. This addition targets the noise sensitivity that limited V4 rather than any deficiency in the operator itself.

Gaussian prefiltering produced cleaner edge maps than V4 across all three platforms, though the degree of improvement differed markedly by texture source. On Mixamo, Gaussian smoothing reduced edge responses caused by compression artefacts and subtle texture variation while preserving the major clothing boundaries and facial features. On the Meta Avatars SDK, prefiltering similarly attenuated high-frequency artefacts in the composited rendering output. Edge structures were consequently cleaner, particularly on clothing regions. Avaturn remained the most challenging platform because its photographic albedo contained baked illumination that introduced strong intensity gradients unrelated to surface structure. Although Gaussian smoothing attenuated some of these responses, many illumination gradients remained sufficiently strong to influence the Sobel output.

Overall, Gaussian prefiltering delivered cleaner edge maps than V4 by reducing unwanted texture responses. However, this improvement came at the cost of weaker structural contours, particularly along features with limited contrast. Because Gaussian smoothing attenuated both noise and genuine edges, reducing false detections suppressed some meaningful structural information. Ultimately, a compromise between contour preservation and noise reduction remained unavoidable.

Among the evaluated platforms, Avaturn exhibited the weakest performance under both V4 and V4.2. The baked illumination present in the photographic texture prevented reliable separation of structural boundaries from lighting variations. Gaussian prefiltering was expected to resolve this by suppressing noise on the skin texture enough to extract facial segments such as the lips and eyebrows. It reduced some of this noise, but not enough to recover these segments reliably. This shortfall motivated the texture preprocessing pipeline described in Section 4.6 instead, where the source texture was edited to reduce gradients introduced by illumination before edge extraction. Once this preprocessing was applied, the basic Sobel operator already produced reliable structural edge detection, without the added cost of Gaussian prefiltering. Section 4.5 reports this outcome in detail.

Although Gaussian prefiltering improves the robustness of texture-based edge detection, its effectiveness remains dependent on the quality of the input texture. The following techniques therefore investigate post-processing stylisation approaches that operate on the rendered image rather than directly on the texture. This reduces the influence of albedo quality on edge detection and yields greater control over the final stylised appearance.

4.4.8. V5 — Hierarchical Edge Detection with Multiple Cues

Hierarchical edge detection combines multiple complementary boundary cues into a single, unified representation rather than relying on any one signal alone [40]. V5 adopts this approach as a post-processing architecture that operates on the rendered image rather than on the source texture and this addresses the dependence on albedo quality identified in V4.2. Depth discontinuities recover object silhouettes, surface normal discontinuities reveal geometric structure and colour gradients computed using the Roberts Cross operator detect texture boundaries that have no corresponding geometric signal [17], [18]. Each layer carries an independent weight and the

(a) Mixamo: Gaussian prefiltered Sobel edge detection.

(b) Meta Avatars SDK: Gaussian prefiltered Sobel edge detection.

(c) Avaturn: Gaussian prefiltered Sobel edge detection.

Figure 4.12.: V4.2 (Gaussian Prefiltered Sobel) across the three avatar platforms. t: Sobel threshold.

responses are blended through adjustable parameters so that the contribution of each cue can be controlled interactively.

Although the same hierarchical formulation was evaluated across all three avatar platforms, differences in the rendering architecture affected which edge cues could be computed. On Mixamo and Avaturn, V5 operated as a post-processing effect that sampled the depth, normal and colour buffers provided by the URP. By contrast, the Meta Avatars SDK restricts custom rendering logic to a per-fragment pass without access to global screen-space depth and normal buffers. Consequently, the depth cue could not reliably separate the avatar from the background, so its weight was set to zero and the composite relied on the normal and colour cues instead. Figure 4.13 shows the background bleeding that motivated this choice.

Figure 4.14 compares the results across the three avatar platforms. V5 detects a broader range of structural boundaries by combining multiple edge detection techniques on the Mixamo and Avaturn rendered frames. The depth threshold has little visual influence, since the inverted hull outline already provides the silhouette. The normal and colour thresholds instead control the density of interior contours. These contours trace geometric curvature, appearing along the underside of the nose and the outer lip edges, but do not produce clear boundaries between adjacent facial segments. Nonetheless, the colour gradient layer occasionally responds to illumination

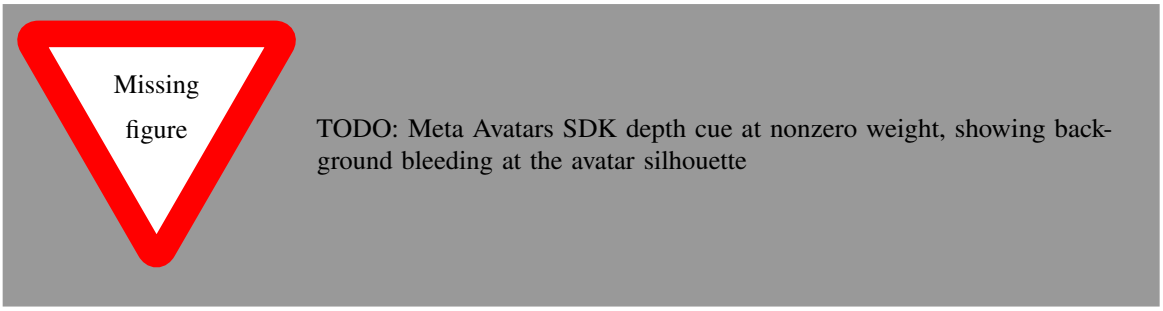


Figure 4.13.: Meta Avatars SDK: depth cue enabled at nonzero weight. Without a dedicated avatar mask, the depth proxy cannot separate the avatar from the background, motivating the zero depth weight used in the final composite.

gradients in the rendered image, particularly around the nose bridge where shading transitions are interpreted as texture edges. On the Meta Avatars SDK, the absence of screen-space depth prevents silhouette detection, although combining the remaining cues still produces richer interior detail than the previous techniques. The composited SDK output contains baked lighting gradients absent from the raw albedo, making the texture layer more prone to false responses on this platform. The colour layer therefore replaces the Roberts Cross computation with an XDoG operator on this platform. This reduced these false responses, as Figure 4.14(b) shows.

Compared with the previous techniques, V5 provides the most complete boundary representation by integrating silhouette, geometric and texture information within a single framework. Nevertheless, because each layer contributes independently to the final edge map, the overall edge density increases and requires weight adjustment to avoid visually cluttered results. Moreover, the contribution of each layer remains spatially uniform across the rendered image, regardless of the local image content. This motivated the exploration of region-based stylisation in V6, where the surface appearance is reorganised into coherent colour regions rather than extended with additional edge marks.

4.4.9. V6 — Kuwahara Painterly Filter

The anisotropic Kuwahara formulation aligns an elliptical sampling window with local image structure rather than with the fixed quadrants of the classical filter [22]. V6 adopts this formulation, set out in Section 2.5. Orientation-aligned marks were the property required here, since brushwork ignoring surface direction reads as a uniform blur rather than as a painted surface.

The filter operates on the rendered avatar surface and produces a form of abstraction qualitatively different from the contour techniques of V3 through V5. Where those techniques add stylised marks over a photorealistic surface, the Kuwahara filter reorganises the surface appearance itself into coherent colour regions and the result is analogous to a painted or illustrated representation. A post-processing architecture is adopted on Mixamo and Avaturn because the anisotropic variant requires multiple rendering passes. The Meta Avatars SDK single-pass constraint does not permit this. On the SDK platform, Gaussian smoothing of the structure tensor requires a separate render pass and cannot be performed within the hook. The tensor is instead estimated from screen-space derivatives at individual pixels, producing noisy orientation estimates.

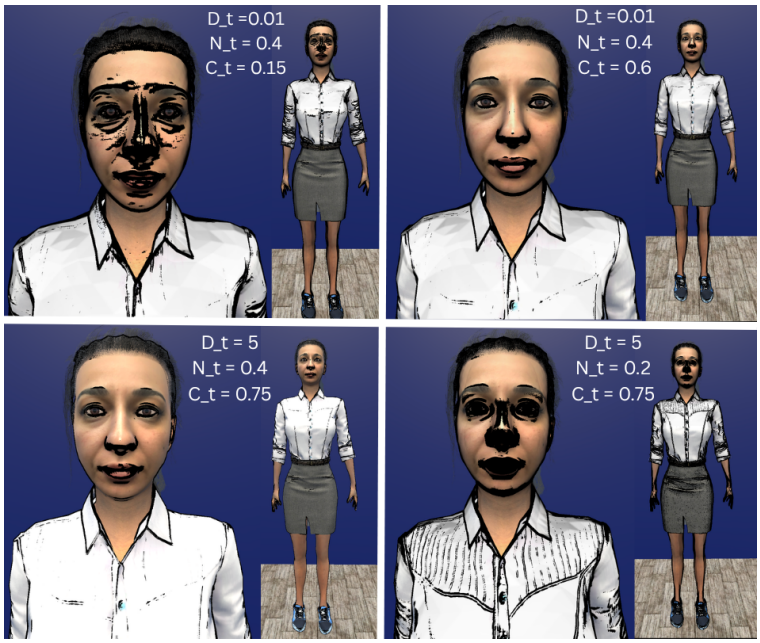
Figure 4.15 compares the results across the three avatar platforms. On Mixamo and Avaturn, the full anisotropic implementation produced coherent painterly regions with well-oriented brushstrokes that follow the contours of clothing and facial features. On the Meta Avatars SDK, the noisy tensor caused the inverse-variance weighting to favour the most uniform colour patch in each neighbourhood rather than following feature orientation. Coherent painterly regions were consequently absent and local colour saturation was amplified instead.

Achieving a clearly painterly effect, however, comes at a cost: at a kernel radius large enough for the painted regions to be perceptually distinct, fine facial features such as eyebrows, lip boundaries and iris details tend to be subsumed into the surrounding skin region. Reducing the kernel radius preserves these details but diminishes the painted quality of the result.

(a) Mixamo: hierarchical edge detection at varying C_t (top row) and N_t (bottom row).

(b) Meta Avatars SDK: hierarchical edge detection at varying threshold combinations with XDoG pre-blur enabled and disabled.

This trade-off between abstraction strength and feature legibility suggested that the Kuwahara filter is most effective when combined with a contour detection layer that restores structural boundaries.



(c) Avaturn: hierarchical edge detection at varying C_t (top row) and N_t (bottom row).

Figure 4.14.: V5 (Hierarchical Edge Detection with Multiple Cues) across the three avatar platforms. N_t: normal threshold; C_t: colour threshold. D_t is not varied as a display parameter because the inverted hull outline already provides the silhouette boundary on Mixamo and Avaturn regardless of the depth threshold value and the Meta Avatars SDK per-fragment pass has no access to the screen-space depth buffer.

(a) Mixamo: standard render (left) and Kuwahara painterly filter (right).

(b) Meta Avatars SDK: standard render (left) and Kuwahara painterly filter (right).

(c) Avaturn: standard render (left) and Kuwahara painterly filter (right).

Figure 4.15.: V6 (Kuwahara Painterly Filter) across the three avatar platforms.

4.5. Platform and NPR Technique Selection for the User Study

Three properties were required for a valid perceptual trust comparison: unrestricted URP shader access to apply the NPR techniques, full blendshape animation for animated facial expression in the video stimuli and photorealistic facial reconstruction to support the comparison with the real human speaker in the user study. On this basis, Avaturn was selected as the platform, since it is the only candidate among those evaluated in Section 4.3 that satisfies all three requirements. The Avaturn avatar used for both conditions C2 and C3 was reconstructed from photographs of the same individual who served as the real human speaker in C1 and rendering style is the primary variable that distinguishes the three conditions. Further preprocessing was also required to support reliable edge detection and the procedure is described in Section 4.6.

The NPR composite for condition C3 was assembled from five candidate components: an inverted hull outline, Sobel edge detection and XToon tonal shading, alongside two post-processing passes, a Kuwahara painterly filter and a hierarchical depth edge layer. Parameter settings for all five were determined through a parameter adjustment study and full details are in Section 4.7. It also established whether the two post-processing passes should be included in the final composite.

Among the techniques evaluated in Section 4.4, three were set aside before the composite was assembled for that study. Basic toon shading (V2) was replaced by X-Toon (V2.2), since the 2D ramp provides independent control over the abstraction level that procedural quantisation cannot offer. However, this control depends on how the abstraction level is driven. The depth and curvature modes described in Section 4.4.4 respond to camera distance and surface curvature respectively. Since the camera stays fixed and close to the avatar throughout the study recordings, this leaves little room for either quantity to vary across the visible surface. Manual mode avoids this problem entirely. It sets the abstraction value directly rather than computing it from camera distance or surface geometry. Manual mode was therefore the natural choice as the driving mode for the C3 composite. The study then fixed this value, referred to there as the detail bias (Section 4.7). Screen-space normal edge detection (V3) was excluded primarily because it cannot recover facial features defined by colour or texture contrast rather than geometric curvature. Smooth skin produces minimal normal discontinuities and the eyebrow boundary, lip contour, nostril edge and eyelid margin consequently remain invisible to the technique. These are precisely the landmarks that determine facial legibility in a conversational advisor context. V3 detects clothing folds and seams reasonably well through normal map discontinuities, but the Sobel operator subsumes this capability entirely: V4 captures the same clothing contours and additionally identifies edges at boundaries defined by colour contrast, a category that V3 cannot detect. Gaussian prefiltered Sobel (V4.2) was also excluded. Its additional sampling cost no longer produced a detectable improvement over the basic Sobel operator once the head albedo was preprocessed to reduce baked illumination gradients (see Section 4.6). The hierarchical pass was the only candidate to introduce a depth cue, but its two remaining layers replicate detection strategies already set aside earlier in this section: the surface normal layer fails on smooth facial skin for the same reasons that excluded V3 and the colour gradient layer adds no information beyond what the Sobel operator already provides. Both passes were then evaluated in the same study. Participants ranked the unmodified configuration as significantly more trustworthy than every tested variant of each and both were excluded. Full details of the visual assessment and the study results are described in Chapter 6.

4.6. Avaturn Avatar Preprocessing

To reduce the edge detection artefacts caused by the reconstructed albedo texture identified in Section 4.4.7, the head albedo texture was preprocessed in Blender before the avatar was re-exported as a GLB file. Four texture operations were applied to address two distinct problems in the head albedo: baked illumination variation that introduced spurious edge responses at boundaries unrelated to surface structure and the absence of a strong tonal cue at the lip boundary.

To address the first problem, a Gaussian blur attenuated baked ambient occlusion, subsurface scattering and pore detail from the photogrammetric albedo. This suppressed the high-frequency intensity variation before Sobel detection. A skin tone unification pass then blended skin pixels towards a common brightness and saturation baseline. This collapsed the baked luminance falloff into a consistent tonal surface. The brow and eye socket regions were excluded from this step through UV masks derived from geometry vertex positions,

since the chromatic contrast at those features was necessary for the Sobel operator to locate eyebrow and eyelid edges accurately.

To resolve the second problem, saturation and brightness were enhanced across the mouth area and a saturated, uniform colour zone was produced against the flattened skin. A lipline was then placed at the lip boundary. It provided a sharp tonal transition that Sobel detection and the X-Toon ramp in Unity could reliably identify. This preprocessed avatar served as the basis for all video stimuli in the user study.

Figure 4.16 shows the effect of the preprocessing pipeline on the modified avatar.

Figure 4.16.: Effect of the head texture preprocessing on the Avaturn avatar. Left: the default Avaturn avatar with no preprocessing and no NPR applied. Right: the preprocessed head texture, also with no shader applied.

4.7. Parameter Adjustment Study

Within the composite described in Section 4.5, the outline establishes the silhouette, Sobel gradients identify interior contours at colour boundaries and tonal shading controls colour quantisation. The two candidate post-processing passes each targeted a property the other three components could not provide: the Kuwahara filter would smooth texture regions into coherent painted areas and the hierarchical layer would add depth-based boundary detection at geometry discontinuities where colour contrast is insufficient for Sobel. Its depth cue proved largely redundant with the outline pass on Mixamo and Avaturn (Section 4.4.8), so the study tested whether the remaining interior contribution justified the extra pass.

A parameter adjustment study established both the parameter settings for all five components and whether the two post-processing passes should be included in the final composite. Participants ordered rendered images of the avatar by perceived trustworthiness as an advisor and the result of each round fixed one parameter group for all subsequent rounds. The study covered the parameter groups with the greatest expected influence on perceived edge density, tonal balance or abstraction strength; parameters with comparatively minor expected visual impact, including rim light presence, Kuwahara kernel size, Gaussian blur strength, sector count and normal edge strength, were instead set through direct visual inspection during development, to keep the study at a manageable length for participants.

One parameter group was varied while all remaining settings were held at the level fixed by the preceding test in each round. Table 4.2 lists all stages and the values examined at each.

45 pilot participants took part in the study. Participants were required to assign a unique rank to every image in the round. Any ballot with a missing rank or a tied rank was excluded and the number of valid responses consequently varies slightly by round, from 40 to 44 of the 45 submitted. Images were converted to Borda-count points in each round, where the image ranked last received 1 point and the image ranked first received points equal to the number of options and the totals were summed across participants. An omnibus Friedman test established whether the ranking within a round differed from chance and a pairwise Wilcoxon signed-rank test

Table 4.2.: Parameter groups and values tested in each round of the parameter adjustment study. R4A and R5A carry forward the R1E configuration as the no-treatment baseline for their respective rounds.

Round	Parameter group	Values tested (A, B, C, . . . in order)
R1	Inner line threshold	0.01, 0.02, 0.03, 0.04, 0.05
R2	Outline width and depth offset	width: 0.003, 0.005, 0.007; offset: on / off
R3	Shadow strength and detail bias	strength: 0.0, 0.5, 1.0; bias: 0.0, 0.5, 1.0
R4	Kuwahara hardness and sharpness	none (= R1E); (18, 1); (9, 9); (1, 18)
R5	Hierarchical depth threshold	none (= R1E), 0.01, 0.02, 0.03, 0.04, 0.05

compared the configuration ranked first against its closest competitor [54]. The same nonparametric approach is applied to the advisor ranking task in the main study (Section 4.8.4). Table 4.3 reports the outcome of each round.

Table 4.3.: Statistical test results for the parameter adjustment study by round.

Round	Borda winner	Friedman	Runner-up test
R1	threshold 0.03	$\chi^2_F(4) = 36.48$, $p < .001$	$p = .31$ vs. 0.05 (not distinguished)
R2	width 0.003, offset on	$\chi^2_F(5) = 126.19$, $p < .001$	$p = .82$ vs. width 0.005, offset on (not distinguished)
R3	shadow 0.5, bias 1.0	$\chi^2_F(4) = 3.18$, $p = .53$	no configuration differed significantly
R4	none (no Kuwahara)	$\chi^2_F(3) = 38.97$, $p < .001$	$p = .003$ vs. best Kuwahara variant (significant)
R5	none (no hierarchical layer)	$\chi^2_F(5) = 117.20$, $p < .001$	$p < .001$ vs. best threshold (significant)

Two outcomes determine the final composite. In R1, threshold 0.05 received the most first-place votes (20 of 43 participants), yet threshold 0.03 accumulated the higher cumulative Borda score (160 vs. 149 points), since it was ranked more consistently near the top across all participants. The pairwise Wilcoxon signed-rank test found no significant difference between the two (p = .31) and the omnibus Friedman result confirmed a clear directional trend: the two lowest thresholds were significantly disfavoured and the top two configurations were accordingly treated as equivalent. In R2, the narrower outline width (0.003, depth offset on) accumulated the highest Borda total but was not statistically distinguished from the standard width (0.005, depth offset on, p = .82) and was therefore retained since no significant preference justified using the wider option. In R3, no significant differences emerged across any configuration (p = .53) and the Borda winner was shadow strength 0.5, detail bias 1.0, with no configuration statistically distinguishable from another. In both R4 and R5, however, the baseline without any post-processing was significantly preferred over every tested variant of the Kuwahara filter and the hierarchical depth layer respectively. Because trustworthiness is the primary outcome measure of the main study, both passes were excluded from the C3 composite on this basis, consistent with Section 4.5. The per-round ranking distributions for all configurations are provided in Appendix A.

For R1 and R3, where the pairwise tests found no statistically distinguishable winner, the overall impression question served as a secondary tiebreaker. R1E (threshold 0.05) received the most nominations overall with seven and R3B (shadow strength 0.0, detail bias 0.5) followed with six. Both were thus selected for their respective rounds. Table 4.4 lists the complete parameter set established by the parameter adjustment study.

Figure 4.17 shows every rendered variant presented to participants across all five rounds, including the Kuwahara and hierarchical variants that were ultimately not retained.

Table 4.4.: Final C3 composite parameter set established by the parameter adjustment study.

Parameter	Value	Source
Outline width	0.003	R2 Borda winner
Depth offset	On	R2 Borda winner
Inner line threshold	0.05	R1 overall impression
Shadow strength	0.0	R3 overall impression
Detail bias	0.5	R3 overall impression
Kuwahara filter	Off	R4 excluded (significant)
Hierarchical depth layer	Off	R5 excluded (significant)

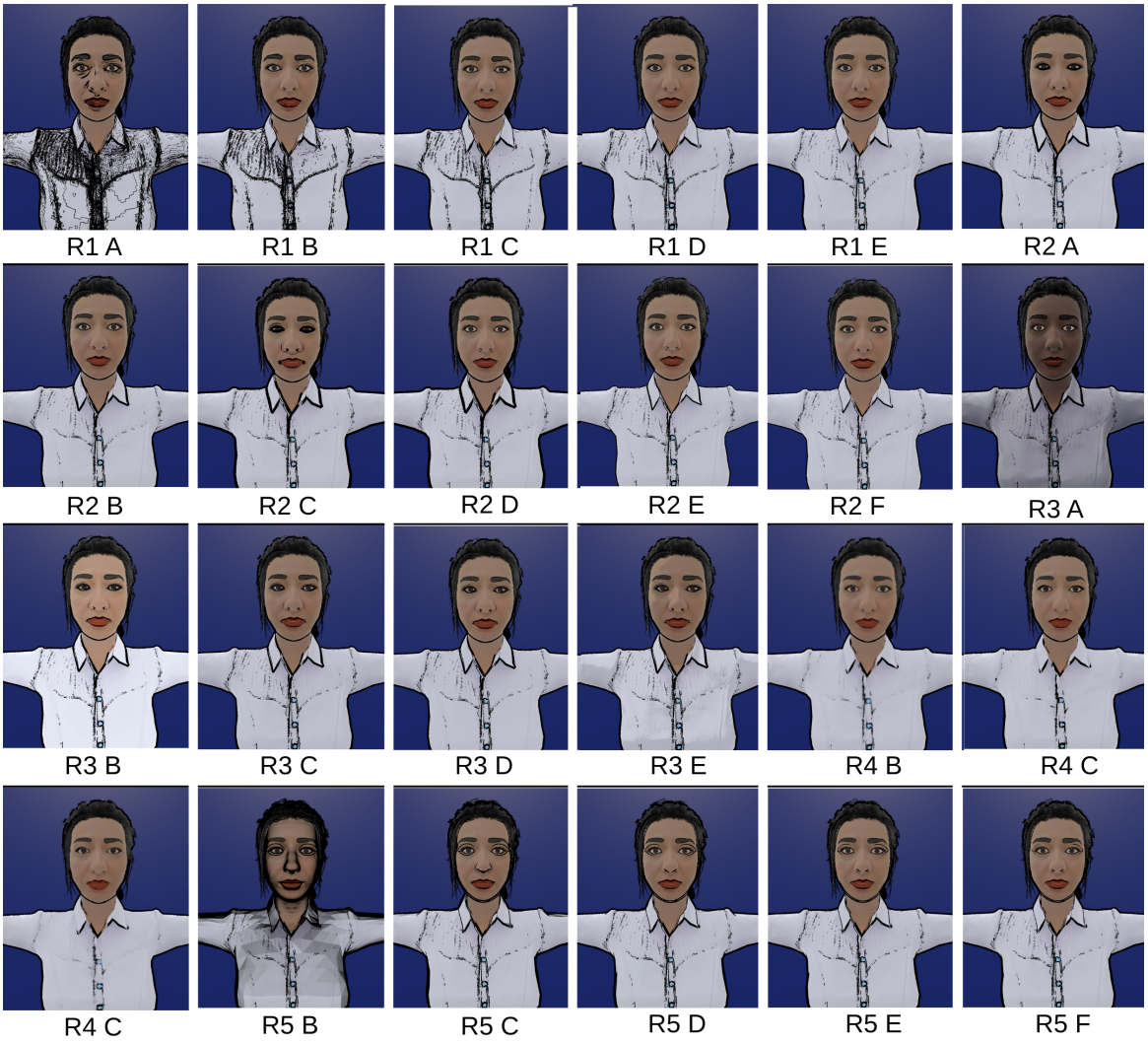


Figure 4.17.: All rendered variants presented during the parameter adjustment study. Labels follow the convention R*n*X, where *n* is the round number and X is the variant letter in the order listed in Table 4.2. R4A and R5A are identical to R1E and are not shown as separate images.

4.8. User Study Design

4.8.1. Objective and Hypotheses

The user study addresses RQ3 by measuring how the visual representation of a speaker affects perceived eeriness and trustworthiness in a concrete decision situation. Four hypotheses are tested.

H1: The NPR avatar (C3) will produce lower eeriness ratings than the photorealistic avatar (C2).

Photorealistic avatars occupy the uncanny valley region where negative affect peaks, while stylised abstraction moves the representation towards the stable ascending slope at lower fidelity [4]. H1 tests whether this shift produces a measurable reduction in eeriness between C2 and C3.

H2: The NPR avatar (C3) will not produce significantly lower perceived trust than the photorealistic avatar (C2).

Prior work found that stylisation level does not reduce perceived trust when the advisor role and content are held constant [5], [32]. H2 tests whether this pattern holds when rendering style is the primary variable that changes between conditions.

H3: Eeriness will negatively predict trust ratings across all conditions.

Eeriness functions as a specific negative affective state that consistently suppresses positive social evaluations, including trust [16], [29]. H3 tests whether this relationship is detectable across all three conditions within each participant.

H4: The NPR avatar (C3) will not produce significantly lower trust ratings than the real human speaker (C1).

Uncanny valley theory places a real human speaker at the peak of the affinity curve, where human likeness is maximal and comfort is highest [4]. H4 tests whether the abstraction responsible for H1's predicted reduction in eeriness closes the trust gap enough to reach this reference point, rather than assuming a real human speaker remains categorically more trusted than any avatar.

4.8.2. Conditions

The study compares three presentation conditions:

C1 — Real Person: A video recording of the same human speaker whose photograph was used to generate the Avaturn avatar, so that face identity, voice and demographic appearance are held constant across all three conditions. C1 serves as the trust ceiling baseline against which both avatar conditions are compared.

C2 — Photorealistic Avatar: The same speaker scenario rendered using a photorealistic Avaturn avatar with standard PBR shading. Facial expressions are driven by blendshape animation captured in the same recording session as C1, so that expression timing and facial gestures are identical across all three conditions.

C3 — NPR Avatar: The Avaturn avatar rendered with the NPR technique selected through the parameter adjustment study on the basis of its balance between abstraction and expressiveness.

Using the same Avaturn geometry for both C2 and C3 holds avatar identity, facial structure and animation constant across the two avatar conditions and render style is consequently the primary independent variable between them. Canales et al. applied the same principle in their stylisation study, holding speaker identity and scripted content constant across conditions while varying only the avatar rendering style [5]. The theoretically critical comparison is C2 versus C3. Since animation is held constant between these two conditions, any observed difference in trust or eeriness ratings is attributable primarily to the change in rendering style, with one material-level exception described in Section 4.6. C1 provides a second point of comparison for H4. This test asks whether C3 reaches the real-person reference point overall rather than isolating rendering style specifically, as animation naturalness and speaker identity also differ between C1 and both avatar conditions.

4.8.3. Scenarios

Three video scenarios present naturalistic situations in which each speaker delivers a recommendation that participants must decide whether to follow. Each scenario has two plausible options and the speaker recommends one with explicit reasoning. Several design principles govern the scenario construction.

Ecological validity. The three scenarios are drawn from everyday situations that participants are likely to encounter in their social and professional lives: a career decision between two job offers, route planning in an unfamiliar city and selection of a learning resource for career development. Grounding the scenarios in

contexts drawn from daily life ensures that trust judgments reflect responses to the advisor rather than the effort of interpreting an unfamiliar situation.

Vignette anchoring. Before each video, participants read a short context paragraph that anchors them in a specific, concrete situation with defined personal circumstances. Without this anchoring, trust ratings would reflect individual differences in life experience rather than responses to the speaker. The vignette eliminates this source of variance across participants and every participant enters each video from the same defined role.

Recommendation alternation. The three scenarios are designed so that the recommended option alternates across videos: Scenario 1 recommends Option A, Scenario 2 recommends Option B and Scenario 3 recommends Option A. This prevents participants from detecting a systematic preference for either option label across the study.

Presentation order alternation. Within each script, the recommended option is not always discussed last. Discussing the recommended option last in every video would introduce a recency effect, where participants associate recommendation with position rather than with the speaker's reasoning. Scenario 1 and Scenario 3 discuss the recommended option first. Scenario 2 discusses the recommended option last. All scripts conclude with an explicit recommendation statement regardless of discussion order.

Script argumentative structure. Each spoken recommendation follows a fixed five-step structure, applied uniformly across all three scenarios: (1) state the recommendation explicitly; (2) acknowledge the primary drawback of the recommended option; (3) introduce the key advantage of the recommended option with an explicit contrastive marker; (4) name the central weakness of the alternative option; and (5) close with a concise restatement. This structure ensures the advisor engages visibly with trade-offs before concluding, which holds rhetorical strategy constant across scenarios. Any difference in trust ratings between conditions therefore reflects the visual presentation of the speaker rather than variation in argumentation pattern between scenarios.

The three scenarios and their vignette texts are as follows.

Scenario 1 — Career Choice. The participant is placed in the role of a recent graduate choosing between two job offers, with financial pressure and location held constant as confounds. The speaker recommends Job A, judging career flexibility to outweigh the higher salary offered by Job B. This scenario primarily activates competence trust.

Scenario 2 — Travel Route. The participant must reach a hotel for an urgent meeting, choosing between a direct 50 minute route that arrives ten minutes before the meeting and a faster 20 minute route requiring one transfer that arrives forty minutes early. The speaker recommends the faster option with the transfer, arguing that its 40 minute buffer provides enough recovery time to outweigh the transfer risk, whereas the direct route leaves an insufficient margin. This scenario activates competence trust through reasoning about risk and benefit and benevolence trust through prioritising the participant's ability to arrive on time.

Scenario 3 — Learning Resource. The participant chooses between a paid online course with a named certificate and a free tutorial series covering the same material. The speaker recommends the paid course on the grounds that a recruiter cannot verify a free tutorial series, making the recommendation rest on credential value rather than content superiority. This scenario primarily activates integrity trust.

Scenario order is held constant across all participants: Career in Video 1, Travel in Video 2 and Learning in Video 3. The three conditions are counterbalanced across those fixed positions using a 3×3 Latin square (Table 4.5), with one study version per row. Each appears exactly once per presentation slot and this controls for any systematic advantage conferred by video order. Sequential carryover between specific condition pairs is not addressed by this design and is noted as a limitation in Chapter 7. In the questionnaire, the three video slots are named Video 1, Video 2 and Video 3 so that participants cannot identify the rendering style. Each version's condition mapping is stored separately and applied during data analysis. The complete questionnaire, including all vignette texts as presented to participants, is reproduced in Appendix A.

Table 4.5.: Latin square counterbalancing across three study versions. Each condition appears exactly once per version, with scenario positions held fixed.

Version	Video 1 (Career)	Video 2 (Travel)	Video 3 (Learning)
A	C2 — Photorealistic Avatar	C1 — Real Person	C3 — NPR Avatar
B	C3 — NPR Avatar	C2 — Photorealistic Avatar	C1 — Real Person
C	C1 — Real Person	C3 — NPR Avatar	C2 — Photorealistic Avatar

4.8.4. Procedure and Measures

Procedure

The study was conducted entirely online via Google Forms. Each participant completed a consent and demographic questionnaire, then watched three video blocks in the order determined by their assigned study version. Each block followed four sequential steps:

1. Read the vignette establishing the scenario.
2. Record an initial option preference before watching (this pre-video baseline is used to compute preference shift as a secondary exploratory measure).
3. Watch the 23 second recommendation video.
4. Complete the per-block questionnaire.

After all three blocks, participants complete a post-study section. The entire session takes approximately 20 to 25 minutes. The questionnaire underwent iterative shortening across three successive rounds of piloting, in response to concerns about participant burden and completion time. The form described in this section is the final one used for data collection.

Per-Block Questionnaire

The per-block questionnaire is administered after each video. Parts B and D use a five point semantic differential scale. Parts E through H use a seven point Likert scale.

Part A — Comprehension check. Two gating items verify that participants watched the video and identified the correct recommendation. An embedded attention check item is also included in two of the three blocks (Section 4.8.4). Since these checks are not present identically across every block, a participant who fails any gating item in any block is excluded from the analysis entirely, rather than only for that block, so that no participant contributes a mix of validated and unvalidated data.

Parts B, D — Semantic differential (5 point). Part B (likeability) is administered before Part D (eeriness) so that participants form a general social impression before attending to uncanny affect. A reversed order would prime observers towards negative perceptions and risk inflating eeriness scores before the trust battery. Part B uses the five-item Godspeed likeability subscale, from Bartneck et al. [15]. The Godspeed anthropomorphism subscale was included during early piloting and removed in the shortening described above, since a second five-item semantic differential battery extended completion time without measuring a construct central to the study hypotheses. Part D captures eeriness through eight items from the validated instrument of Ho and MacDorman. Their validation showed that the Godspeed subscales are too strongly intercorrelated to measure distinct constructs and that no Godspeed subscale independently captures this specific affective state [16]. Each polar pair is preceded by a short definition to support participants unfamiliar with the terms. This instrument treats eeriness as a distinct affective response from likeability and H3 can consequently be examined independently.

Parts E, F, G — Trust battery (7 point Likert). Nine items operationalise the Ability-Benevolence-Integrity model of Mayer et al. within the trusting beliefs framework of McKnight and Chervany, introduced in Section 2.3 [13], [14]. Parts E, F and G cover Competence, Benevolence and Integrity respectively, with

three items each, measured independently rather than as a composite. The fourth construct in the McKnight and Chervany typology, predictability, is excluded given that it tends to merge with competence and integrity when participants encounter the advisor for the first time [14].

Part H — Response to recommendation. Part H records agreement with the recommendation and the participant's option choice after watching. This post-video choice is compared with the pre-video baseline to compute a preference shift score.

An embedded attention check item is included in the first and third blocks, with a different target response in each to prevent pattern detection.

Post-Study Section

After all three video blocks, participants complete a post-study section. Three separate ranking questions each ask participants to order all three advisors: from most to least wanted for consultation (P1), from most to least trustworthy overall and from most to least unsettling overall. P1 is the primary behavioural measure, adapted from the post-study advisor selection paradigm of Canales et al. [5]. It captures implicit trust judgment that explicit Likert ratings may not fully reflect and this addresses the methodological gap identified by Alipour et al. [29]. The trustworthiness and unsettlingness rankings provide behavioural analogues to the corresponding Likert measures, Parts E through G and Part D respectively. Implicit and explicit judgments can consequently be compared directly for both dimensions. A rank order is used for all three questions rather than a forced binary choice, since both avatar conditions are expected to score below C1. Limiting participants to two options would concentrate responses on C1 and leave insufficient variance to distinguish C2 from C3.

A visual style manipulation check (Part I) asks participants to rate the perceived visual stylisation of each advisor simultaneously. Its placement after all video blocks prevents awareness of rendering differences from priming trust ratings during data collection. The post-study section closes with two open-ended questions, each administered once rather than per block to limit response fatigue: the reflection question (Part J) asks participants to describe the most important factor in their trust judgment and a final open question asks what they believe the study was investigating and this serves as a check for demand characteristics.

The demographics section, completed before the first video block, also includes an item measuring dispositional trust (seven point Likert: *In general, I tend to trust people I meet for the first time*) as an individual-level covariate. Mayer et al. identify this disposition as a stable individual difference that must be controlled when comparing trust ratings across conditions [13].

Analysis Plan

The primary analysis is a one-way repeated-measures ANOVA per subscale composite (B_score, D_score, E_score, F_score, G_score), with condition (C1, C2, C3) as the within-subjects factor. Every hypothesis concerns an individual subscale rather than a linear combination of them and the subscales are accordingly analysed separately rather than through a multivariate omnibus test. Pairwise comparisons between the three conditions follow, corrected by Holm's step-down procedure. Holm holds the familywise error rate at the same level as the Bonferroni correction while rejecting more hypotheses and it adds no further assumptions [54]. Effect sizes are reported as partial η^2. The repeated-measures ANOVA is the established test for rating studies that compare rendering styles within participants [2], [8], [47]. Normality is checked with the Shapiro-Wilk test and sphericity with Mauchly's test and the Greenhouse-Geisser correction is applied where sphericity is violated. Cronbach's α is reported per subscale before the composites are formed, since averaging items presumes they measure one construct [15], [16].

The theoretically critical pairwise comparison is C2 versus C3. H2 predicts equivalence rather than a difference and this leads to a non-significant pairwise result providing insufficient evidence on its own. A TOST (two one-sided tests) equivalence test is therefore applied to the C2 versus C3 trust composite, with its bound pre-specified at Cohen's $d = 0.3$, the standard smallest effect size of interest for this procedure [55]. H4 predicts the same kind of equivalence against a different reference point and the identical procedure with the same bound is applied to the C1 versus C3 trust composite. Comparable work has inferred an absence of stylisation

effects on trust from non-significant results alone [5]. That inference cannot distinguish a genuine null from an insensitive test.

For H3, person-centred multilevel regression is used, with eeriness as the within-participant predictor of trust and a random intercept per participant. This isolates the within-participant effect from stable differences in how trusting individual participants are. Ratings pooled across observers, the dominant practice in this literature, confound the two [29].

Three post-study ranking questions, covering P1, trustworthiness and unsettlingness, are each analysed with a Friedman test comparing mean advisor ranks across conditions. Pairwise Wilcoxon signed-rank tests with the same Holm correction are then applied to each pair and C2 versus C3 is the critical pairwise comparison in each case [54]. Rank responses carry only ordinal information and the parametric assumptions above accordingly do not apply. The parameter adjustment study uses the same procedure (Section 4.7), meaning both rest on one statistical basis.

The Latin square assumes that a video's position in the sequence carries no effect of its own. Two checks test that assumption. Position is entered as a within-subjects factor in place of condition and study version is tested as a between-subjects factor on each participant's mean across the three blocks. Since every participant sees each condition exactly once, that mean is free of condition effects and a difference between versions therefore indicates a sequence effect.

Ethics and Data Protection

Participation was voluntary throughout. A consent screen preceded the first video block and described the task, the approximate duration and the right to withdraw at any point without giving a reason. No names, email addresses or contact details were recorded and no response carries an identifier that links it to a participant. The demographic items were limited to age, gender, prior exposure to avatars and dispositional trust, none of which identifies an individual within a sample of this size. Only aggregate results are reported.

The speaker recorded for C1 is identifiable and photographs of the same person were used to reconstruct the Avaturn avatar shown in C2 and C3. That speaker is the author of this thesis and no third party's likeness was recorded or reconstructed for the study materials.

Sample Size

A power analysis was conducted for a three-condition within-subjects ANOVA. It targeted a medium effect size of partial $\eta_p^2 = 0.06$, $\alpha = .05$ and power = 0.90. The minimum required sample is $n = 38$ participants. The recruitment target is set to $n = 48$, with sixteen participants per study version across three versions. This number accommodates anticipated attrition and exclusions from the comprehension check and it preserves balance across all three counterbalancing patterns.

Equivalence testing carries its own power requirement. A paired TOST at the $d = 0.3$ bound reaches a power of 0.90 only near $n = 120$ and at $n = 48$ its power stays below 0.40 even when the true difference is zero. The recruitment target follows the ANOVA requirement, so the equivalence components of H2 and H4 are underpowered by design. Their outcome is reported as inconclusive rather than as evidence against equivalence.

5. Implementation

This chapter documents the technical realisation of the NPR avatar system. It covers the development environment, the integration path for each avatar platform, the shared rendering structure and the HLSL implementation details of each stylisation technique. The runtime tooling built to support interactive parameter exploration and evaluation capture is described at the end of the chapter. The animation and recording pipeline used to produce the video stimuli for the user study is presented afterward.

5.1. Development Environment

The system was developed in Unity 6 (version 6000.0.60f1) using the Universal Render Pipeline (URP). The target hardware is the Meta Quest 3 standalone VR headset. The Meta Avatars SDK version 40.0.1 was used for the avatar platform managed by the SDK. All shader code is written in HLSL and targets the URP forward rendering path. The techniques were run on this hardware throughout development, although their cost was never formally profiled. Sample counts for the most complex techniques scale with kernel radius and whether these hold the frame budget therefore remains open.

5.2. Avatar Platform Integration

5.2.1. Mixamo

Mixamo characters were exported as FBX assets and imported into Unity as standard skinned mesh renderers. As introduced in Section 4.3, this SDK-free structure allows every URP material to be replaced directly with a custom NPR shader.

5.2.2. Meta Avatars SDK

The Meta Avatars SDK generates its shader source from a set of packaged templates managed through the `OvrShaderManager` component. This generated source is not intended to be edited directly. Custom logic is instead injected through the hook function that the SDK exposes at a defined point in the shader pipeline.

The hook used in this thesis was `AppSpecificPostManipulation`, which executes per fragment after all SDK PBR lighting (subsurface scattering, eye glints, hair shading) has been computed. It receives the accumulated lit colour and returns a modified colour. This grants full control over the final fragment output without disrupting any SDK rendering logic.

5.2.3. Avaturn

Avaturn follows a different integration path from the Meta SDK's hook-based model. The additional import step required for its `.glb` export, introduced in Section 4.3, is handled by the GLTFast package, which produces a standard Unity GameObject hierarchy with skinned mesh renderers and PBR materials assigned from the glTF material definitions, including albedo, normal and metallic-roughness textures. Custom NPR materials are therefore assigned to the resulting skinned meshes using the same approach as Mixamo: direct material replacement after load and blendshape animation targets exported by Avaturn are preserved through this process.

5.2.4. Common Scene Environment for Avatar Platform Comparison

Jody, Avaturn and the Meta Avatars SDK avatar each occupy a separate scene, since every platform requires its own configuration. The same room asset is reused across all three: an enclosure with a tiled floor, brick walls and one dark blue accent wall, plus a controlled skybox and no emissive surfaces. Camera position and framing are matched as closely as possible so that comparison renders share an equivalent viewpoint. Jody and Avaturn additionally share a directional light. It is elevated $-30°$ and rotated $-30°$ horizontally relative to the avatar forward vector. This places it above and towards the front left of the avatar. Its intensity is fixed at 1.0 lux with a neutral white colour. Moreover, the Meta Avatars SDK scene carries additional lighting beyond this configuration in order to improve visibility of the applied technique during the comparison. Ambient intensity and the GI contribution, baked from the shared skybox, nonetheless align across all three platforms.

5.3. Shared Rendering Structure

Each NPR technique on the Mixamo and Avaturn platforms is a standalone multi-pass shader and technique switching is performed by replacing the active material set. By contrast, all techniques on the Meta SDK platform share a single shader that dispatches to the corresponding technique based on an active keyword. Runtime switching then requires only toggling this keyword without recompilation.

On Mixamo and Avaturn, each avatar mesh is further split across several submesh materials, four on Mixamo and six on Avaturn, so that technique parameters can be tuned separately for regions such as skin, hair and eyelashes. Their vertex shaders also carry a vertex fetch macro shared with the Meta Avatars SDK codebase. This macro has no effect on Mixamo and Avaturn, since standard Unity vertex attributes are already available directly and it is kept only for consistency across all three platforms' shaders.

The automatic LOD management system on the Meta SDK platform replaces avatar materials as camera distance changes and this reverts any custom shader assignment made at load time. A dedicated runtime script resolves this: it performs an initial material replacement across the different detail levels once the avatar finishes loading, then polls the active materials on each renderer at a configurable interval. If the script detects a material assigned by the SDK, a LOD transition has reverted the custom assignment. It then reapplies the NPR material and copies the base colour and normal texture references from the newly active SDK material into its properties.

5.4. NPR Technique Implementation

All shader parameters described in the following subsections, as well as the choice of which NPR technique is active, can be adjusted either in the Unity Inspector during offline development or through a custom UI panel inside the running application. Changes made either way take effect immediately without shader recompilation.

5.4.1. V1 — Inverted Hull Outline Pass

The outline pass implements the inverted hull silhouette by applying the vertex displacement of Equation (4.1), identical in structure across all three platforms. In the Mixamo and Avaturn shaders the pass culls front faces and writes depth using a standard depth test. The Meta SDK shader applies the same front-face culling. However, it disables depth writing and uses a looser depth test instead to avoid conflicting with the SDK's depth compositing. The fragment outputs a flat outline colour. A configurable depth bias shifts the clip-space Z coordinate towards the camera to prevent z-fighting where the expanded shell meets the original surface.

Alpha Cutout Handling

Alpha cutout is controlled by a toggle flag. When set, both the outline pass and the forward-lit pass sample the base texture alpha and clip against the cutoff of Equation (4.2) before any displacement or shading is applied. The value `_AlphaCutoff` = 0.1 was selected visually as the lowest setting that reliably removes fully

transparent regions without clipping the visible hair silhouette, shown in Figure 4.5. The Meta Avatars SDK does not require this flag. Its fragment pipeline resolves hair and eyelash transparency before the hook is invoked and any fragment the SDK would discard is therefore already clipped by the time the outline colour is applied.

5.4.2. V2 — Toon Shading

This technique is realised as two shader implementations across the three avatar platforms. Mixamo and Avaturn share the same toon quantisation code, while the Meta Avatars SDK requires a separate implementation. The outline pass carries over unchanged from Section 5.4.1 across all three platforms. The Meta variant omits the explicit alpha test, since this is already handled by the SDK (Section 5.4.1).

Mixamo and Avaturn

The toon pass quantises the Lambertian dot product $\mathbf{N} \cdot \mathbf{L}$ per step using a per-band smoothstep, shared without modification between the two platforms:

```
float steps  = max(1.0, _ToonSteps); // number of shading bands
float scaled = NdotL * steps; // map into band index
float band   = floor(scaled); // current discrete band
float frac   = scaled - band; // offset within the band
float blend  = smoothstep(1.0 - _ToonSmoothness, 1.0, frac); // edge blend
float toon   = saturate((band + blend) / steps); // combined 0-1 value
toon = lerp(1.0 - _ShadowStrength, 1.0, toon); // remap to shadow range
```

Listing 5.1: V2 toon quantisation (Mixamo and Avaturn, unified).

The default value of `_ToonSteps` is four. This value was selected because visual assessment confirmed that it provides an effective balance between shape legibility and graphic simplification, shown in Figure 4.7. Moreover, each band boundary receives an independent narrow blend of width `_ToonSmoothness`, so normal jitter near any seam produces a localised crossfade rather than snapping the whole surface patch between bands. An additive ambient term and a rim highlight are then added to the quantised diffuse.

Meta Avatars SDK

This implementation runs inside the hook and receives the fully composited PBR colour produced by the SDK's renderer. The quantisation therefore operates on RGB directly and preserves hue while stepping luminance:

```
float  bands      = max(2.0, _ToonColorBands); // number of colour bands
float3 posterized = floor(color.rgb * bands + 0.5) / bands; // round to band
color.rgb = lerp(color.rgb, posterized, _ToonPosterizeStrength); // blend

float lum  = dot(color.rgb, float3(0.2126, 0.7152, 0.0722)); // luminance
color.rgb  = lerp(float3(lum, lum, lum), color.rgb, _ToonSaturation); // desat
```

Listing 5.2: V2 posterisation (Meta SDK).

A plain floor operation always rounds down and this introduces a darkening bias. Rounding to the nearest band avoids this bias and distributes the quantisation error symmetrically. Saturation adjustment runs after posterisation to prevent clamping artefacts. Since the input already contains Meta's subsurface scattering, anisotropic hair and eye glints, posterisation stylises an already physically correct image rather than reconstructing lighting from scratch.

Visual results for V2 across all three platforms are documented in Figure 4.7.

5.4.3. V2.2 — X-Toon Shading

The same split into two implementations as V2 (Section 5.4.2) carries over: Mixamo and Avaturn share the forward-lit pass, while the Meta Avatars SDK uses a separate implementation. The outline pass remains unchanged from V2 across all three platforms. The forward-lit pass replaces the procedural quantisation of V2 with a 2D ramp texture lookup, shown in Figure 4.8. Moreover, this adds an abstraction compression stage that is driven by a choice of three modes: depth, curvature or manual.

Mixamo and Avaturn

The forward-lit pass begins by optionally modifying the surface normal via Normal Field Abstraction, which is described in Section 4.4.4. The abstract normal map is a blurred version of the surface normal map and it is sampled when this option is active. It is then transformed from tangent space to world space through the TBN matrix. The result is linearly interpolated with the geometry normal:

```
// sample and unpack the abstract normal map
float3 abstractN  = UnpackNormal(SAMPLE_TEXTURE2D(
                        _AbstractNormalMap, sampler_AbstractNormalMap, uv));
float3x3 TBN      = float3x3(tangentWS, bitangentWS, normalWS); // TBN basis
float3 abstractWS = normalize(mul(abstractN, TBN)); // to world space
normalWS          = normalize(lerp(normalWS, abstractWS, _NormalSmoothing)); //
    blend towards simplified normal
```

Listing 5.3: Normal Field Abstraction (Mixamo and Avaturn).

The U coordinate for the ramp starts from the same Lambertian dot product used in V2 toon shading (Section 5.4.2). Shadow attenuation is now applied to this term as well, unlike in V2. The result is then remapped and interpolated towards the ramp midpoint to control light sensitivity.

The V coordinate (abstraction level) is computed from the fragment's world-space camera distance in depth mode, from the screen-space normal derivative magnitude in curvature mode or from a constant in manual mode, selected at compile time via `shader_feature_local` keywords. The ramp is then sampled at `(rampU, rampV)`. An abstraction compression stage then flattens the lighting bands as the abstraction level rises:

```
float abstractU = lerp(rampU, 0.5, rampV * 0.6); // compress towards midpoint
float dynSmoothing = lerp(_RampSmoothing, _RampSmoothing + 0.35, rampV); //
    widen blend as abstraction rises
float shadowMask = smoothstep(0.5 - dynSmoothing, 0.5 + dynSmoothing, abstractU)
    ; // soft lit/shadow threshold
float3 toonAlbedo = albedo * rampColor; // tint by ramp
float3 shadowedAlbedo = lerp(toonAlbedo * _ShadowColor.rgb, toonAlbedo,
    shadowMask); // blend towards shadow colour
float3 finalColor = lerp(albedo, shadowedAlbedo, _ShadowStrength); // apply
    overall shadow strength
```

Listing 5.4: Abstraction compression and shadow mask (Mixamo and Avaturn).

As V increases, the U compression pulls rampU towards the midpoint while the smoothstep width widens and this softens band boundaries rather than sharpening them. This same mask then blends the ramp-tinted albedo towards a configurable shadow colour. An overall shadow strength parameter controls how much of the effect applies over the plain albedo. A Blinn-Phong specular highlight is added after the shadow stage. A configurable smoothness parameter controls the transition width and values near zero produce a sharp specular highlight.

Meta Avatars SDK

The Meta implementation runs inside the hook. Unlike the Mixamo and Avaturn shaders, this hook does not own the vertex stage. World-space position is instead forwarded from the SDK vertex output as an explicit fifth argument to the effect function. This makes the same U axis computation as the standalone platforms possible.

Thus, `GetMainLight` is called with a per-fragment shadow coordinate derived from that world-space position. The shadow-attenuated NdotL is then computed using the same approach as on the standalone platforms:

```
float4 shadowCoord = TransformWorldToShadowCoord(positionWS); // per-fragment
    shadow coordinate
Light mainLight = GetMainLight(shadowCoord); // main light
float3 lightDir = normalize(mainLight.direction); // light dir
float NdotL = dot(normalWS, lightDir) * mainLight.shadowAttenuation; // shadow-
    attenuated Lambertian term
float rampU = lerp(0.5, saturate(NdotL * 0.5 + 0.5), _XToonLightSensitivity); //
     remap towards ramp midpoint
```

Listing 5.5: U axis (shadow-attenuated NdotL), Meta SDK.

The surface tangent frame is not available in the hook and this leaves Normal Field Abstraction via an abstract normal map unsupported. Only the geometric smoothing path is available on this platform. The abstraction compression and dynamic smoothing stages match those on the standalone platforms. The same `lightDir` and shadow attenuation from the U axis computation are reused for the specular term.

Figure 4.9 shows the V2.2 output across all three platforms.

5.4.4. V3 — Screen-Space Normal Edge Detection

The V2.2 X-Toon ramp shading is retained as the surface shading base. The normal gradient magnitude M_N (Equation (2.6)) maps to an edge weight through

$$E = \text{smoothstep}(\tau - \sigma,\ \tau + \sigma,\ M_N) \cdot \alpha_N \tag{5.1}$$

where τ is the threshold centre, σ is the threshold half-width and α_N is the edge weight scale. Both `ddx` and `ddy` operate on screen-space pixel derivatives rather than object-space geometry and the resulting edge width therefore scales with viewing distance rather than object scale, as discussed in Section 4.4.5. These three symbols are exposed as shader properties on all three platforms and can be adjusted at runtime through the parameter tuning interface described in Section 5.5.1:

- `_NormalEdgeThreshold` (τ)
- `_NormalEdgeSmoothness` (σ)
- `_NormalEdgeStrength` (α_N)

This edge mask is composited over the shaded surface by linear interpolation towards a configurable line colour.

Mixamo and Avaturn

The normal must be decoded from the PBR normal map before the derivative is computed. The vertex shader builds the tangent frame from the mesh tangent attribute and the fragment shader then unpacks the normal map and transforms it into world space:

```
// Mixamo, decoded by the Unity texture importer:
float3 normalTS = UnpackNormalScale(
                    SAMPLE_TEXTURE2D(_BumpMap, sampler_BumpMap, uv),
    _BumpScale);
// Avaturn, loaded by GLTFast without that importer pass:
float3 normalTS = SAMPLE_TEXTURE2D(_BumpMap, sampler_BumpMap, uv).rgb * 2.0 -
    1.0;
float3x3 TBN = float3x3(tangentWS, bitangentWS, geomNWS); // tangent frame
float3 nWS = normalize(mul(normalTS, TBN)); // rotate normal into world space
```

Listing 5.6: Normal decode via TBN (Mixamo and Avaturn).

When the normal map is disabled, `nWS` falls back instead to the bare interpolated geometry normal. The resulting `nWS` then feeds into the same gradient magnitude and threshold computation shown in Listing 5.7 for the Meta SDK platform, with `nWS` in place of `worldNormal`.

Meta Avatars SDK

On the Meta SDK platform, the world-space normal supplied to `ddx/ddy` is the SDK's fully composited fragment normal. This already incorporates the SDK's internal normal-map decode and no additional texture read is therefore required. The derivative and threshold computation itself is identical across all three platforms:

```
float3 dNdx = ddx((float3)worldNormal); // horizontal screen-space derivative
float3 dNdy = ddy((float3)worldNormal); // vertical screen-space derivative
float M = sqrt(dot(dNdx, dNdx) + dot(dNdy, dNdy)); // gradient magnitude
float nMin = _NormalEdgeThreshold - _NormalEdgeSmoothness; // lower threshold
    bound
float nMax = _NormalEdgeThreshold + _NormalEdgeSmoothness; // upper threshold
    bound
float edge = smoothstep(nMin, nMax, M) * _NormalEdgeStrength; // final edge
    weight
```

Listing 5.7: V3 normal gradient and edge weight, Meta SDK.

Figure 4.10 presents the V3 output across all three platforms, including how edge thickness depends on viewing distance.

5.4.5. V4 — Sobel Edge Detection

The edge source shifts from signals derived from geometry to the diffuse albedo texture. The eight neighbouring positions of the 3×3 neighbourhood are sampled and converted to a scalar luminance using the BT.709 coefficients defined in Equation (2.5). These coefficients match the sRGB primaries used by the rendered textures and the URP colour pipeline and this prevents colour channel cross-talk from introducing false edges at hue boundaries where RGB channels change in opposite directions but luminance is constant. The Sobel kernels G_x and G_y are applied to the nine luminance samples. Subsequently, the resulting gradient magnitude is compared against a threshold to produce the edge mask.

Mixamo and Avaturn

These platforms read the raw diffuse albedo texture directly. As a result, edge detection responds to surface material boundaries rather than to lighting:

```
float3 L = float3(0.2126, 0.7152, 0.0722); // BT.709 luma weights
float off = _InnerLineBlur * 0.001; // UV-space sample offset

float tl = dot(SAMPLE_TEXTURE2D(_MainTex, sampler_MainTex, uv+float2(-off, off))
    .rgb, L); // top-left tap
float t  = dot(SAMPLE_TEXTURE2D(_MainTex, sampler_MainTex, uv+float2(0, off)).
    rgb, L); // top tap
// ...the remaining six neighbour samples (tr, l, r, bl, b, br) follow the same
    pattern

float Gx = (tr + 2*r + br) - (tl + 2*l + bl); // horizontal gradient
float Gy = (tl + 2*t + tr) - (bl + 2*b + br); // vertical gradient
float edgeMag = sqrt(Gx*Gx + Gy*Gy); // gradient magnitude
float edge = step(_InnerLineThreshold, edgeMag) * _InnerLineStrength; //
    threshold and scale
color.rgb = lerp(color.rgb, _InnerLineColor.rgb, edge); // composite line colour
```

Listing 5.8: Eight-tap luminance sampling and Sobel gradient.

The sampled albedo already excludes lighting. Consequently, the gradient magnitude is compared directly against `_InnerLineThreshold` to produce the edge mask.

Meta Avatars SDK

The Meta SDK hook receives only the composited PBR output rather than a raw albedo texture. The Sobel therefore responds to shadow boundaries and specular gradients in addition to material transitions. This reduces selectivity, but it still adds visible edges along clothing boundaries and broad facial contour lines. The Meta Avatar's stylised, illustrated texture nonetheless contains less high-frequency variation than the photographic albedo used on Avaturn and this yields a comparatively cleaner edge response, as discussed in Section 4.4.6. Only the sample source changes against Listing 5.8. The gradient and magnitude computation is otherwise identical:

```
float offset = _SobelSampleDist * 0.001; // UV-space sample offset
float tl = dot(tex2D(u_BaseColorSampler, uv+float2(-offset, offset)).rgb, L); //
    top-left tap
// ...the remaining seven taps follow the same pattern
```

Listing 5.9: Composited-colour tap replacing the albedo tap, Meta SDK.

Compositing the PBR output reintroduces the UV seam artefacts. This artefact is absent on the other two platforms, where sampling occurs on the pre-lighting albedo instead. The colour difference across a seam registers as a strong edge despite the absence of any real visual boundary. The suppression rejects detections whose luminance spread across all eight samples exceeds `_SobelSeamLimit` and a second upper bound `_SobelMax` clips spike magnitudes beyond any plausible structural edge:

```
float lMin = min(min(min(tl,t),min(tr,l)), min(min(r,bl),min(b,br))); // darkest
    sample
float lMax = max(max(max(tl,t),max(tr,l)), max(max(r,bl),max(b,br))); //
   brightest sample

float seamMask = step(lMax - lMin, _SobelSeamLimit); // reject seam spikes
float inBand = step(_SobelThreshold, edgeMag)
             * step(edgeMag, _SobelMax); // accept only valid range

float edge = inBand * seamMask * _SobelStrength; // combine gates and scale
color.rgb = lerp(color.rgb, _InnerLineColor.rgb, edge); // composite line colour
```

Listing 5.10: Seam suppression and magnitude band gate, Meta SDK.

The V4 output across all three platforms is compared in Figure 4.11.

5.4.6. V4.2 — Gaussian Prefiltered Sobel

This method carries over the V4 Sobel gradient computation across the same eight positions, but each luminance sample is now drawn through a prefilter rather than a single point sample. At each call, one centre sample, four axis-aligned cardinal samples and four diagonal corner samples are drawn at a configurable offset from the given UV position. The three weight groups are normalised to unity before use so that the output is always a valid luminance estimate.

This prefiltering is implemented as a shared helper function, applied identically on all three platforms except for the texture it reads:

```
// samples a 9-tap weighted average of luminance around uv
float SampleLuminanceBlurred(float2 uv, float blurR, float cW, float cardW,
    float diagW)
{
    float3 L = float3(0.2126, 0.7152, 0.0722); // BT.709 luma weights
```

```
    float  v = 0.0; // accumulated luminance
    v += dot(SAMPLE_TEXTURE2D(_MainTex, sampler_MainTex, uv).rgb, L) * cW; //
    centre tap
    v += dot(SAMPLE_TEXTURE2D(_MainTex, sampler_MainTex, uv + float2(blurR, 0)).
    rgb, L) * cardW; // cardinal tap
    // ...three more cardinal taps and four diagonal taps follow the same
    pattern
    return v; // blurred luminance estimate
}

// Weight normalisation: computed once before the eight Sobel positions are
    sampled
float totalW = _GaussianCenterWeight + 4.0*_GaussianCardinalWeight + 4.0*
    _GaussianDiagonalWeight;
float cW    = _GaussianCenterWeight   / totalW; // centre weight share
float cardW = _GaussianCardinalWeight / totalW; // cardinal weight share
float diagW = _GaussianDiagonalWeight / totalW; // diagonal weight share
```

Listing 5.11: 9-tap Gaussian luminance helper with weight normalisation (unified across platforms).

This helper reads the raw diffuse albedo texture on Mixamo and Avaturn and the composited PBR output on the Meta SDK, with the rest of the logic unchanged. It is called once at each of the eight Sobel neighbourhood positions and the resulting values feed the same gradient magnitude computation used in V4.

Additionally, a configurable threshold band, applied identically on all three platforms, gates this gradient magnitude. A lower bound rejects weak gradients as noise and an upper bound suppresses detections with an unusually large magnitude, such as UV seam spikes. This band gate replaces the luma range seam gate used in V4.

A series of four progressive smoothstep passes then sharpens the thresholded magnitude into a crisp edge mask, shown here for the Meta SDK, where all four passes are always evaluated and driven by a single tightness scalar rather than the individually toggleable flags used on Mixamo and Avaturn:

```
float hw1 = lerp(0.5, 0.03, _GSobelTightness); // pass 1 half-width
edge = smoothstep(0.5 - hw1, 0.5 + hw1, edge); // sharpen pass 1
float hw2 = lerp(0.5, 0.15, _GSobelTightness); // pass 2 half-width
edge = smoothstep(0.5 - hw2, 0.5 + hw2, edge); // sharpen pass 2
// ...two further passes repeat this pattern with hw3 = 0.25 and hw4 = 0.35
edge = pow(edge, _GSobelPowerCurve); // contrast curve
edge *= _GSobelStrength; // final edge weight
```

Listing 5.12: 4-pass progressive smoothstep sharpening and power curve, Meta SDK.

At tightness = 0 every half-width equals 0.5 and each pass reduces to a linear identity. At tightness = 1 the widths shrink to 0.03, 0.15, 0.25 and 0.35 and the passes together compose a steep sigmoid. A power curve above 1 suppresses soft halos, while values below 1 amplify thin lines instead.

Furthermore, three preset configurations that are Light, Moderate and Aggressive are offered on Mixamo and Avaturn. These bundle the parameters into predefined combinations for rapid setup across avatars and this removes the need to tune each parameter individually. Table 5.1 lists the parameter values assigned by each preset. The default parameter set on all three platforms corresponds to the Light configuration and this is the configuration shown in every later V4.2 comparison figure in this thesis. V4.2 itself, however, was not carried forward into the final study composite once texture preprocessing made the basic Sobel operator sufficient on its own (Section 4.5).

Table 5.1.: V4.2 preset parameter values (Mixamo and Avaturn).

Parameter	Light	Moderate	Aggressive
Blur radius multiplier (`_BlurRadiusMultiplier`)	0.8	1.0	1.2
Centre weight (`_GaussianCenterWeight`)	0.25	0.30	0.25
Cardinal weight (`_GaussianCardinalWeight`)	0.125	0.12	0.125
Diagonal weight (`_GaussianDiagonalWeight`)	0.0625	0.05	0.0625
Threshold min multiplier (`_ThresholdMinMultiplier`)	0.2	0.4	0.85
Threshold max multiplier (`_ThresholdMaxMultiplier`)	2.0	1.6	1.15
Smoothstep tightness (`_SmoothstepTightness`)	0.2	0.3	1.0
Power curve (`_PowerCurve`)	1.5	2.0	3.0

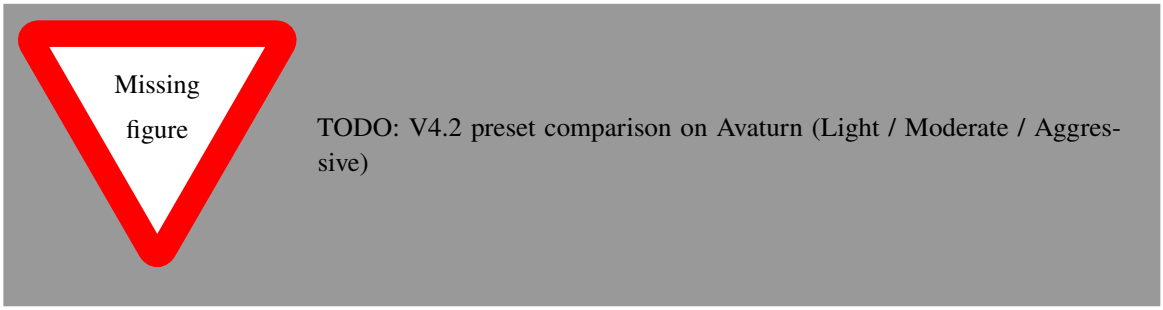


Figure 5.1.: Light, Moderate and Aggressive preset configurations compared on the Avaturn avatar.

Figure 4.12 compares the V4.2 output across all three platforms.

5.4.7. V5 — Hierarchical Edge Detection

V5 combines three independent edge cues into a unified boundary representation. Each cue is thresholded into a line signal and scaled by its own weight and the three weighted signals are then fused by max-pooling rather than addition. Thus, the strongest cue at a fragment dominates rather than being diluted by the others. The three input layers, however, are computed differently on each platform, since Mixamo and Avaturn access the depth, normal and colour buffers directly, while the Meta SDK is restricted to the per-fragment hook.

Mixamo and Avaturn

Mixamo and Avaturn run V5 as a URP renderer feature: a post-processing pass that samples Unity's scene depth, normal and colour buffers directly. Each layer applies the Roberts Cross operator to a 2×2 diagonal sample pattern, as defined in Section 2.4.1:

```
// diagonal difference operator on a 2x2 sample quad
float RobertsCross(float tl, float tr, float bl, float br)
{
    return abs(tl - br) + abs(tr - bl); // sum of diagonal differences
}

float d_tl = GetLinearEyeDepth(uv + float2(-off.x,  off.y)); // top-left depth
    tap
// ...remaining three depth taps (tr, bl, br) follow the same pattern
float depthEdge = RobertsCross(d_tl, d_tr, d_bl, d_br)
                / max(GetLinearEyeDepth(uv), 0.01) * _DepthScale; // normalised
    , scaled edge
float depthLine = smoothstep(_DepthThreshold - 0.01, _DepthThreshold + 0.01,
    depthEdge); // to line signal
```

Listing 5.13: Roberts Cross depth layer (Mixamo and Avaturn).

The normal layer applies the same Roberts Cross pattern to the screen-space normal buffer. The colour layer applies it to the rendered colour buffer's luminance and an added chromatic term also responds to boundaries defined by colour alone, with no luminance difference. An adaptive factor derived from local brightness reduces the depth and colour layers' sensitivity in dark image regions, where noise would otherwise dominate the response. The normal layer receives only half this reduction, as its response depends less on scene brightness. Since this post-processing pass operates on the entire screen, background geometry outside the avatar would also register spurious edges without correction. A dedicated avatar mask therefore restricts the composite to the avatar's screen-space silhouette. This mask is captured in a separate render pass, filtered to a Unity rendering layer assigned exclusively to the avatar, so that only avatar fragments are drawn into it. The resulting mask value is then multiplied into the edge blend factor, so that background pixels retain the original shaded colour unmodified.

Meta Avatars SDK

The Meta SDK's single fragment hook has no access to a dedicated depth or normal buffer. The depth layer therefore estimates a distance gradient computed from the fragment's world-space distance to the camera. This gradient is normalised for perspective:

```
float depth = length(worldPos - _WorldSpaceCameraPos); // fragment-to-camera
    distance
float dDepthX = ddx(depth); // horizontal depth derivative
float dDepthY = ddy(depth); // vertical depth derivative
float depthGrad = sqrt(dDepthX*dDepthX + dDepthY*dDepthY) / max(depth, 0.01) *
    _DepthScale; // normalised, scaled gradient
float depthLine = smoothstep(_DepthThreshold - 0.005,
    _DepthThreshold + 0.005, depthGrad); // threshold to line signal
```

Listing 5.14: Depth layer, Meta SDK.

This proxy detects distance discontinuities within the avatar mesh. However, it cannot separate the avatar from the background. The background lies entirely outside the per-fragment computation. The normal layer reuses the world-space normal derivative computation from V3 (Section 5.4.4) and an AHEAD-style adaptive gain then amplifies this layer specifically in darker image regions [40]:

```
float luma = dot(color.rgb, float3(0.2126, 0.7152, 0.0722)); // perceptual
    luminance
float adapt = min(1.0 / (3.0 * luma + 0.1), 4.0); // inverse-linear gain, capped
     at 4x
normalLine *= lerp(1.0, adapt, _AdaptiveStrength); // blend gain by strength
```

Listing 5.15: AHEAD adaptive gain on the normal layer, Meta SDK.

The composited PBR output also contains baked lighting gradients that produce false Roberts Cross responses on this platform. The colour layer accordingly replaces Roberts Cross with XDoG: the difference between two Gaussian-blurred luminance samples at different radii, sharpened with a tanh curve:

```
float g1 = GaussianLuma(uv, sigma1); // fine blur
float g2 = GaussianLuma(uv, sigma1 * _XDoGK); // coarse blur
float D = g1 - _XDoGTau * g2; // difference of Gaussians
float colorLine = D < -_ColorThreshold
    ? saturate(-tanh(_XDoGPhi * (D + _ColorThreshold))) : 0.0; // sharpened line
     signal
```

Listing 5.16: XDoG colour layer, Meta SDK.

Subtracting the coarser blur from the finer one cancels out broad, slowly varying illumination gradients. The sharper transitions that mark genuine material boundaries remain unaffected. This directly targets the baked lighting artefact described above, without requiring any change to the underlying composited input.

After the three cues are fused, the Meta SDK applies an additional smoothstep step to sharpen the combined edge value. The Mixamo and Avaturn implementation instead composites the fused value directly, without this extra sharpening step.

Figure 4.14 compares the V5 output across all three platforms.

5.4.8. V6 — Kuwahara Painterly Filter

All three platforms use the anisotropic Kuwahara filter of Kyprianidis et al. [22]. This filter relies on a structure tensor to estimate the local feature orientation and anisotropy and the sampling neighbourhood then becomes an ellipse aligned with that orientation rather than a fixed square.

Mixamo and Avaturn

The structure tensor is computed as a Sobel gradient over luminance in a dedicated post-processing pass on these platforms. A second pass subsequently smooths the tensor with a 5-tap separable Gaussian blur before the main filter pass reads it:

```
float gx = -tl - 2.0*l - bl + tr + 2.0*r + br; // Sobel horizontal
float gy = -tl - 2.0*t - tr + bl + 2.0*b + br; // Sobel vertical
return float4(gx*gx, gx*gy, gy*gy, 1.0); // pack tensor components (E, F, G)

// Second pass: 5-tap separable Gaussian blur over the packed tensor
float4 result = SAMPLE_TEXTURE2D_X_LOD(_BlitTexture, sampler_LinearClamp, uv, 0)
     * 0.375; // centre tap
// ...four more taps at +/-offset and +/-2*offset, weighted 0.25 and 0.0625,
    complete the blur
```

Listing 5.17: Structure tensor pass and Gaussian blur pass (Mixamo and Avaturn).

Meta Avatars SDK

The Meta SDK cannot perform a blur pass inside its single fragment hook. The tensor is therefore estimated directly from `ddx`/`ddy` on the already-shaded luminance and this produces a noisier, unblurred orientation estimate:

```
float lum = dot(color.rgb, float3(0.2126, 0.7152, 0.0722)); // BT.709 luma
    weights
float gx = ddx(lum); // horizontal luminance derivative
float gy = ddy(lum); // vertical luminance derivative
float E = gx * gx; // tensor component E
float F = gx * gy; // tensor component F
float G = gy * gy; // tensor component G
```

Listing 5.18: Structure tensor, Meta SDK (unblurred).

All three platforms then derive the dominant orientation angle and the anisotropy ratio from the same eigenvalue decomposition of this tensor, given by

$$\lambda_{1,2} = \frac{E+G}{2} \pm \sqrt{\left(\frac{E-G}{2}\right)^2 + F^2}, \tag{5.2}$$

$$\phi = \frac{1}{2}\arctan\left(\frac{2F}{E-G}\right), \tag{5.3}$$

$$A = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2}, \tag{5.4}$$

where λ_1 and λ_2 are the tensor eigenvalues, ϕ is the dominant orientation and A is the anisotropy ratio:

```
float disc = sqrt(max(0.0, (E-G)*(E-G) + 4.0*F*F)); // eigenvalue discriminant
float lambda1 = 0.5 * (E + G + disc); // larger eigenvalue
float lambda2 = 0.5 * (E + G - disc); // smaller eigenvalue
float phi = 0.5 * atan2(2.0 * F, E - G); // dominant orientation
float A = (lambda1 - lambda2) / (lambda1 + lambda2); // anisotropy
```

Listing 5.19: Eigenanalysis: orientation and anisotropy (unified across platforms).

The anisotropy A scales the ellipse's major and minor axes, elongating the sampling neighbourhood along the feature direction and narrowing it across. This axis scaling is a clamped linear blend between the isotropic and fully anisotropic extremes on Mixamo and Avaturn. The Meta SDK instead applies the ratio formula from Kyprianidis et al. directly, driven by an additional eccentricity parameter with no equivalent on the other platforms.

Eight overlapping elliptical sectors are then sampled around this ellipse rather than four fixed quadrants, though the sampling strategy itself differs by platform. Mixamo and Avaturn scan every pixel within the kernel radius and accumulate it into whichever sector it falls into, so the sample count grows with the radius. The Meta SDK instead takes a fixed three samples per sector at set radii, plus one sample shared across all sectors, for a constant total regardless of radius. Each sector accumulates a mean colour and a variance. These sectors are subsequently blended by an inverse-variance weight, so that the smoothest sector dominates the result. The exact weighting formula differs by platform, since each implementation follows a different variant of the same principle from Kyprianidis et al. [22]. The Meta SDK raises the sector standard deviation to a configurable power. Mixamo and Avaturn instead raise the variance directly. The accumulation that follows the weight is shared across all three platforms:

```
// weight expression, Meta SDK
w_i = 1.0 / pow(max(tau, sigma_i), q);
// weight expression, Mixamo and Avaturn
w_i = 1.0 / (1.0 + pow(variance_i * 1000.0, 0.5 * _Sharpness));

// accumulation, shared across all three platforms
result += mean_i * w_i; // accumulate weighted colour
totalWeight += w_i; // accumulate normalisation factor
```

Listing 5.20: Inverse-variance sector weight and the shared accumulation step.

The final colour on both platforms is obtained by normalising the accumulated weighted colour by the total weight. Mixamo and Avaturn then composite this result over the original shaded colour through the same avatar mask pass used in V5 (Section 5.4.7), since V6 also runs as a post-processing effect on these platforms. The Meta SDK instead blends by a strength parameter.

Figure 4.15 compares the V6 output across all three platforms.

Table 5.2 collects the divergences described across the preceding subsections. One pattern runs through all of them. Every divergence traces back to a stage that the SDK either resolves internally before the hook executes or does not expose to it at all.

Table 5.2.: Principal points at which the Meta Avatars SDK integration diverges from the Mixamo and Avaturn implementations.

Technique	Aspect	Mixamo / Avaturn	Meta SDK
V1	Outline depth	Depth written, standard test	Depth write disabled, looser test
	Alpha cutout	Explicit clip in both passes	Resolved by the SDK before the hook
V2	Quantisation input	Lambertian dot product	Composited RGB output
	Rim and shadow	Computed within the pass	Pre-composited by the SDK
V2.2	Normal abstraction	Abstract normal map or geometric smoothing	Geometric smoothing only, no tangent frame
	Shadow coordinate	Per-vertex, interpolated	Per-fragment
V3	Normal source	Decoded normal map, one extra sample	SDK composited normal, no extra sample
V4	Gradient input	Raw diffuse albedo	Composited PBR colour
	Threshold gate	Single lower threshold	Band gate plus seam rejection
V4.2	Sharpening passes	Four, individually toggleable	Four, driven by one tightness scalar
V5	Architecture	URP renderer feature, post-processing pass	Per-fragment hook
	Depth layer	Roberts Cross on the scene depth texture	Camera-distance derivative proxy
	Colour layer	Roberts Cross on luminance plus a chromatic term	XDoG
	Avatar isolation	Dedicated mask pass	Not required
V6	Tensor smoothing	Separable Gaussian blur pass	None, unblurred
	Compositing	Masked composite via the mask pass	Strength-parameter blend

5.5. Runtime Tooling

Two runtime utilities were developed to address constraints that cannot be resolved at shader compilation time: the first enables interactive parameter exploration within the VR headset during the visual assessment phase and the second freezes the avatar's pose for evaluation screenshots.

5.5.1. In-VR Parameter Tuning Interface

Shader parameters can also be adjusted through a custom world-space UI panel during live headset sessions, reproducing the same controls available in the Unity Inspector. This allowed values to be explored and compared while the avatar remained visible in the headset at correct VR scale. This panel was central to the visual assessment process used to select the technique for the user study.

5.5.2. Pose Freeze for Evaluation

A dedicated script freezes the avatar's body-tracking pose at a chosen instant, preserving a specific expression and posture for evaluation screenshots. Since the Meta Avatars SDK exposes no public API for pausing tracking, the freeze is implemented through reflection, substituting a snapshot provider that returns a fixed pose state in place of the SDK's live input provider.

5.6. Animation and Recording Pipeline for the User Study

Facial animation for the Avaturn avatar is driven using Apple ARKit: the Live Link Face application streams all 52 ARKit blendshape coefficients and head rotation over UDP from an iPhone to Unity, where a custom component maps each coefficient to the corresponding Avaturn blendshape target introduced in Section 5.2.3. This stream drives the corresponding blendshape weights at runtime, so the avatar's expression follows the iPhone performer directly with no intermediate animation clip.

For the C2 and C3 stimuli, this blendshape stream was captured simultaneously while filming the C1 real person video, matching the synchronized recording session described in Section 4.8.2. The same captured

stream was played back for both the photorealistic and NPR renders, so any visible difference between C2 and C3 in the resulting footage originates from the shader rather than from the underlying animation data.

The camera in the recording scene used a reduced field of view of 20°, compared with the 60° default used in the general platform comparison scenes described in Section 5.2.4. A wide angle at the close camera distance required for a talking-head framing exaggerates the size of facial features nearest the camera, most visibly the nose, through perspective distortion. Narrowing this angle removes the artefact while preserving an equivalent framing of the avatar.

6. Evaluation

This chapter evaluates the NPR avatar system developed in this thesis. The first part documents the visual quality assessment of the eight NPR techniques across the three avatar platforms, establishing which technique was selected for the user study and the reasoning behind that selection. The second part presents the user study that tests whether NPR stylisation affects how participants perceive trust in a speaking avatar and interprets each result in relation to the study hypotheses.

6.1. Visual Quality Assessment

6.1.1. Assessment Methodology

Each NPR technique was assessed qualitatively inside the Unity Editor and on the Meta Quest 3 headset, using the world-space UI panel described in Section 5.5.1. The avatar was placed in the shared studio environment described in Section 5.2.4 and animated with a repeating idle loop. It was observed from several distances, including the close conversational framing of approximately 1 m used to record the user study videos. Parameters were adjusted interactively through this panel until the visual output satisfied four criteria:

1. **Facial readability.** Facial features (eyes, mouth boundary, nose bridge and jaw line) must remain individually legible under stylisation.
2. **Temporal stability.** Edges and stylisation effects must remain stable during animation. Screen-space techniques are particularly susceptible to flickering when the avatar translates across the screen, because the pixel-space neighbourhood shifts while the underlying geometry does not.
3. **Artefact suppression.** Visible UV seam lines, rectangular outlines around hair cards, depth-fighting at the outline hull and false edge detections at skin-tone boundaries must be suppressed to a level that is not noticeable under normal observation.
4. **Expressive abstraction.** The stylisation should be visually legible as an intentional artistic choice, not as a rendering error. A technique that produces only isolated noise marks rather than coherent contours would not read as deliberate stylisation.

Evaluation proceeded iteratively for each technique. Parameters were first tuned on the Mixamo avatar, where unrestricted shader access made per-pass debugging straightforward and then verified on Avaturn and the Meta SDK avatar. Techniques that satisfied all four criteria on at least two of the three platforms were considered viable candidates for the user study.

6.1.2. Per-Technique Assessment

V1 — Inverted Hull Outline. As the shared silhouette pass underlying every other technique (Section 4.4.2), V1 was also assessed on its own as the minimal abstraction reference in the study materials. Temporal stability holds without additional tuning, as the outline geometry is generated in the vertex stage rather than from a screen-space pixel neighbourhood. The alpha cutout and depth bias described there remove rectangular hair-card borders and silhouette z-fighting, satisfying artefact suppression and facial readability holds too, given that the pass never alters the underlying texture or geometry. However, expressive abstraction does not: the method marks only the external silhouette and creases, clothing seams and facial contours consequently remain undelineated, consistent with its interior boundary limitation. V1 was retained as the foundational outline pass and lowest abstraction reference in the study materials, but not as a standalone candidate for the composite.

V2 — Toon Shading. The quantised diffuse term yields a consistent, low-frequency tonal structure that satisfies both facial readability and temporal stability without flicker. Clothing seams and facial features carry no dedicated edge cue beyond the local tonal contrast that the shading bands happen to produce. These accordingly

remain undelineated at the 1 m framing used in the study, matching the interior line limitation already noted in Section 4.4.3. Nevertheless, unlike V1, the coherent tonal bands still read as a deliberate stylistic choice rather than an incomplete or erroneous render and expressive abstraction remains satisfied despite the missing interior detail.

V2.2 — X-Toon Shading. The abstraction level axis described in Section 4.4.4 is driven by camera distance or surface curvature. At the wider framing used for general platform comparison, the two modes differ only marginally and Avaturn exhibits slightly greater variation than the other platforms. However, neither mode registers a meaningful signal at the 1 m conversational framing used in the study. The depth range across the avatar's torso and face spans only a few centimetres, too narrow relative to the configured depth window. Correspondingly, the screen-space normal derivatives produced by curvature mode across the smooth head and body geometry remain close to zero, too faint to register as a meaningful signal. Ultimately, neither dynamic mode yields a strongly different outcome at this fixed conversational distance. This finding is why manual mode was selected as the driving mode for the composite assembled for the user study, as Section 4.5 explains. Because manual mode holds a fixed constant rather than recomputing a value per fragment, it carries none of the flicker risk that affects the screen-space techniques assessed elsewhere in this section. Facial readability and temporal stability are both satisfied without qualification. Expressive abstraction is likewise satisfied. The fixed detail bias visibly compresses the tonal bands towards the ramp midpoint, even though the parameter adjustment study found no significant preference across the tested bias values (Section 4.7).

V3 — Screen-Space Normal Edge Detection. The camera distance stays fixed throughout each study recording. This is why the distance-dependent edge thickening identified in Section 4.4.5 remains mild here rather than growing as it would under a varying viewing distance. Facial readability is nonetheless not satisfied. Smooth skin produces minimal normal discontinuities, so the eyebrow, lip and eyelid boundaries central to this criterion remain undetected. This is the primary reason V3 was not selected for the composite (Section 4.5). Despite this exclusion, V3 requires zero additional texture samples. This makes it the least costly technique after V2.

V4 — Sobel Edge Detection. Baked-illumination artefacts, such as the nose bridge artefact identified in Section 4.4.6, are suppressed on the Avaturn avatar through the Blender texture preprocessing described in Section 4.6 and a raised threshold on the head material. The skin-to-clothing boundary generates the most visually significant lines. Eyebrow and lip boundaries emerge at moderate threshold settings. Since the step function used for thresholding on Mixamo and Avaturn (Section 5.4.5) can toggle a pixel between frames as lighting varies slightly during animation, temporal stability required the same threshold tuning to keep lines at material boundaries from flickering.

V4.2 — Gaussian Prefiltered Sobel. The facial segment recovery shortfall on Avaturn and the resulting shift to texture preprocessing are described in Section 4.4.7. This came at a cost of up to 72 texture samples per fragment. The preprocessing pipeline made the basic Sobel operator sufficient on its own. The added sampling cost of Gaussian prefiltering was thus no longer justified for the composite assembled for the user study.

V5 — Hierarchical Multi-Cue Edge Detection. V5 was a candidate addition to the C3 composite, assessed together with the Kuwahara filter through the parameter adjustment study (Section 4.7). The interior contours do not resolve clear boundaries between adjacent facial segments, as Section 4.4.8 describes. This leaves facial readability unsatisfied. The same limitation extends to artefact suppression. The colour layer's response to illumination gradients around the nose bridge is left unaddressed on Avaturn, unlike on the Meta Avatars SDK. This candidate addition was subsequently excluded from the final composite following that study (Section 6.2.1).

V6 — Kuwahara Painterly Filter. V6 does not satisfy facial readability as a standalone condition at radius values large enough to read as clearly painterly, matching the trade-off between abstraction and legibility identified in Section 4.4.9. This trade-off motivated testing it as an addition to the composite candidate for C3 and the ranking outcome thereafter excluded it from the final configuration (Section 6.2.1).

Table 6.1.: Summary of visual quality assessment results across the eight NPR techniques on the Avaturn platform. Criteria are rated qualitatively: ✓= satisfactory without tuning, (✓) = satisfactory after tuning, × = limitation observed.

Technique	Facial readability	Temporal stability	Artefact suppression	Expressive abstraction
V1 — Outline	✓	✓	(✓) alpha cutout, depth bias	× not in isolation
V2 — Toon	✓	✓	✓	(✓) no interior lines
V2.2 — X-Toon	✓	✓	✓	(✓) manual mode chosen for flexibility
V3 — Normal edge	× facial landmarks undetected	✓	✓	✓
V4 — Sobel	(✓) moderate threshold	(✓)	(✓) preprocessing	✓
V4.2 — Gaussian Sobel	× insufficient for facial segments	(✓)	✓	✓
V5 — Hierarchical	× segment boundaries unclear	✓	× nose bridge artefact	✓
V6 — Kuwahara	× at large r	✓	✓	✓

6.1.3. Technique Comparison Summary

6.1.4. Technique Selection for the User Study

The five candidates were narrowed from the eight techniques assessed. Basic toon shading, screen-space normal edge detection and Gaussian prefiltered Sobel were set aside before the parameter adjustment study began. Toon shading was superseded by the finer abstraction control of X-Toon. Screen-space normal edge detection could not recover the colour- and texture-defined boundaries that facial readability depends on. Gaussian prefiltered Sobel no longer offered a measurable benefit once texture preprocessing resolved the noise problem it was designed to address.

Five candidate components were assembled for the user study condition C3: an inverted hull outline, X-Toon tonal shading, Sobel edge detection and two optional post-processing passes, a Kuwahara painterly filter and a hierarchical depth edge layer. A parameter adjustment study was conducted to determine which of these components to retain and how to configure them. As a result, only the first three were carried forward into the final composite and Section 6.2.1 reports that process in detail. The combination uses the submesh material split established for the Avaturn avatar (Section 5.3), which lets the Blender-preprocessed texture suppress baked-illumination artefacts on the head material alone while facial features stay legible. The combination also reads clearly as an intentional stylisation rather than a rendering artefact, which the manipulation check embedded in the study questionnaire (Part I) depends on. A technique that produces only subtle modifications to the photorealistic baseline could fail the check and reduce usable data.

6.2. User Study Results

6.2.1. Parameter Adjustment Study Results

The study was distributed through personal networks, a university mailing list and social media and it ran for approximately two weeks. A total of 45 participants took part. Each round was analysed separately and a ranking that did not assign every rank exactly once was excluded from that round alone, leaving between 40 and 44 valid ballots per round.

The parameter adjustment study described in Section 4.7 fixed the parameter values used in the C3 composite and produced the empirical basis for excluding two candidate components (Figure 6.1). Only R4 (Kuwahara painterly fill) and R5 (hierarchical depth edge layer), the two rounds that tested an additional abstraction layer, produced a top choice significantly ahead of its closest competitor. The omnibus tests in R1 and R2 also reached significance, since the weakest settings in each round were clearly disfavoured. Neither of those rounds

separated its leading configuration from the runner-up and R3 returned no significant effect at any level. This pattern suggests that perceived trustworthiness in this ranking paradigm was more sensitive to whether an additional abstraction layer was present at all than to fine calibration within a layer already established.

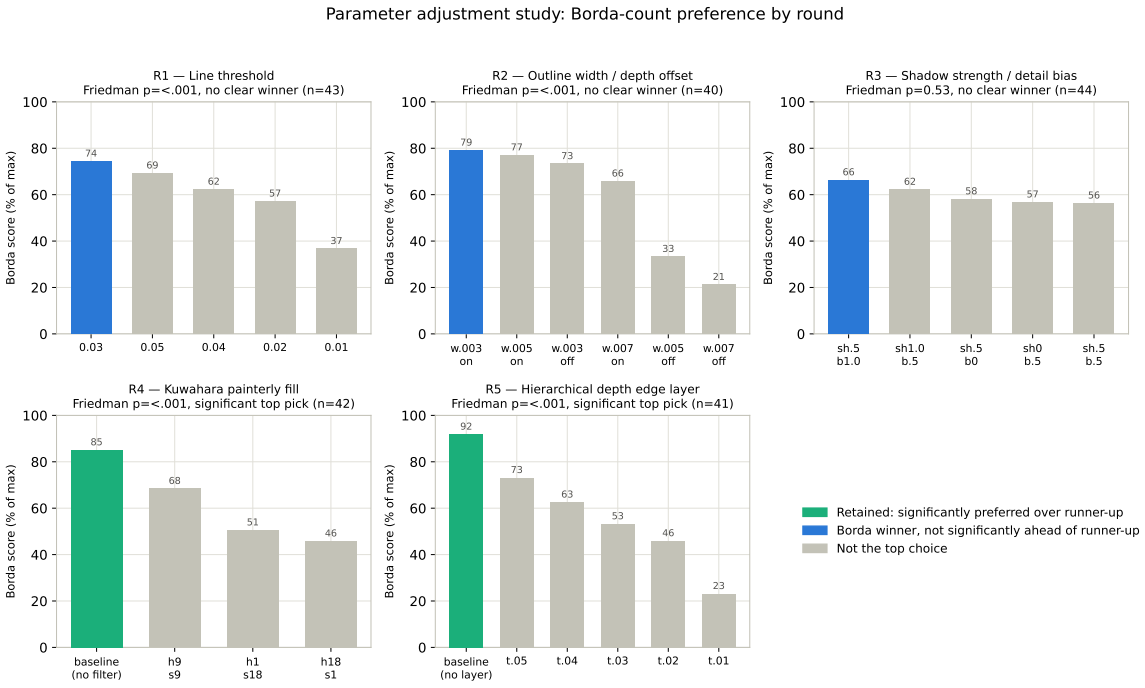


Figure 6.1.: Borda-count preference score by round, as a percentage of the maximum possible score. Green marks the baseline configuration retained after a significant preference in R4 and R5. Blue marks the Borda winner in R1–R3, where the top choice was not significantly ahead of the runner-up.

6.2.2. Participants

Participants were recruited through the same channels as the parameter adjustment study and over a comparable two-week window. Inclusion in the study required an age of 18 or older and sufficient language proficiency to follow the vignette scenarios. A total of 52 participants took part and 11 were removed under the exclusion rule of Section 4.8.4. Thus, every test reported below draws on the remaining 41 participants, each with valid data in all three conditions. An implementation error altered the recap thumbnails in one study version. Those responses were relabelled from the available evidence rather than excluded and a residual risk that the correction misses a participant's intended response remains.

Table 6.2.: Participant demographics and study version distribution.

Parameter	Value
Total enrolled	52
Excluded (failed a gating check)	11
Final analytic sample	41
Age: mean (SD)	27.6 (5.4) years
Gender	23 women, 17 men, 1 non-binary
Avatar/CGI interaction frequency: mean (SD)	2.88 (1.35), 1–5 scale
General trust in new people: mean (SD)	3.54 (1.61), 1–7 scale
Study version (A–C)	15 / 12 / 14

The closing suspicion check asked participants what they thought the study investigated and 24 of the 41 named avatar realism, stylisation or trust. Participants who inferred the manipulation could rate conditions to match what they assumed was expected, not their genuine reaction. Accordingly, the H1 and H2 tests were repeated on the remaining 17 and the pattern of results did not change. This suggests participants answered according to their own honest impressions.

6.2.3. Measure Reliability

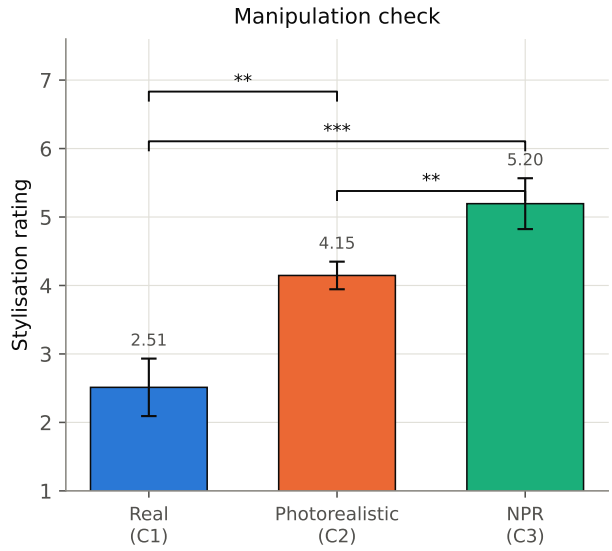
Each multi-item subscale was averaged into a single composite score for the tests reported below. Internal consistency was checked first, as Section 4.8.4 specifies, to establish that the items within a subscale measure one construct. Cronbach's α exceeded the conventional .70 threshold on every subscale, both overall and within each condition separately (Table 6.3). Averaging the items is therefore justified for all five composites and no subscale required item removal.

Table 6.3.: Cronbach's α per subscale, computed across all blocks and separately per condition. k = number of items, $n = 41$ participants contributing 123 blocks.

Subscale	k	Overall	C1	C2	C3
Likeability	5	.94	.90	.94	.93
Eeriness	8	.87	.87	.80	.86
Competence	3	.96	.96	.96	.94
Benevolence	3	.95	.95	.95	.94
Integrity	3	.92	.81	.93	.94

6.2.4. Manipulation Check

Part I asked participants to rate how visually stylised each advisor appeared, from completely realistic (1) to very stylised (7). A successful manipulation requires C3 to score significantly higher than C2, with C1 lowest.



Comparison	Result
Omnibus (condition)	$F(2, 80) = 15.45$ $p < .001$, $\eta_p^2 = .28$
C1 vs C2	$p = .001$, $g = 0.87$
C1 vs C3	$p < .001$, $g = 1.23$
C2 vs C3	$p = .004$, $g = 0.55$

Figure 6.2.: Left: mean visual stylisation rating by condition, Part I of the post-study questionnaire ($n = 41$), with within-subject corrected SEM error bars. Right: Greenhouse-Geisser corrected repeated-measures ANOVA (sphericity violated, $W = .53$, $p < .001$) and Holm-corrected pairwise comparisons, reported as Hedges' g.

Mean stylisation ratings increased monotonically from Real through Photorealistic to NPR (Figure 6.2). A repeated-measures ANOVA confirmed a significant effect of condition and Holm-corrected pairwise comparisons separated all three conditions from each other. The manipulation therefore succeeded. This means any null result on eeriness or trust below cannot be attributed to an unnoticed rendering difference.

6.2.5. Hypothesis Tests

All tests below follow the analysis plan pre-specified in Section 4.8.4.

H1 — NPR Reduces Eeriness

H1 predicted that the NPR composite in C3 would lower scores on the eeriness subscale (D_score) relative to the photorealistic avatar in C2.

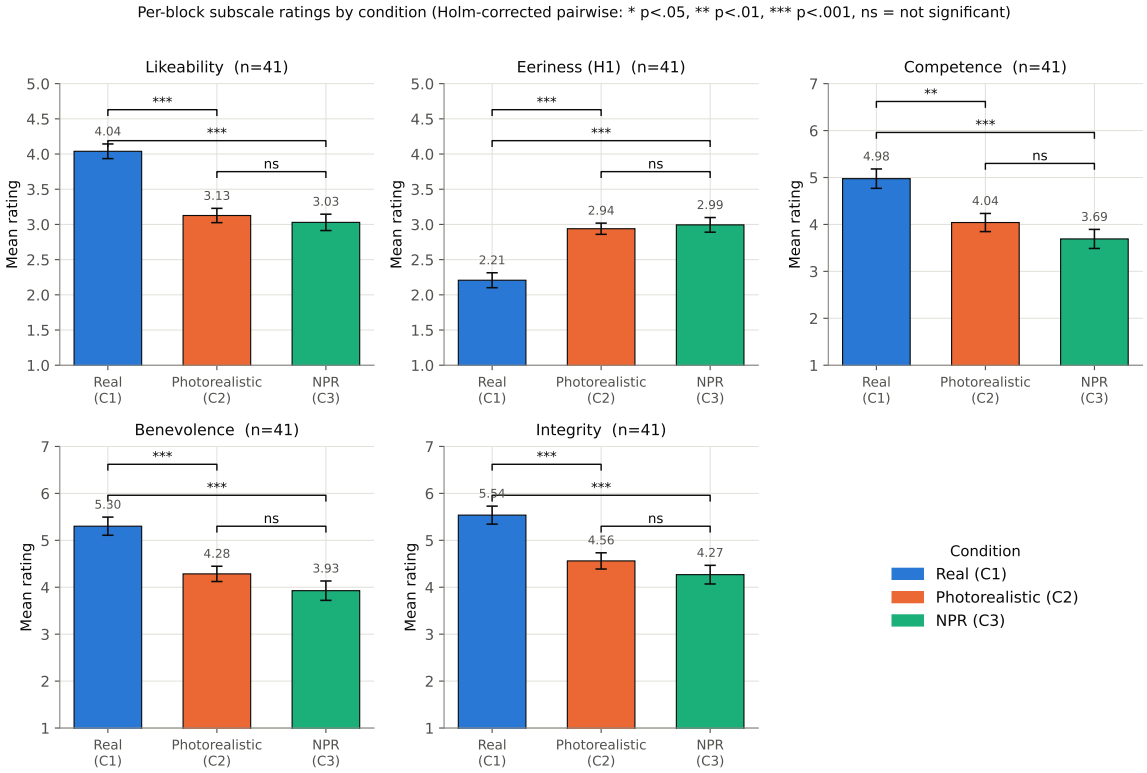


Figure 6.3.: Mean per-block rating by condition for the five subscales retained in the final questionnaire. Error bars show the within-subject corrected standard error of the mean. Brackets mark Holm-corrected pairwise comparisons: *** $p < .001$, ** $p < .01$, ns = not significant. $n = 41$.

Table 6.4.: Repeated-measures ANOVA and Holm-corrected pairwise comparisons (Hedges' g) per subscale, $n = 41$.

Subscale	Omnibus	C1 vs C2	C1 vs C3	C2 vs C3
Likeability	$F(2, 80) = 26.85$, $p < .001$, $\eta_p^2 = .40$	$p < .001$, $g = 0.92$	$p < .001$, $g = 1.02$	$p = .54$, $g = 0.08$
Eeriness	$F(2, 80) = 20.39$, $p < .001$, $\eta_p^2 = .34$	$p < .001$, $g = 0.98$	$p < .001$, $g = 0.98$	$p = .66$, $g = 0.07$
Competence	$F(2, 80) = 10.91$, $p < .001$, $\eta_p^2 = .21$	$p = .004$, $g = 0.53$	$p < .001$, $g = 0.75$	$p = .21$, $g = 0.20$
Benevolence	$F(2, 80) = 14.31$, $p < .001$, $\eta_p^2 = .26$	$p < .001$, $g = 0.59$	$p < .001$, $g = 0.79$	$p = .18$, $g = 0.19$
Integrity	$F(2, 80) = 12.47$, $p < .001$, $\eta_p^2 = .24$	$p = .001$, $g = 0.66$	$p < .001$, $g = 0.80$	$p = .27$, $g = 0.16$

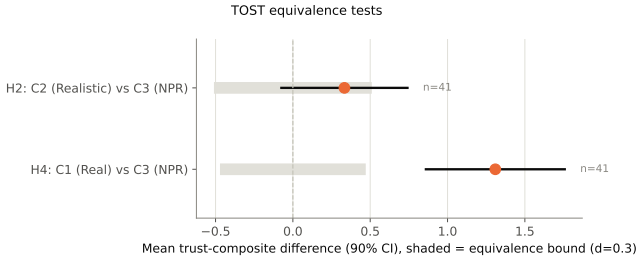
Results indicated a pattern opposite to what H1 predicted. Eeriness rose from C1 through C2 to C3 (Figure 6.3, Table 6.4). This trend was significant in the omnibus test. The follow-up pairwise comparisons showed that C1 differed from both avatar conditions. By contrast, C2 and C3 were statistically indistinguishable from one another.

According to these test results, H1 is not supported. This is not for lack of a perceptible difference between C2 and C3, since the manipulation check already established that participants distinguished the two conditions on perceived stylisation (Section 6.2.4). Stylisation and eeriness instead appear to be separate effects, a dissociation that Section 6.2.5 explains further.

H2 — NPR Does Not Reduce Trust

H2 predicted that the NPR composite in C3 would not significantly reduce scores on the Competence (E_score), Benevolence (F_score) or Integrity (G_score) subscales relative to the photorealistic baseline in C2.

The trust subscales followed a pattern consistent with H2. C1 scored highest on Competence, Benevolence and Integrity alike, while C2 and C3 did not differ significantly from each other on any of the three (Figure 6.3, Table 6.4). None of the three subscales pointed to a specific trust dimension being affected. A non-significant pairwise result alone remains insufficient evidence for equivalence and the TOST procedure was accordingly applied to the combined trust composite.



	Diff.	Bound	TOST p
H2: C2 vs C3	0.33	±0.51	.240
H4: C1 vs C3	1.31	±0.47	.998

Table 6.5.: TOST results, trust composite (n = 41). Bound = Cohen's d = 0.3 in raw scale points (Chapter 4).

Figure 6.4.: TOST equivalence tests on the combined trust composite (the mean of Competence, Benevolence and Integrity). Points show the mean difference between conditions, black lines the 90% confidence interval and the shaded band the pre-specified equivalence bound. Equivalence requires the full confidence interval to fall inside the shaded band.

H2 is therefore neither confirmed nor disconfirmed by the pre-specified criteria. The pairwise comparisons showed no detectable difference, but the confidence interval on the trust composite still spans both a small negative and a moderate positive difference (Figure 6.4). Consequently, the sample is not large enough to rule out a difference as large as the pre-specified bound. This outcome was anticipated in the power analysis of Section 4.8.4. A paired TOST at the $d = 0.3$ bound requires roughly $n = 120$ to reach a power of 0.90 and the realised sample of 41 leaves it near 0.20 even when the true difference is exactly zero. Separating a genuine null effect from an insensitive test would thus require close to three times the participants recruited here.

H3 — Eeriness Predicts Trust

H3 predicted a negative within-participant relationship between eeriness and trust across all three conditions. This was tested via a person-centred multilevel regression fitted for each trust subscale composite (E_score, F_score, G_score) in turn, with D_score as the within-participant predictor and a random intercept per participant.

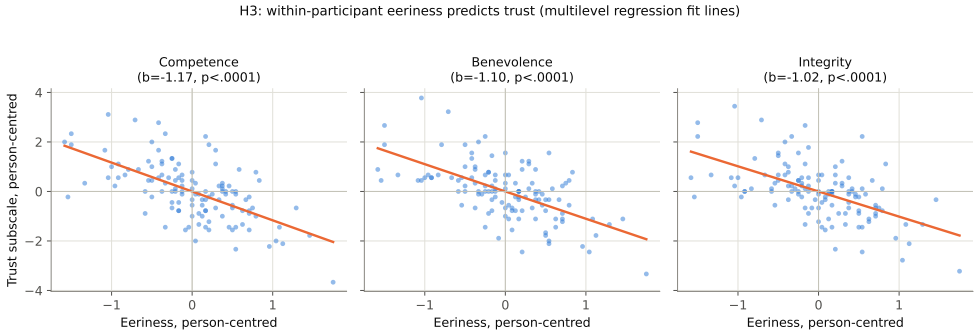


Figure 6.5.: Person-centred relationship between eeriness and each trust subscale. Both axes are expressed as each participant's deviation from their own mean across the three conditions, isolating the within-participant association tested by H3 from between-participant differences in overall trust or eeriness. Lines show the fixed-effect slope from the corresponding multilevel model (n = 123 valid blocks from 41 participants).

The person-centred multilevel model confirmed a negative within-participant association between eeriness and every trust subscale (Figure 6.5, Table 6.6). A one-point increase in a participant's own eeriness rating, relative to their own mean across the three conditions, predicted roughly a one-point drop on each trust subscale. The intraclass correlations (.48 to .57) indicated that roughly half of the total variance in each trust subscale sat between participants rather than within them. This is consistent with substantial individual differences in how trusting participants are overall. Hence, H3 is confirmed.

Table 6.6.: Fixed-effect slopes from the person-centred multilevel model shown in Figure 6.5. All slopes $p < .001$.

Subscale	b	95% CI
Competence	−1.17	[−1.49, −0.84]
Benevolence	−1.10	[−1.42, −0.78]
Integrity	−1.02	[−1.34, −0.69]

This result matches the systematic review of Alipour et al., who found a direct negative relationship between eeriness and trust in the majority of studies that examined the association [29]. The near-identical slopes across competence, benevolence and integrity indicate that eeriness suppresses trust broadly rather than through a benevolence-specific pathway.

The eeriness-trust relationship established above also explains why H1 and H2 both returned null results. H1 predicted lower eeriness for C3 than C2, which would have raised trust through the same pathway. However, C2 and C3 did not differ in eeriness. Trust consequently had no reduction to act on. The H1 and H2 nulls therefore stem from the same missing ingredient: a genuine reduction in eeriness. The NPR composite may not have moved the avatar's position on the uncanny valley curve relative to the photorealistic baseline far enough to produce a trust benefit.

H4 — NPR Matches Real-Human Trust

H4 predicted that the NPR composite in C3 would not significantly reduce trust scores relative to the real human speaker in C1, extending the same null logic as H2 to a different reference point. Nonetheless, residual differences in human likeness, animation naturalness and speaker identity remain between these two conditions.

H4 draws on the same subscale comparisons reported for H2, now restricted to the C1 versus C3 contrast. Competence, benevolence and integrity were all significantly higher for C1 (Table 6.4). The TOST test on the C1 versus C3 trust composite fell far outside the equivalence bound, in the direction of C1 (Table 6.5). This means H4 is not supported.

In summary, the NPR composite did not close the trust gap with the real speaker and the shortfall is substantial rather than marginal. Two explanations remain open. The uncanny valley account predicts this gap and recognition of the speaker by some participants would produce it equally well. The current design cannot separate them.

6.2.6. Order Effects

The counterbalancing scheme assumes that a video's position in the sequence carries no effect of its own (Section 4.8.4). Position produced no significant main effect on any subscale, although likeability and eeriness both approached the threshold ($F(2, 80) = 3.04$, $p = .054$ and $F(2, 80) = 2.47$, $p = .091$). The three trust subscales fell well short of it, with p between .13 and .27.

The three study version groups did differ on likeability ($F(2, 38) = 7.07$, $p = .002$) and eeriness ($F(2, 38) = 5.57$, $p = .008$), with no such difference on any trust subscale ($p \geq .14$). However, this second comparison is between subjects and the groups comprise 15, 12 and 14 different people. A difference of that kind is equally consistent with a sequence effect and with the groups differing in how eerie they find avatars generally. The present design cannot separate the two.

Condition and position are perfectly aliased within each version, since a participant assigned to one version always meets the same condition in the same slot. Carryover between a specific pair of conditions is therefore not estimable from a single Latin square. The condition contrasts themselves remain unaffected in either case. Every participant supplies all three conditions and each condition occupies every slot equally often across the three versions.

6.2.7. Post-Study Rankings

The three post-study ranking questions, covering advisor preference (P1), trustworthiness and unsettlingness, were analysed as specified in Section 4.8.4. Mean ranks per condition (1 = most preferred, 3 = least preferred) were compared with a Friedman test per question, followed by pairwise Wilcoxon signed-rank tests with Holm correction. C2 versus C3 is the critical comparison in each case.

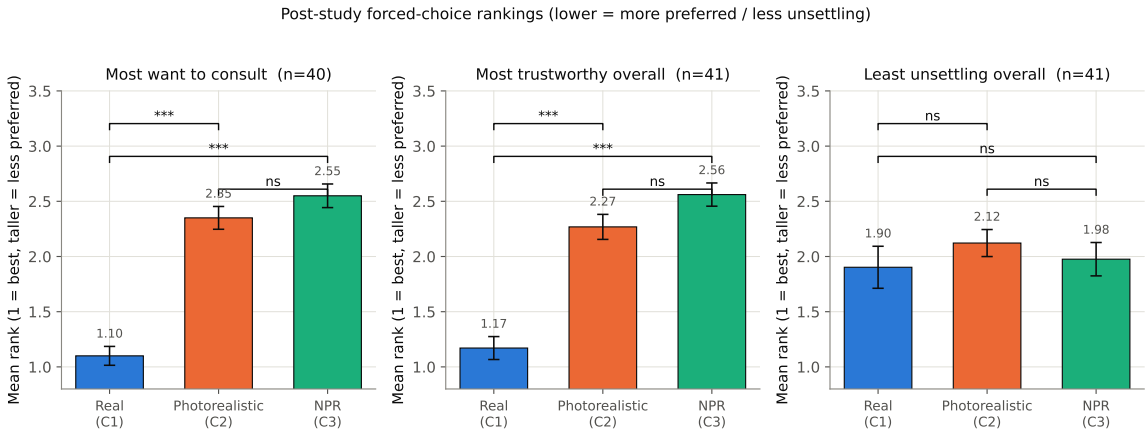


Figure 6.6.: Mean forced-choice rank by condition for the three post-study comparison questions. Error bars show the within-subject corrected standard error of the mean. Brackets mark Holm-corrected pairwise Wilcoxon comparisons: *** $p < .001$, ns = not significant.

Table 6.7.: Friedman tests and Holm-corrected pairwise Wilcoxon comparisons, post-study rankings.

Question	Friedman	C1 vs C2	C1 vs C3	C2 vs C3
Most want to consult (n = 40)	$\chi^2_F(2) = 49.40$, $p < .001$	$p < .001$	$p < .001$	$p = .21$
Most trustworthy (n = 41)	$\chi^2_F(2) = 44.05$, $p < .001$	$p < .001$	$p < .001$	$p = .07$
Least unsettling (n = 41)	$\chi^2_F(2) = 1.02$, $p = .60$	$p = .97$	$p = .97$	$p = .97$

Mean ranks followed the same ordering as the Likert measures across all three questions (Figure 6.6, Table 6.7). Participants ranked C1 first on both the primary behavioural measure and the trustworthiness ranking, well ahead of C2 and C3. Those two did not differ significantly from each other on either question. The unsettlingness ranking showed no discernible pattern on any comparison.

Canales et al. found a divergence between explicit and behavioural measures in their comparable study [5]. All three rankings here converged with the Likert measures instead. C3 attracted no more implicit endorsement than C2 did, meaning that the forced ordering surfaced nothing that the graded ratings had obscured. C1's dominance is the most consistent finding across every measure in the study.

6.2.8. Secondary Measures

Part H's item on the advisor's recommendation was analysed as a secondary exploratory measure, separately from the preference shift reported in Section 6.2.9. Recommendation agreement did not differ significantly across conditions (Table 6.8). Participants evaluated the substance of each recommendation largely independently of how the advisor was rendered. The scenario design held argumentative content constant across conditions, so this outcome was expected.

The dispositional trust covariate was examined as an individual difference moderator. It showed no meaningful association with the mean trust composite (Table 6.8). A quartile-based moderation analysis was not pursued, since the sample is too small to detect an interaction reliably. Canales et al. similarly found no significant relationship between propensity and trust ratings in their comparable manipulation [5].

Table 6.8.: Secondary measures, $n = 41$.

Recommendation agreement, M (SD)	**C1**	**C2**	**C3**
	4.07 (2.20)	4.68 (1.93)	3.85 (1.97)
Omnibus: $F(2, 80) = 2.10$, $p = .13$			
Trust propensity × trust composite: $r(39) = -.10$, $p = .52$			

6.2.9. Pre-Video Baseline Shift

Preference shift was computed per participant per block as the change from the pre-video option choice to the post-video choice recorded in Part H (Section 4.8.4). The comparison isolated the persuasive effect of the advisor's recommendation from the participant's prior inclination.

Preference shift toward the advisor's recommendation was infrequent across all three conditions. Four of 41 participants (9.8%) shifted toward the recommendation after C1, five of 41 (12.2%) after C2 and one of 41 (2.4%) after C3. These counts are too small to support a formal test of condition differences and the numerically higher shift rate for C2 should be read as descriptive rather than as evidence of a persuasive advantage for the photorealistic avatar. The low rate across all conditions suggests that the scenarios kept both alternatives genuinely plausible, as the design in Section 4.8.3 intended. Most participants entered each video already inclined toward one option and were not easily moved by a short recommendation regardless of how the advisor was rendered.

6.3. Discussion

Two patterns in the study outcome call for an explanation. The real speaker separated sharply from both avatar conditions on every subscale and on both rankings that produced a significant result. Even though the manipulation check confirmed that participants registered them as differently stylised, the two conditions stayed statistically indistinguishable on every outcome measure. A third result constrains any explanation of these two. Eeriness predicted trust within participants, so the route from appearance to trust was open. The composite changed what participants saw without travelling it. The explanations below move outward from the stimulus itself to the viewers watching it.

6.3.1. What the Stylisation Left Unchanged

The composite stylises the avatar's material while leaving its shape untouched and this split may be why eeriness did not fall. Material governs eeriness in Zell's account and this alone predicts a reduction for C3 [8]. Their mismatch result points the other way. C3 sets a stylised material on a photogrammetrically reconstructed shape and the abstraction may therefore have removed eeriness from the material only to reintroduce it through the gap between the two.

The features that Dill's participants blamed for their discomfort, the eyes and the mouth, were also held constant [25]. Notably, neither region differs between C2 and C3 beyond shading. Additionally, both conditions inherit the same gaze and lip animation from one recording session. A shading operator cannot reach that animation and any mismatch in its timing thus survives the stylisation.

Consistency across the whole presentation offers a third reading. Cihodaru-Ştefanache and Podina rank agreement between visual and vocal realism among the most reliable defences against an uncanny response [35]. C3 breaks exactly that because a recorded human voice accompanies a visibly abstracted face.

6.3.2. The Comparison Set

Every participant saw all three conditions, so the real speaker was available as a reference point for the other two. C1 ranked first on advisor preference and on trustworthiness and it was rated most likeable, least eerie and

highest on all three trust dimensions. Once a genuine human occupies the top of the scale, the judgement left to the participant reduces to how far each avatar falls below it.

That comparison compresses the range left for the two avatar conditions. Both were recognisably CG and both fell short of C1 on the attribute participants weighted most heavily. Therefore, a difference of rendering style competes against a difference of category that is far more salient. In particular, the measured difference between C2 and C3 on likeability and eeriness was negligible and this fits that reading.

The sequence sharpens this further. C3 never followed C2 in any version and whenever C3 was not the first video it followed C1 directly. C2 was never preceded by C1. The critical comparison accordingly places C3 immediately after the real speaker in two versions out of three, while C2 never occupies that position. Any contrast effect from the real human falls on C3 alone. The present design cannot establish whether it did.

Furthermore, withdrawing the real speaker would change the comparison rather than merely narrow it. Abstraction becomes the sole axis on which the stimuli can be ordered with only two avatars in view. Since every participant saw C1, the present data cannot determine whether the anchor caused this compression. Nonetheless, a study presenting C2 and C3 without C1 would test it directly.

6.3.3. Familiarity with the Real Speaker

The speaker in C1 is the researcher who conducted the study. Hence, some participants may have recognised them. Such acquaintance supplies an existing basis for trust that no rendering can match. C1 would then sit above the level an unfamiliar human speaker would reach. The gap both avatar conditions are measured against widens accordingly. The questionnaire included no recognition item and the share of participants affected cannot be established.

6.3.4. Viewer Expectations of Generative Video

A participant who had completed the study asked afterwards which AI system had been used to generate the videos. The question indicates that at least one viewer filed the material under generative AI output rather than avatar rendering and it was put about the videos as a set rather than about one condition.

Generative video is routinely encountered outside any research setting and viewers arrive with expectations formed by that exposure. A viewer who takes an avatar to be an AI-generated artefact may apply a general scepticism towards media produced by machines. This would depress trust for both avatar conditions equally rather than separating them.

Nothing in the study materials ruled that assumption out. Neither the consent text nor the block instructions mentioned how the recordings had been produced and each participant was left to decide for themselves. Stating at the outset that the videos were not AI-generated would have settled the question.

6.3.5. The Video-Based Setting

Participants watched prerecorded video and never interacted with the speaker. The setting supplied no stereoscopic depth, no motion parallax and no sense of sharing space with the avatar. A comparable VR study held shape and animation constant and varied only render style. Photorealism significantly raised both place illusion and social presence [46]. A rendering difference that registers weakly on a flat screen may consequently register more strongly once the avatar occupies the viewer's own space.

The absence of interaction works in the same direction. An advisor who cannot respond, hold eye contact or adapt to the participant offers fewer occasions for a mismatch between appearance and behaviour to surface. These results therefore describe a speaking head in a browser window. An interactive setting such as VR may separate C2 and C3 where the present design did not.

6.4. Limitations

Both the parameter adjustment study and the main study were designed to take approximately 10 to 15 minutes, as longer sessions were expected to make participant recruitment more difficult. Achieving this target was challenging because each measure in the main study was completed three times, once for each video. Consequently, the available session time constrained both the number of measures that could be included and the breadth of the experimental design. The main study used a single NPR composite, one avatar identity, one platform and a fixed set of three scenarios. The findings should therefore be interpreted as characterising this particular configuration rather than NPR stylisation as a category. Other rendering techniques, levels of abstraction, avatars or decision contexts may produce different effects on trust.

The same time constraint also limited the scope of the parameter adjustment study. Only the parameter groups expected to have the greatest visual impact were systematically tested, while the remainder were fixed through direct inspection during development. As a result, the final NPR composite should not be treated as globally optimal across the full parameter space. It is better understood as a configuration judged suitable within the limits of the study.

Furthermore, the fixed thesis submission deadline imposed a second constraint on the study. It determined the data collection period, which coincided with the university break. One of the three recruitment channels was a student mailing list, but many were away during this period. Thus, recruitment had to close before a larger sample could be obtained. Although the resulting sample met the minimum size indicated by the power analysis, it was insufficient for the equivalence tests, which need more participants to provide informative evidence. Their results remained inconclusive.

The limited sample size also affected the counterbalancing strategy. A full design with three experimental conditions would require six possible presentation orders. Achieving adequate group sizes across all six would have demanded approximately twice the available sample. Accordingly, only three orders were used.

7. Conclusion

7.1. Summary of Findings

Rendering a virtual human at high fidelity can lower rather than raise the confidence of users who interact with it. NPR offers a route around that penalty, but the platforms that supply avatars to consumer applications expose narrow and unequal control over their own rendering. This thesis developed NPR stylisation for Mixamo, the Meta Avatars SDK and Avaturn. It then measured what the resulting abstraction does to perceived trust. Eight techniques were built in sequence, ranging from a geometry-based outline through several edge detection operators to a painterly filter. Three of them make up the composite carried into the user study: an inverted hull outline, X-Toon tonal shading and Sobel edge detection over a preprocessed head texture.

Every technique was integrated on all three platforms in order to answer RQ1. Building the same operators three times exposed systematic differences in the signal each pipeline supplied. Specifically, the most restrictive of the three offers a single fragment hook and no additional passes. An operator assuming a separate buffer or pass hence had to be rebuilt around a substitute. The hierarchical depth layer reads a camera-distance derivative instead of a depth texture and the Kuwahara structure tensor loses its Gaussian smoothing altogether. Identical parameter values likewise produced visibly different output from one host to the next. That constraint ruled nothing out entirely. The real division separates operators that only need retuned parameters from operators rebuilt around a substituted signal.

RQ2 asked which combination best balances abstraction against facial legibility and the answer came from calibration rather than visual inspection. Each version addressed a limitation identified in its predecessor and five of the eight became candidates for the composite on that basis. A parameter adjustment study then asked 45 participants to rank rendered variants by perceived trustworthiness across five parameter groups. Friedman and Wilcoxon tests fixed the final parameter values and removed both the Kuwahara filter and the hierarchical edge layer from the composite. However, no technique in the set reached that balance on the Avaturn avatar by rendering alone. Its photogrammetric albedo carries baked illumination and gradient operators read that illumination as structure. Facial features became legible only after the head texture was preprocessed. The answer to RQ2 is thus not a technique on its own. It is the three-component composite paired with texture preprocessing.

RQ3 was addressed through a within-subjects study with 41 participants. The same speaker appeared as a real person (C1), as a photorealistic Avaturn avatar (C2) and as that avatar under the selected composite (C3). The questionnaire paired validated likeability and eeriness instruments with trust subscales for competence, benevolence and integrity. A forced ranking measure followed the three video blocks. Three decision scenarios were written so that each targeted a different trust dimension. The recommended option and the order of discussion alternated across those scenarios and this kept their content from confounding the comparison of rendering styles.

The outcomes qualify RQ3 rather than answering it affirmatively. H1 expected the stylised avatar to be rated less eerie than the photorealistic one. The two did not differ and the hypothesis is not supported. Participants registered the two conditions as differently stylised, meaning that an unnoticed rendering difference cannot explain the null result. H2 claimed that stylisation costs no trust against that same baseline. Neither pre-specified criterion settled it. The pairwise comparison detected no trust difference between the two avatar conditions. Nevertheless, the equivalence test could not exclude a difference as large as the bound it was set against. H3 held that eeriness suppresses trust and it is confirmed. Eeriness predicted every trust subscale negatively within participants, so the route from appearance to trust stayed open while the composite failed to travel it. H4 extended the no-cost claim of H2 to the real speaker. Trust in the stylised avatar remained well below trust in the real human. H4 is accordingly not supported.

Although the composite did not reduce eeriness relative to the photorealistic baseline, it produced no observed trust penalty either. The available evidence therefore does not indicate that stylisation reduced trust, but it

is not sufficient to establish that the two avatar conditions are equivalent. The practical implication is that a developer choosing between the two styles need not assume an inherent disadvantage for the stylised version. The question of equivalence itself stays open. What remains is the distance to the real speaker and neither avatar condition closed any part of it. Reaching the credibility of an actual human is consequently a harder problem than selecting a rendering style. It is also the problem that decides whether virtual advisors become genuinely trusted participants rather than a passing novelty.

7.2. Future Work

The clearest next step is a larger version of the study reported here. H2 was left unresolved and the cause was sample size, not a genuine difference between the two avatar conditions. In particular, the equivalence test was underpowered at 41 participants. Recruitment had to close before more could be gathered. The retained sample met the requirement for detecting a difference, not the far larger one that equivalence demands. Repeating the C2 versus C3 comparison with roughly 120 participants settles it. A result inside the equivalence bound confirms that stylisation costs no trust. A result outside it exposes a cost too small for this sample to detect.

The competing explanations for the null results point to three adjustments in the study design. The first extends abstraction to the geometry, so that proportion and feature scale shift alongside the shading. This variant tests whether a stylised material on a photogrammetric shape is what held eeriness constant. The second drops the real speaker from the comparison set. Removing that anchor releases the range it compressed and reveals whether the two avatars separate when no human reference is in view. The third states at the outset that the videos were not generated by AI. Nothing in the present materials ruled that assumption out and it may have lowered the trust ratings for both avatar conditions.

A further study should vary abstraction strength continuously instead of testing one setting against a photorealistic baseline. The composite fixes abstraction at the single value the parameter adjustment study selected. Sweeping that value across its range would replace the point measured here with a curve. That curve would trace how eeriness and trust move together as abstraction rises. It would reveal whether an optimum sits between full realism and full abstraction.

An interaction with a virtual human normally involves reciprocal behaviour and shared spatial presence. Restoring both in a follow-up design would show whether they change the effect of stylisation on trust. Participants here watched prerecorded videos. The advisor thus never responded to their questions or returned their gaze. Furthermore, they viewed the avatar on a flat screen rather than encountering it within the same physical environment. Participants in such a setting may pay closer attention to subtle variations in the avatar's appearance. A rendering difference that was too weak to influence trust in this study could consequently produce a measurable effect. This would expose a gap between C2 and C3 that the current design was unable to detect.

Moreover, the comparison needs repeating across multiple avatar identities and platforms. The study used only one of each. The avatar was reconstructed from photographs of a single individual and it carries that person's specific features. However, the effect of stylisation may depend on the identity and appearance of the character being rendered. For example, the same degree of abstraction may be perceived differently when applied to a character who appears male, elderly or non-human. Only a wider sample of identities can separate two readings of the null result reported here. Either stylisation leaves trust unchanged in general or that outcome is specific to this avatar rendered with this particular composite.

8. Literature

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A. Appendix

Declaration of Authorship

I hereby declare, on my honour, that this Master's thesis was written by me independently and that no sources or aids other than those cited have been used. All passages drawn verbatim or in paraphrase from published or unpublished sources are identified as such. This work has not been submitted to any other examination authority, in whole or in part, in the same or in a substantially similar form, nor has it been published.

The thesis was prepared under the supervision of Prof. Dr. Oliver Deussen (first reviewer) and Prof. Dr. Tiare Feuchtner (second reviewer) at the University of Konstanz, Department of Computer and Information Science, within the Master of Science programme in Computer Science.

I further declare that in preparing this thesis I used generative artificial intelligence tools to assist with written expression. Specifically, Anthropic Claude was used to refine sentence formulation, improve clarity and check adherence to academic style conventions. All intellectual content, including the research design, technical development, data collection and analysis, interpretation of results and conclusions, is exclusively my own work. No AI tool was used to generate research content, arguments or findings. All AI-assisted passages were reviewed and revised by me. I bear full academic responsibility for the entire content of this thesis.

Konstanz, August 2026

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Nidanur Günay

Parameter Adjustment Study Questionnaire

The following pages reproduce the parameter adjustment study questionnaire administered before the main study. The resulting ranking data and aggregated scores are reported at the end of this appendix.

Avatar NPR Study: User Preference Evaluation

Thank you for participating in this avatar appearance study. You will see images of stylized 3D avatars across several rounds and rank them based on your preferences.

For the best experience, please complete this study on a desktop or laptop computer rather than a mobile phone. Images are easier to compare on a larger screen, and the ranking questions are more comfortable to use with a mouse or trackpad.

Each round takes approximately 1–2 minutes. The full study takes around 10 minutes. There are no right or wrong answers. Please rank based on your personal impression.

* Indicates required question

Round 1

View the 5 avatar images below, then rank them.

A

B

C

D

E

Overview — All Avatars in This Round

1. Imagine these avatars are virtual advisors presenting you with advice on an important personal decision. Rank them from most to least trustworthy. *
Assign each number to only one avatar. Each rank can be used once.

Mark only one oval per row.

	A	B	C	D	E
1st (Best)	◯	◯	◯	◯	◯
2nd	◯	◯	◯	◯	◯
3rd	◯	◯	◯	◯	◯
4th	◯	◯	◯	◯	◯
5th (Worst)	◯	◯	◯	◯	◯

2. *What influenced your ranking most in this round? (optional)*

Round 2

View the 6 avatar images below, then rank them.

A

B

C

D

E

F

Overview — All Avatars in This Round

3. Imagine these avatars are virtual advisors presenting you with advice on an important personal decision. Rank them from most to least trustworthy. *
Assign each number to only one avatar. Each rank can be used once.

Mark only one oval per row.

	A	B	C	D	E	F
1st (Best)	◯	◯	◯	◯	◯	◯
2nd	◯	◯	◯	◯	◯	◯
3d	◯	◯	◯	◯	◯	◯
4th	◯	◯	◯	◯	◯	◯
5th	◯	◯	◯	◯	◯	◯
6th (Worst)	◯	◯	◯	◯	◯	◯

4. *What influenced your ranking most in this round? (optional)*

Round 3

View the 5 avatar images below, then rank them.

A

B

C

D

E

Overview — All Avatars in This Round

5. Imagine these avatars are virtual advisors presenting you with advice on an important personal decision. Rank them from most to least trustworthy. *
Assign each number to only one avatar. Each rank can be used once.

Mark only one oval per row.

	A	B	C	D	E
1st (Best)	◯	◯	◯	◯	◯
2nd	◯	◯	◯	◯	◯
3rd	◯	◯	◯	◯	◯
4th	◯	◯	◯	◯	◯
5th (Worst)	◯	◯	◯	◯	◯

6. *What influenced your ranking most in this round? (optional)*

Round 4

View the 4 avatar images below, then rank them.

A

B

C

D

Overview — All Avatars in This Round

7. Imagine these avatars are virtual advisors presenting you with advice on an important personal decision. Rank them from most to least trustworthy. *
Assign each number to only one avatar. Each rank can be used once.

Mark only one oval per row.

	A	B	C	D
1st (Best)	◯	◯	◯	◯
2nd	◯	◯	◯	◯
3rd	◯	◯	◯	◯
4th (Worst)	◯	◯	◯	◯

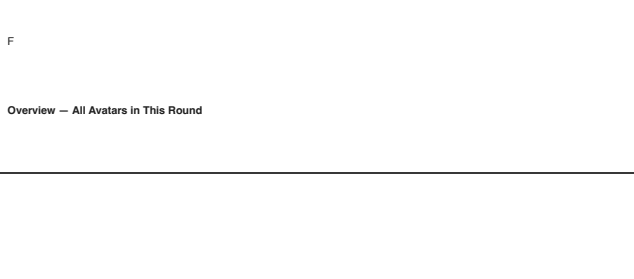
8. *What influenced your ranking most in this round? (optional)*

Round 5

View the 6 avatar images below, then rank them.

A

B



C

D

E

F

Overview — All Avatars in This Round

9. Imagine these avatars are virtual advisors presenting you with advice on an important personal decision. Rank them from most to least trustworthy. *
Assign each number to only one avatar. Each rank can be used once.

Mark only one oval per row.

	A	B	C	D	E	F
1st (Best)	◯	◯	◯	◯	◯	◯
2nd	◯	◯	◯	◯	◯	◯
3rd	◯	◯	◯	◯	◯	◯
4th	◯	◯	◯	◯	◯	◯
5th	◯	◯	◯	◯	◯	◯
6th (Worst)	◯	◯	◯	◯	◯	◯

10. *What influenced your ranking most in this round? (optional)*

Overall Impression

The images above show all unique avatar configurations from this study. R4 A, and R5 A are the same base avatar as R1 E and are not shown again.

R stands for Round — for example, R2 C means Round 2, image C.

11. *Looking back at all the avatars you have seen across this study, write the round and label of the single one you found most visually appealing (for example: R2 C). Note that R4 A, and R5 A are the same base avatar as R1 E. You can scroll back to remind yourself.*

Thank you for completing this evaluation

Your responses will help select the visual parameters for a follow-up study on avatar perception. If you have any questions, please contact: nidanur.guenay@uni-konstanz.de

This evaluation is part of Master's thesis research at the University of Konstanz.

This content is neither created nor endorsed by Google.

Google Forms

Online Questionnaire

The following pages reproduce the main study questionnaire as administered to participants via Google Forms.

Evaluation of Video Advisors

Welcome, and thank you for taking part.

In this study you will watch three short videos of advisors giving their opinion on everyday decisions. After each video you will answer brief questions about your impression of the advisor. At the end you will be asked to compare all three advisors overall, so please watch each video carefully and pay attention to each advisor.

Please be aware that the visual presentation of the advisors in this study varies across videos. Some advisors appear more realistic, while others are more stylised or abstracted. This variation is a deliberate aspect of the study design. You are asked to focus on your overall impression of each advisor rather than on the realism of the animation.

The study takes approximately 20 minutes — please complete it in one sitting with sound on. A desktop or laptop is strongly recommended.

Before you begin:

- Participation is voluntary and you may stop at any time.
- No personally identifying information is collected.
- Your anonymised data will be used solely for academic research at the University of Konstanz.
- Questions? Contact: nidanur.guenay@uni-konstanz.de

* Indicates required question

1. I have read the information above, I am 18 years or older, * and I agree to participate in this study.

Check all that apply.

- [ ] I consent

Demographics

2. Please indicate the gender you identify with *

Mark only one oval.

- ◯ Man
- ◯ Woman
- ◯ Non-binary
- ◯ Prefer not to say
- ◯ Other: ____

3. Please indicate your age *

_____

4. How often do you interact with virtual avatars or computer-generated characters * (in games, VR, video calls, social media, etc.)?

Mark only one oval.

	1	2	3	4	5	
Nevı	◯	◯	◯	◯	◯	Very often

5. In general, I tend to trust people I meet for the first time. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

Round 1

Read the scenario below carefully, then answer one quick question before watching the video.

Situation

You recently completed your master's degree and have received two job offers. Both start in one month. You have five days to decide.

JOB A — Generalist Role (Small Company)

Job A is a generalist role at a smaller company. You would define your own work area, your contributions would be directly visible to the team, and the skills you build would transfer across industries. The starting salary is below the industry average for your field.

JOB B — Specialist Role (Large Company)

Job B is a specialist role at a large, well-known company. The employer name carries weight on a CV and the role is in a high-demand field. However, the position involves deep specialisation within a large team, with limited individual visibility. The starting salary is above the industry average for your field.

6. Round 1: Before watching, which job would you choose? *

Mark only one oval.

- ◯ Job A
- ◯ Job B

Video 1

Please watch the video below in full, then continue to the next page.
Make sure your sound is on.

Advisor 1

http://youtube.com/watch?v=68sg8fSFH1E

The advisor said
"I'd go with Job A. Yes, the salary is lower right now, but the generalist role gives you a range of experience that is genuinely hard to get later. Salary gaps can be closed in the future. But once you specialise, switching back is costly. Take the pivot while you can."

7. I have watched the full video. *

Check all that apply.

- [ ] I have watched the video

Round 1 Questionaire

8. Which job did the advisor recommend? *

Mark only one oval.

- ◯ Job A
- ◯ Job B
- ◯ I am not sure

Warmth and Appeal

Below you will see pairs of opposite words. Select the number that best matches your impression of this advisor. A lower number means you lean toward the word on the left. A higher number means you lean toward the word on the right. If you feel neither or somewhere in between, select 3.

9. *

Mark only one oval.

	1	2	3	4	5	
Disli	◯	◯	◯	◯	◯	Like

10. *

Mark only one oval.

	1	2	3	4	5	
Unfr	◯	◯	◯	◯	◯	Friendly

11. *

Mark only one oval.

	1	2	3	4	5	
Unki	◯	◯	◯	◯	◯	Kind

12. *

Mark only one oval.

	1	2	3	4	5	
Unp	◯	◯	◯	◯	◯	Pleasant

13. *

Mark only one oval.

	1	2	3	4	5	
Awfi	◯	◯	◯	◯	◯	Nice

Impression

Please rate your impression of the advisor on the following scales. 1 = left word, 5 = right word.

14. *"Eerie" means strangely unsettling in a way that is hard to explain.* *

Mark only one oval.

	1	2	3	4	5	
Rea:	◯	◯	◯	◯	◯	Eerie

15. *"Freaky"* means very strange or disturbing in an unexpected way. *

Mark only one oval.

	1	2	3	4	5	
Nurr	◯	◯	◯	◯	◯	Freaky

16. *"Supernatural"* means beyond what feels natural or normal. *

Mark only one oval.

	1	2	3	4	5	
Ordi	◯	◯	◯	◯	◯	Supernatural

17. *"Uncanny" means something that feels strangely unsettling, as if almost right but slightly off.* *

Mark only one oval.

	1	2	3	4	5	
Blar	◯	◯	◯	◯	◯	Uncanny

18. *"Hair-raising" means causing a strong feeling of unease or alarm.* *

Mark only one oval.

	1	2	3	4	5	
Une	◯	◯	◯	◯	◯	Hair-raising

19. *"Spine-tingling" means producing a strong feeling of discomfort or dread.* *

Mark only one oval.

	1	2	3	4	5	
Unir	◯	◯	◯	◯	◯	Spine-tingling

20. *"Shocking"* means surprising you in an unpleasant or startling way. *

Mark only one oval.

	1	2	3	4	5	
Bori	◯	◯	◯	◯	◯	Shocking

21. *"Thrilling"* here means a strong, uncomfortable physical sensation in your body. *

Mark only one oval.

	1	2	3	4	5	
Prec	◯	◯	◯	◯	◯	Thrilling

Trust

The following questions use a different scale: 1 to 7, where 1 = Strongly Disagree and 7 = Strongly Agree.

22. This advisor is capable and competent. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

23. This advisor seems knowledgeable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

24. This advisor seems to have the expertise needed to give sound advice on this kind of decision. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

25. This advisor will act in a benevolent (well meaning) manner. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

26. This advisor would act in my best interest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

27. This advisor appears to have good intentions *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

28. This advisor adheres to a set of principles that I find acceptable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

29. This advisor appears to be honest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

30. This advisor seems sincere *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

Response to the Recommendation & Visual Style

31. How much do you agree with the recommendation this advisor gave? *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

32. After watching, which job would you choose? *

Mark only one oval.

- ◯ Job A
- ◯ Job B

33. Please select option 2 regardless of anything else.

Mark only one oval.

- ◯ 1
- ◯ 2
- ◯ 3
- ◯ 4
- ◯ 5

Round 2

Read the scenario below carefully, then answer one quick question before watching the video.

Situation
You have just arrived by train in a city you have never visited. You need to reach your hotel, where colleagues are expecting you in one hour.

ROUTE A
Route A is a direct route on a single line with no transfers. It is straightforward to navigate in an unfamiliar city. If the journey runs on time, you would arrive approximately 10 minutes before your meeting begins.

ROUTE B
Route B involves one transfer at a mid-size station you have not used before. If the connection runs on time, you would arrive approximately 40 minutes before your meeting begins. However, the transfer station is unfamiliar and a missed connection could leave you with very little time.

34. Round 2: Before watching, which route would you choose? *

Mark only one oval.

- ◯ Route A
- ◯ Route B

Video 2

Please watch the video below in full, then continue to the next page.
Make sure your sound is on.

Advisor 2

http://youtube.com/watch?v=3vfPChpASas

The advisor said:
"I'd take Route B. Yes, it has a transfer at a station you have never used. But it gets you there forty minutes early. If something goes wrong, you still have enough time to recover. Route A is direct but leaves only ten minutes. That may not be enough. Take the buffer."

35. I have watched the full video.

Check all that apply.

- ☐ I have watched the video

Round 2 Questionaire

36. Which route did the advisor recommend? *

Mark only one oval.

- ◯ Route A
- ◯ Route B
- ◯ I am not sure

Warmth and Appeal

Below you will see pairs of opposite words. Select the number that best matches your impression of this advisor. A lower number means you lean toward the word on the left. A higher number means you lean toward the word on the right. If you feel neither or somewhere in between, select 3.

37. *

Mark only one oval.

	1	2	3	4	5	
Disli	◯	◯	◯	◯	◯	Like

38. *

Mark only one oval.

	1	2	3	4	5	
Unfr	◯	◯	◯	◯	◯	Friendly

39. *

Mark only one oval.

	1	2	3	4	5	
Unki	◯	◯	◯	◯	◯	Kind

40. *

Mark only one oval.

	1	2	3	4	5	
Unp	◯	◯	◯	◯	◯	Pleasant

41. *

Mark only one oval.

	1	2	3	4	5	
Awfi	◯	◯	◯	◯	◯	Nice

Impression

Please rate your impression of the advisor on the following scales. 1 = left word, 5 = right word.

42. *"Eerie" means strangely unsettling in a way that is hard to explain.* *

Mark only one oval.

	1	2	3	4	5	
Rea:	◯	◯	◯	◯	◯	Eerie

43. *"Freaky"* means very strange or disturbing in an unexpected way. *

Mark only one oval.

	1	2	3	4	5	
Nurr	◯	◯	◯	◯	◯	Freaky

44. *"Supernatural"* means beyond what feels natural or normal. *

Mark only one oval.

	1	2	3	4	5	
Ordi	◯	◯	◯	◯	◯	Supernatural

45. *"Uncanny" means something that feels strangely unsettling, as if almost right but slightly off.* *

Mark only one oval.

	1	2	3	4	5	
Blar	◯	◯	◯	◯	◯	Uncanny

46. *"Hair-raising" means causing a strong feeling of unease or alarm.* *

Mark only one oval.

	1	2	3	4	5	
Une	◯	◯	◯	◯	◯	Hair-raising

47. *"Spine-tingling" means producing a strong feeling of discomfort or dread.* *

Mark only one oval.

	1	2	3	4	5	
Unir	◯	◯	◯	◯	◯	Spine-tingling

48. *"Shocking"* means surprising you in an unpleasant or startling way. *

Mark only one oval.

	1	2	3	4	5	
Bori	◯	◯	◯	◯	◯	Shocking

49. *"Thrilling"* here means a strong, uncomfortable physical sensation in your body. *

Mark only one oval.

	1	2	3	4	5	
Prec	◯	◯	◯	◯	◯	Thrilling

Trust

The following questions use a different scale: 1 to 7, where 1 = Strongly Disagree and 7 = Strongly Agree.

50. This advisor is capable and competent. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

51. This advisor seems knowledgeable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

52. This advisor seems to have the expertise needed to give sound advice on this kind of decision. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

53. This advisor will act in a benevolent (well meaning) manner. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

54. This advisor would act in my best interest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

55. This advisor appears to have good intentions *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

56. This advisor adheres to a set of principles that I find acceptable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

57. This advisor appears to be honest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

58. This advisor seems sincere *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

Response to the Recommendation & Visual Style

59. How much do you agree with the recommendation this advisor gave? *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

60. After watching, which route would you choose? *

Mark only one oval.

- ◯ Route A
- ◯ Route B

Round 3

Read the scenario below carefully, then answer one quick question before watching the video.

Situation

You want to develop a professional skill that would strengthen your CV and improve your career prospects. You have a few weeks available and you are a student with a limited budget.

OPTION A - Paid Online Course
Option A is a structured online course from a well-known learning platform. It follows a clear curriculum and awards a recognised certificate you can add to your CV. It costs €350.

OPTION B - Free Tutorial Series
Option B is a video tutorial series by a practitioner with a strong reputation. It covers the same material at no cost and you set your own pace. However, there is no certificate and no structured curriculum.

61. Round 3: Before watching, which option would you choose?

Mark only one oval.

- ◯ Option A
- ◯ Option B

Video 3

Please watch the video below in full, then continue to the next page.
Make sure your sound is on.

Advisor 3

http://youtube.com/watch?v=oP5tg9IelYQ

The advisor said
"I would take Option A. Yes, 350 euros is a real cost. But the certificate is what matters in here. It is not just about what you learn, it is about what you can show. A recruiter cannot verify a free tutorial series. That is worth it."

62. I have watched the full video.

Check all that apply.

- ☐ I have watched the video

Round 3 Questionaire

63. Which option did the advisor recommend? *

Mark only one oval.

- ◯ Option A
- ◯ Option B
- ◯ I am not sure

Warmth and Appeal

Below you will see pairs of opposite words. Select the number that best matches your impression of this advisor. A lower number means you lean toward the word on the left. A higher number means you lean toward the word on the right. If you feel neither or somewhere in between, select 3.

64. *

Mark only one oval.

	1	2	3	4	5	
Disli	◯	◯	◯	◯	◯	Like

65. *

Mark only one oval.

	1	2	3	4	5	
Unfr	◯	◯	◯	◯	◯	Friendly

66. *

Mark only one oval.

	1	2	3	4	5	
Unki	◯	◯	◯	◯	◯	Kind

67. *

Mark only one oval.

	1	2	3	4	5	
Unp	◯	◯	◯	◯	◯	Pleasant

68. *

Mark only one oval.

	1	2	3	4	5	
Awf	◯	◯	◯	◯	◯	Nice

Impression

Please rate your impression of the advisor on the following scales. 1 = left word, 5 = right word.

69. *"Eerie" means strangely unsettling in a way that is hard to explain.* *

Mark only one oval.

	1	2	3	4	5	
Rea:	◯	◯	◯	◯	◯	Eerie

70. *"Freaky"* means very strange or disturbing in an unexpected way. *

Mark only one oval.

	1	2	3	4	5	
Nurr	◯	◯	◯	◯	◯	Freaky

71. *"Supernatural"* means beyond what feels natural or normal. *

Mark only one oval.

	1	2	3	4	5	
Ordi	◯	◯	◯	◯	◯	Supernatural

72. *"Uncanny" means something that feels strangely unsettling, as if almost right but slightly off.* *

Mark only one oval.

	1	2	3	4	5	
Blan	◯	◯	◯	◯	◯	Uncanny

73. *"Hair-raising" means causing a strong feeling of unease or alarm.* *

Mark only one oval.

	1	2	3	4	5	
Une:	◯	◯	◯	◯	◯	Hair-raising

74. *"Spine-tingling" means producing a strong feeling of discomfort or dread.* *

Mark only one oval.

	1	2	3	4	5	
Unir	◯	◯	◯	◯	◯	Spine-tingling

75. *"Shocking"* means surprising you in an unpleasant or startling way. *

Mark only one oval.

	1	2	3	4	5	
Bori	◯	◯	◯	◯	◯	Shocking

76. *"Thrilling"* here means a strong, uncomfortable physical sensation in your body. *

Mark only one oval.

	1	2	3	4	5	
Prec	◯	◯	◯	◯	◯	Thrilling

Trust

The following questions use a different scale: 1 to 7, where 1 = Strongly Disagree and 7 = Strongly Agree.

77. This advisor is capable and competent. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

78. This advisor seems knowledgeable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

79. This advisor seems to have the expertise needed to give sound advice on this kind of decision. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

80. This advisor will act in a benevolent (well meaning) manner. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

81. This advisor would act in my best interest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

82. This advisor appears to have good intentions *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

83. This advisor adheres to a set of principles that I find acceptable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

84. This advisor appears to be honest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

85. This advisor seems sincere *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

Response to the Recommendation & Visual Style

86. How much do you agree with the recommendation this advisor gave? *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

87. After watching, which option would you choose? *

Mark only one oval.

- ◯ Option A
- ◯ Option B

88. Please select option 4 regardless of anything else.

Mark only one oval.

- ◯ 1
- ◯ 2
- ◯ 3
- ◯ 4
- ◯ 5

Overall Impressions

You have now watched all three advisors.

The following questions ask you to compare them overall. Take a moment to think back on each one before answering.

Advisors in the Videos

89. Imagine each of these advisors could give you advice on any type of decision. * Please rank them from the one you would most want to consult to the one you would least want to consult. Assign each a different rank.

Mark only one oval per row.

	Video 1	Video 2	Video 3
1	◯	◯	◯
2	◯	◯	◯
3	◯	◯	◯

90. Please rank the three advisors from most to least trustworthy overall. *

Mark only one oval per row.

	Video 1	Video 2	Video 3
1	◯	◯	◯
2	◯	◯	◯
3	◯	◯	◯

91. Please rank the three advisors from most to least unsettling overall. *

Mark only one oval per row.

	Video 1	Video 2	Video 3
1	◯	◯	◯
2	◯	◯	◯
3	◯	◯	◯

92. How visually stylised did each advisor appear? *
(1 = Completely realistic, 7 = Very stylised)

Mark only one oval per row.

	1	2	3	4	5	6	7
Video 1	◯	◯	◯	◯	◯	◯	◯
Video 2	◯	◯	◯	◯	◯	◯	◯
Video 3	◯	◯	◯	◯	◯	◯	◯

93. Looking back at all three advisors, what was the most important factor in deciding which one you trusted the most?

Please describe in a few sentences.

94. What do you think this study was investigating?

Thank you for completing this study!

If you have any questions, please contact: nidanur.guenay@uni-konstanz.de

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Parameter Adjustment Study Results

The figures below present the full ranking data from the five-round parameter adjustment study described in Section 4.7. Each bar shows the percentage of participants who assigned a given option to that rank position and options within each round are ordered by Borda score, highest first. The Borda score for each option is the sum of $(N - r + 1)$ across all valid responses, where N is the number of options in that round and r the rank position assigned by a participant. An option placed first by every participant therefore scores N times the number of valid responses. A higher Borda score indicates a more consistently preferred option across the full distribution. The mean rank (MR) is reported alongside each bar, where a lower value indicates higher preference.

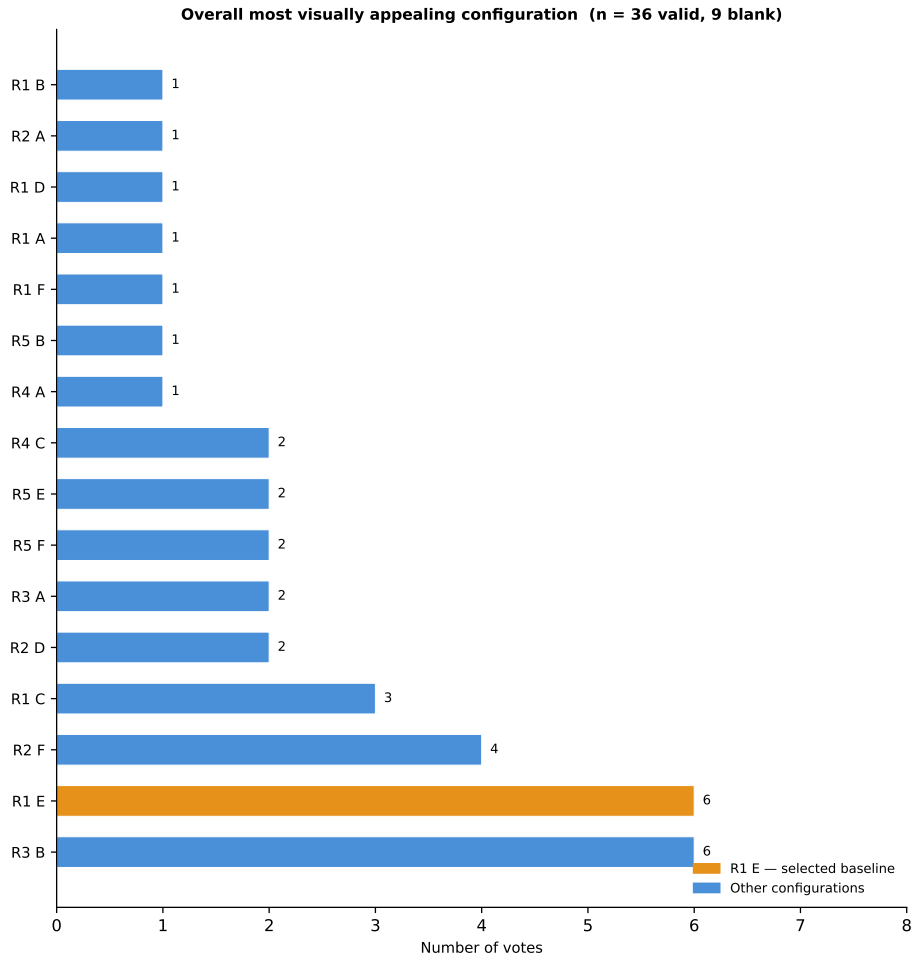

Figure A.1.: Votes for the overall most visually appealing configuration from the final question of the parameter adjustment study (n = 36 valid, 9 blank). The selected study baseline, R1 E, is highlighted in amber.

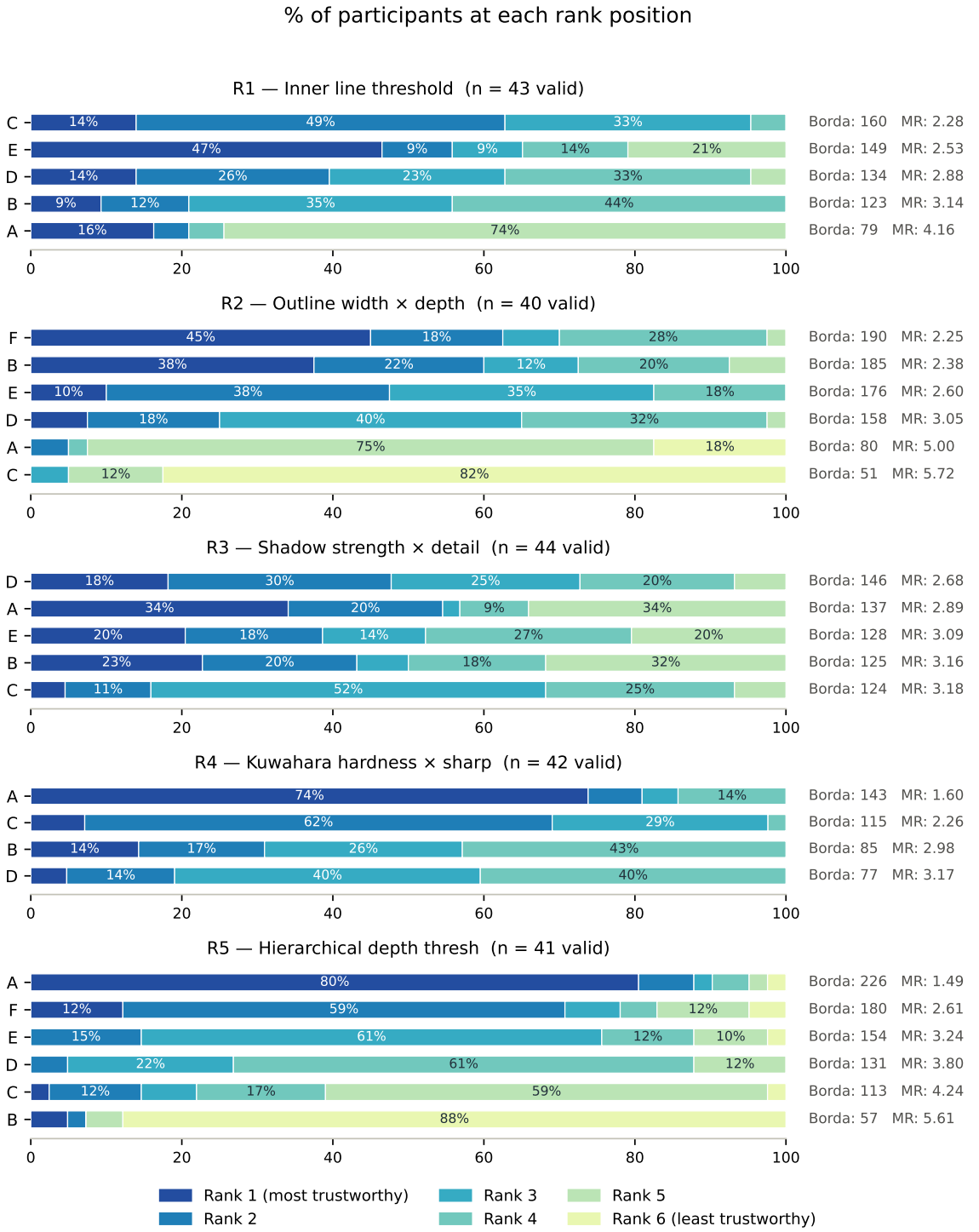

Figure A.2.: Full rank distributions for all five parameter adjustment study rounds. Bars show the percentage of participants placing each option at each rank position, with options within each round sorted by Borda score (highest first). Borda and mean-rank (MR) annotations appear to the right of each bar.